# LANDSCAPE DASHBOARD — TIER-1 & TOP-N GENOMIC REPORT DIGEST

**Cohort Subject:** Melinda Ehrle (Melinda_Ehrle) · **Mean Called Depth:** 29.11x (Median 30.0x) · **Date:** October 13, 2026

Pre-expanded static digest of the **Tier-1 / ClinVar Pathogenic genes** and **Top-N highest-scoring genes** across all 8 Dashboard Report views (Visual Explorer, ClinVar, AlphaGenome Atlas, CADD, SpliceAI, AlphaMissense, REVEL, and Rare Variants ≤1%).

**Total Cohort Catalog:** 1653 Variants / 1129 Genes
**Output Formats:** .HTML · .MHTML (RFC 2557) · .PDF

## 1. Visual Explorer — Tier-1 & Top Clinical Impact Genes

Master genomic catalog filtered strictly to genes carrying Tier-1 / ClinVar Pathogenic variants and highest multi-engine clinical impact scores.

**Selected Scope:** 43 Genes (43 Tier-1/Pathogenic)
Filtered from 1129 matching genes in view

### VWA3B VWA3B chr2:98,162,976 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 46.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an intracellular protein that contains a von Willebrand factor type A domain. Intracellular proteins with VWA domains are thought to function in transcription, DNA repair, ribosomal and membrane transport and the proteasome. Mutations in this gene are associated with Spinocerebellar ataxia, autosomal recessive 22. Alternatively spliced transcript variants have been found for this gene. [provided by RefSeq, May 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs62154921 p.Glu372Ter | STG, p.Glu372Ter | Heterozygous | Unphased | Benign | Tier1 | 46.0 | 0.85 | 50.8 (Protein Termination) | – / – | chr2:98162976 AF: 4.2196% |

#### 1. INHERITANCE & OMIM PATHOLOGY

**VWA3B-related disorder**
Complex / Multifactorial

Quadrant: **Curated**

#### 2. TRANSCRIPT, PHASING & DEPTH

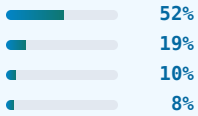
| | |
|---|---|
| Transcript: | ENST00000477737.6 |
| cDNA (HGVS): | c.1114G>T |
| Protein (HGVS): | p.Glu372Ter |
| VAF / Read Depth: | VAF 0.44 · DP – |
| Phased GT: | 0/1 (Unphased) |

#### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.05 DL:0.85 |
| ClinVar Review: | no assertion criteria provided |
| ClinVar VCV: | VCV#3056040 ↗ |
| gnomAD v4 AF: | 4.2196% |

#### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 50.8 (Raw: 2.870 · Top 0.00%)
Primary Driver: Protein Termination




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TAT TAT chr16:71,576,247 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 36.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This nuclear gene encodes a mitochondrial protein tyrosine aminotransferase which is present in the liver and catalyzes the conversion of L-tyrosine into p-hydroxyphenylpyruvate. Mutations in this gene cause tyrosinemia (type II, Richner-Hanhart syndrome), a disorder accompanied by major skin and corneal lesions, with possible cognitive disability. A regulator gene for tyrosine aminotransferase is X-linked. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs118203914 p.Arg57Ter | STG, p.Arg57Ter | Heterozygous | Maternal | Pathogenic | Tier1 | 36.0 | — | 38.8 (Protein Termination) | — / — | chr16:71576247 AF: 0.0017% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Tyrosinemia type II**

Complex / Multifactorial OMIM #276600 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000355962.5 |
| cDNA (HGVS): | c.169C>T |
| Protein (HGVS): | p.Arg57Ter |
| VAF / Read Depth: | VAF 0.46 · DP — |
| Phased GT: | 0\|1 (Maternal) |

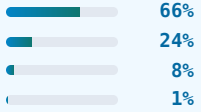
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#402 ↗ |
| gnomAD v4 AF: | 0.0017% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.8 (Raw: 2.285 · Top 0.01%)

Primary Driver: Protein Termination




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs572677025 c.*409G>A | UT3 | Heterozygous | Unphased | Uncertain significance | Tier2 | 0.8 | — | 5.8 (Splicing) | — / — | chr16:71567735 AF: 0.4009% |

## CEP63 CEP63 chr3:134,537,268 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 34.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with six coiled-coil domains. The protein is localized to the centrosome, a non-membraneous organelle that functions as the major microtubule-organizing center in animal cells. Several alternatively spliced transcript variants have been found, but their biological validity has not been determined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114108011 p.Gln185His | MIS, p.Gln185His | Heterozygous | Maternal | Benign | Tier1 | 34.0 | 0.93 | 38.6 (Splicing) | 0.28 / 0.54 | chr3:134537268 AF: 2.0823% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000675561.1 |
| cDNA (HGVS): | c.555G>C |
| Protein (HGVS): | p.Gln185His |
| VAF / Read Depth: | VAF 0.74 · DP — |
| Phased GT: | 0\|1 (Maternal) |

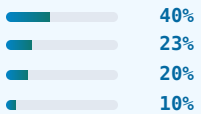
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.93 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#128709 ↗ |
| gnomAD v4 AF: | 2.0823% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.6 (Raw: 2.279 · Top 0.01%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

**OGFOD3** OGFOD3 chr17:82,411,454 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 34.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable several functions, including L-ascorbic acid binding activity; dioxygenase activity; and iron ion binding activity. Located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs150938774<br>c.380+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 34.0 | 0.96 | 33.8 (Splicing) | – / – | chr17:82411454<br>AF: 0.6286% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000313056.10 |
| cDNA (HGVS): | c.380+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

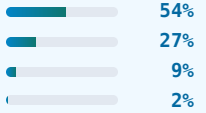
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.42 DL:0.96 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4429500 ↗ |
| gnomAD v4 AF: | 0.6286% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.882 · Top 0.04%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB5 HLA-DRB5 chr6:32,520,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 33.0 Showing 4 of 36 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB5 belongs to the HLA class II beta chain paralogues. This class II molecule is a heterodimer consisting of an alpha (DRA) and a beta (DRB) chain, both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa and its gene contains 6 exons. Exon one encodes the leader peptide, exons 2 and 3 encode the two extracellular domains, exon 4 encodes the transmembrane domain and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. There are multiple pseudogenes of this gene. [provided by RefSeq, Feb 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs201576424 c.101-1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.93 | 28.0 (Splicing) | – / – | chr6:32522175 AF: 0.0063% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

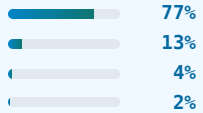
Transcript: **ENST00000374975.4**
cDNA (HGVS): **c.101-1G>A**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.44 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.65 AL:0.93 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4326162 ↗
gnomAD v4 AF: **0.0063%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **28.0** (Raw: 1.444 · Top 0.16%)
Primary Driver: **Splicing**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs796105724 c.371-392G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 8.4 | 1.00 | 25.7 (Splicing) | – / – | chr6:32520043 AF: 1.4415% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

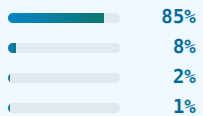
Transcript: **ENST00000374975.4**
cDNA (HGVS): **c.371-392G>A**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.62 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:1.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **1.4415%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.7** (Raw: 1.223 · Top 0.27%)
Primary Driver: **Splicing**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs774541765 c.100+17G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.7 | 0.68 | 18.9 (Splicing) | – / – | chr6:32530108 AF: 0.0027% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

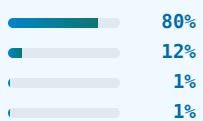
Transcript: **ENST00000374975.4**
cDNA (HGVS): **c.100+17G>A**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.46 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.68 DL:0.01**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0027%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **18.9** (Raw: 0.503 · Top 1.28%)
Primary Driver: **Splicing**



| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | 🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗ | | |
| rs796784269<br>c.371-393T>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 2.8 | 0.94 | 16.6 (Splicing) | – / – | chr6:32520044<br>AF: 0.1426% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000374975.4 |
| cDNA (HGVS): | c.371-393T>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.94 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.1426% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 16.6 (Raw: 0.386 · Top 2.21%)
Primary Driver: Splicing

87%
5%
3%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## IL17RC IL17RC chr3:9,932,859 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 33.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a single-pass type I membrane protein that shares similarity with the interleukin-17 receptor (IL-17RA). Unlike IL-17RA, which is predominantly expressed in hemopoietic cells, and binds with high affinity to only IL-17A, this protein is expressed in nonhemopoietic tissues, and binds both IL-17A and IL-17F with similar affinities. The proinflammatory cytokines, IL-17A and IL-17F, have been implicated in the progression of inflammatory and autoimmune diseases. Multiple alternatively spliced transcript variants encoding different isoforms have been detected for this gene, and it has been proposed that soluble, secreted proteins lacking transmembrane and intracellular domains may function as extracellular antagonists to cytokine signaling. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs148575246<br>c.1522+1G>C | SPL | Heterozygous | Unphased | Benign/Likely benign | Tier1 | 33.0 | 0.97 | 25.9 (Splicing) | – / – | chr3:9932859<br>AF: 1.0744% |

**1. INHERITANCE & OMIM PATHOLOGY**

**IL17RC-related disorder**
Complex / Multifactorial OMIM #616445 ↗

**Candidiasis, familial, 9**
Complex / Multifactorial OMIM #616445 ↗

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000403601.8 |
| cDNA (HGVS): | c.1522+1G>C |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.55 · DP – |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.01 DL:0.97 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#542537 ↗ |
| gnomAD v4 AF: | 1.0744% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)
Primary Driver: Splicing

69%
14%
12%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs115461448<br>p.Ala324Thr | MIS, p.Ala324Thr | Heterozygous | Unphased | Benign | Tier3 | 23.2 | – | 22.0 (Cactus) | 0.09 / 0.17 | chr3:9928397<br>AF: 1.4147% |

## NIPSNAP2 NIPSNAP2 chr7:55,994,989 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 33.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the NipSnap family of proteins that may be involved in vesicular transport. The encoded protein is localized to mitochondria and plays a role in oxidative phosphorylation. A pseudogene of this gene is located on the long arm of chromosome 2. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs748801121<br>c.712+1G>T | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.97 | 33.8 (Splicing) | – / – | chr7:55994989<br>AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000322090.8 |
| cDNA (HGVS): | c.712+1G>T |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.50 · DP – |
| Phased GT: | 0/1 (Unphased) |

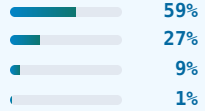
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.44 DL:0.97 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0075% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.881 · Top 0.04%)

Primary Driver: Splicing




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLIT2 SLIT2 chr4:20,567,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the slit family of secreted glycoproteins, which are ligands for the Robo family of immunoglobulin receptors. Slit proteins play highly conserved roles in axon guidance and neuronal migration and may also have functions during other cell migration processes including leukocyte migration. Members of the slit family are characterized by an N-terminal signal peptide, four leucine-rich repeats, nine epidermal growth factor repeats, and a C-terminal cysteine knot. Proteolytic processing of this protein gives rise to an N-terminal fragment that contains the four leucine-rich repeats and five epidermal growth factor repeats and a C-terminal fragment that contains four epidermal growth factor repeats and the cysteine knot. Both full length and cleaved proteins are secreted extracellularly and can function in axon repulsion as well as other specific processes. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2015]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1020555095<br>p.Gly909Asp | MIS, p.Gly909Asp | Heterozygous | Unphased | Not reviewed | Tier1 | 32.0 | – | 23.4 (Cactus) | 0.74 / 0.35 | chr4:20567262<br>AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000504154.6 |
| cDNA (HGVS): | c.2726G>A |
| Protein (HGVS): | p.Gly909Asp |
| VAF / Read Depth: | VAF 0.54 · DP – |
| Phased GT: | 0/1 (Unphased) |

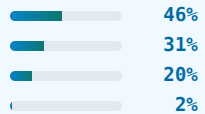
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0009% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.4 (Raw: 0.859 · Top 0.46%)

Primary Driver: Cactus




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TMEM43 TMEM43 chr3:14,139,244 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 32.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to the TMEM43 family. Defects in this gene are the cause of familial arrhythmogenic right ventricular dysplasia type 5 (ARVD5), also known as arrhythmogenic right ventricular cardiomyopathy type 5 (ARVC5). Arrhythmogenic right ventricular dysplasia is an inherited disorder, often involving both ventricles, and is characterized by ventricular tachycardia, heart failure, sudden cardiac death, and fibrofatty replacement of cardiomyocytes. This gene contains a response element for PPAR gamma (an adipogenic transcription factor), which may explain the fibrofatty replacement of the myocardium, a characteristic pathological finding in ARVC. [provided by RefSeq, Oct 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs199526104 p.Trp316Ser | MIS, p.Trp316Ser | Heterozygous | Paternal | Conflicting classifications of pathogenicity | Tier1 | 32.0 | – | 26.5 (AlphaMissense) | 0.53 / 0.76 | chr3:14139244 AF: 0.0126% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**
Complex / Multifactorial OMIM #604400 ↗

**TMEM43-related disorder**
Complex / Multifactorial OMIM #604400 ↗

**Auditory neuropathy, autosomal dominant 3**
Autosomal Dominant OMIM #604400 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

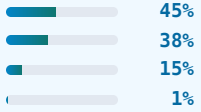
Transcript: ENST00000306077.5
cDNA (HGVS): c.947G>C
Protein (HGVS): p.Trp316Ser
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#46155 ↗
gnomAD v4 AF: 0.0126%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.5 (Raw: 1.335 · Top 0.22%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1043943 c.*115T>C | UT3 | Heterozygous | Paternal | Benign/Likely benign | Tier2 | 1.6 | – | 9.2 (ATAC-seq) | – / – | chr3:14141910 AF: 62.2376% |

## AMPD1 AMPD1 chr1:114,677,435 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Adenosine monophosphate deaminase 1 catalyzes the deamination of AMP to IMP in skeletal muscle and plays an important role in the purine nucleotide cycle. Two other genes have been identified, AMPD2 and AMPD3, for the liver- and erythocyte-specific isoforms, respectively. Deficiency of the muscle-specific enzyme is apparently a common cause of exercise-induced myopathy and probably the most common cause of metabolic myopathy in the human. Alternatively spliced transcript variants encoding different isoforms have been identified in this gene.[provided by RefSeq, Feb 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs764950035 p.Glu435Gly | MIS, p.Glu435Gly | Heterozygous | Unphased | Uncertain significance | Tier1 | 32.0 | – | 28.2 (AlphaMissense) | 0.95 / 0.92 | chr1:114677435 AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Muscle AMP deaminase deficiency**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

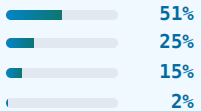
Transcript: ENST00000520113.7
cDNA (HGVS): c.1304A>G
Protein (HGVS): p.Glu435Gly
VAF / Read Depth: VAF 0.49 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#2173226 ↗
gnomAD v4 AF: 0.0075%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.2 (Raw: 1.454 · Top 0.15%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## LDHAL6A LDHAL6A chr11:18,475,640 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 31.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables L-lactate dehydrogenase (NAD+) activity. Predicted to be involved in lactate metabolic process and pyruvate catabolic process. Located in cytoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs145833201<br>c.592+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 31.0 | 0.69 | 23.3 (Splicing) | – / – | chr11:18475640<br>AF: 0.1631% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000280706.3 |
| cDNA (HGVS): | c.592+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.36 · DP – |
| Phased GT: | 0/1 (Unphased) |

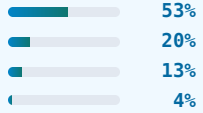
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.69 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4370113 ↗ |
| gnomAD v4 AF: | 0.1631% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.3 (Raw: 0.845 · Top 0.47%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## DMAP1 DMAP1 chr1:44,218,751 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 29.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a subunit of several, distinct complexes involved in the repression or activation of transcription. The encoded protein can independently repress transcription and is targeted to replication foci throughout S phase by interacting directly with the N-terminus of DNA methyltransferase 1. During late S phase, histone deacetylase 2 is added to this complex, providing a means to deacetylate histones in transcriptionally inactive heterochromatin following replication. The encoded protein is also a component of the nucleosome acetyltransferase of H4 complex and interacts with the transcriptional corepressor tumor susceptibility gene 101 and the pro-apoptotic death-associated protein 6, among others. Alternatively spliced transcript variants encoding the same protein have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| chr1:44218751<br>p.Glu239Gly | MIS, p.Glu239Gly | Heterozygous | Unphased | Not reviewed | Tier1 | 29.7 | – | 23.4 (Cactus) | 0.38 / 0.38 | chr1:44218751<br>AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000372289.7 |
| cDNA (HGVS): | c.716A>G |
| Protein (HGVS): | p.Glu239Gly |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0/1 (Unphased) |

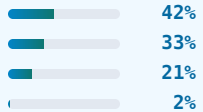
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0001% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.4 (Raw: 0.854 · Top 0.46%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

**PLK4** PLK4 chr4:127,881,151 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 28.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polo family of serine/threonine protein kinases. The protein localizes to centrioles, complex microtubule-based structures found in centrosomes, and regulates centriole duplication during the cell cycle. Three alternatively spliced transcript variants that encode different protein isoforms have been found for this gene. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs149003893<br>p.Gly6Glu | MIS, p.Gly6Glu | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 28.5 | – | 29.8 (AlphaMissense) | 0.36 / 0.99 | chr4:127881151<br>AF: 0.1341% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Inborn genetic diseases**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000270861.10 |
| cDNA (HGVS): | c.17G>A |
| Protein (HGVS): | p.Gly6Glu |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

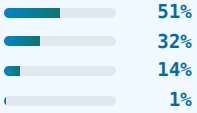
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#436338 ↗ |
| gnomAD v4 AF: | 0.1341% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.8 (Raw: 1.555 · Top 0.11%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB1 HLA-DRB1 chr6:32,582,135 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 28.0 Showing 3 of 35 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB1 belongs to the HLA class II beta chain paralogs. The class II molecule is a heterodimer consisting of an alpha (DRA) and a beta chain (DRB), both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa. It is encoded by 6 exons. Exon one encodes the leader peptide; exons 2 and 3 encode the two extracellular domains; exon 4 encodes the transmembrane domain; and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Hundreds of DRB1 alleles have been described and some alleles have increased frequencies associated with certain diseases or conditions. For example, DRB1*1302 has been related to acute and chronic hepatitis B virus persistence. There are multiple pseudogenes of this gene. [provided by RefSeq, Jul 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28723976<br>c.371-297A>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 7.4 | 0.58 | 12.9 (Splicing) | – / – | chr6:32582135<br>AF: 0.7402% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.371-297A>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.41 · DP – |
| Phased GT: | 0/1 (Unphased) |

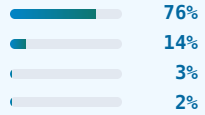
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.58 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.7402% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 12.9 (Raw: 0.248 · Top 5.08%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs16822805<br>p.Ala103Pro | MIS, p.Ala103Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 0.0 | – | 20.2 (AlphaMissense) | 0.21 / 0.78 | chr6:32584172<br>AF: 0.1762% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.307G>C |
| Protein (HGVS): | p.Ala103Pro |
| VAF / Read Depth: | VAF 0.53 · DP – |
| Phased GT: | 0/1 (Unphased) |

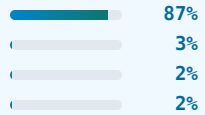
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#691505 ↗ |
| gnomAD v4 AF: | 0.1762% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 20.2 (Raw: 0.573 · Top 0.95%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs11554463<br>p.Tyr107Ter | STG, p.Tyr107Ter | Heterozygous | Unphased | Not reviewed | Tier2 | 28.0 | – | 27.2 (Protein Termination) | – / – | chr6:32584158<br>AF: 0.0001% |

## ACSM2B ACSM2B chr16:20,540,665 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 27.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables benzoate-CoA ligase activity. Predicted to be involved in acyl-CoA metabolic process and fatty acid biosynthetic process. Located in mitochondrion. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs139253186 p.Tyr540Asn | MIS, p.Tyr540Asn | Heterozygous | Unphased | Not reviewed | Tier1 | 27.6 | — | 24.4 (AlphaMissense) | 0.40 / 0.58 | chr16:20540665 AF: 0.4705% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000329697.10 |
| cDNA (HGVS): | c.1618T>A |
| Protein (HGVS): | p.Tyr540Asn |
| VAF / Read Depth: | VAF 0.54 · DP — |
| Phased GT: | 0/1 (Unphased) |

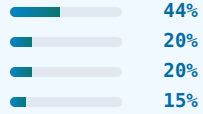
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.4705% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.4 (Raw: 1.012 · Top 0.37%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAHD1 FAHD1 chr16:1,827,542 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 27.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables hydrolase activity, acting on carbon-carbon bonds, in ketonic substances; oxaloacetate decarboxylase activity; and oxaloacetate tautomerase activity. Involved in oxaloacetate metabolic process. Located in cytosol; mitochondrion; and nucleoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs200357596 p.Ala102Ser | MIS, p.Ala102Ser | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | — | 22.1 (Cactus) | 0.71 / 0.24 | chr16:1827542 AF: 0.0037% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000427358.5 |
| cDNA (HGVS): | c.304G>T |
| Protein (HGVS): | p.Ala102Ser |
| VAF / Read Depth: | VAF 0.49 · DP — |
| Phased GT: | 0/1 (Unphased) |

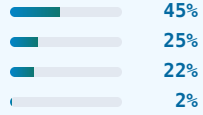
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0037% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.1 (Raw: 0.710 · Top 0.62%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLC27A3 SLC27A3 chr1:153,776,015 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 27.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to a family of integral membrane proteins and encodes a protein that is involved in lipid metabolism. The increased expression of this gene in human neural stem cells derived from induced pluripotent stem cells suggests that it plays an important role in early brain development. Naturally occurring mutations in this gene are associated with autism spectrum disorders. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Feb 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs756978807 p.Leu173Pro | MIS, p.Leu173Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 26.9 (AlphaMissense) | 0.41 / 0.94 | chr1:153776015 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000624995.4 |
| cDNA (HGVS): | c.518T>C |
| Protein (HGVS): | p.Leu173Pro |
| VAF / Read Depth: | VAF 0.23 · DP – |
| Phased GT: | 0/1 (Unphased) |

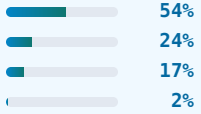
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0001% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.9 (Raw: 1.371 · Top 0.21%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## POLG POLG chr15:89,323,460 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 26.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Mitochondrial DNA polymerase is heterotrimeric, consisting of a homodimer of accessory subunits plus a catalytic subunit. The protein encoded by this gene is the catalytic subunit of mitochondrial DNA polymerase. The encoded protein contains a polyglutamine tract near its N-terminus that may be polymorphic. Defects in this gene are a cause of progressive external ophthalmoplegia with mitochondrial DNA deletions 1 (PEOA1), sensory ataxic neuropathy dysarthria and ophthalmoparesis (SANDO), Alpers-Huttenlocher syndrome (AHS), and mitochondrial neurogastrointestinal encephalopathy syndrome (MNGIE). Two transcript variants encoding the same protein have been found for this gene. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs121918054 p.Gly737Arg | MIS, p.Gly737Arg | Heterozygous | Paternal | Pathogenic/Likely pathogenic | Tier1 | 26.3 | – | 30.3 (AlphaMissense) | 0.94 / 0.94 | chr15:89323460 AF: 0.1619% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Spinocerebellar ataxia with epilepsy**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Sensory ataxic neuropathy, dysarthria, and ophthalmoparesis**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 1**
Autosomal Dominant OMIM #203700; 607459; 613662; 157640; 258450 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000268124.11 |
| cDNA (HGVS): | c.2209G>C |
| Protein (HGVS): | p.Gly737Arg |
| VAF / Read Depth: | VAF 0.61 · DP – |
| Phased GT: | 1\|0 (Paternal) |

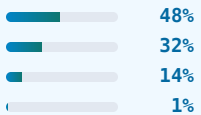
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#13513 ↗ |
| gnomAD v4 AF: | 0.1619% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.3 (Raw: 1.582 · Top 0.09%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## MID1IP1 MID1IP1 chrX:38,805,491 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 26.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable enzyme activator activity. Predicted to be involved in several processes, including negative regulation of microtubule depolymerization; positive regulation of fatty acid biosynthetic process; and positive regulation of ligase activity. Predicted to be located in microtubule cytoskeleton. Predicted to be active in cytosol and nucleus. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs143008624 p.Gly182Asp | MIS, p.Gly182Asp | Heterozygous | Unphased | Benign | Tier1 | 26.1 | – | 27.1 (AlphaMissense) | 0.63 / 0.87 | chrX:38805491 AF: 0.3491% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000614558.3**
cDNA (HGVS): **c.545G>A**
Protein (HGVS): **p.Gly182Asp**
VAF / Read Depth: **VAF 0.48 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#711083 ↗
gnomAD v4 AF: **0.3491%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **27.1** (Raw: 1.393 · Top 0.19%)
Primary Driver: **AlphaMissense**

50%
31%
16%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## WDHD1 WDHD1 chr14:55,002,157 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 25.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene contains multiple N-terminal WD40 domains and a C-terminal high mobility group (HMG) box. WD40 domains are found in a variety of eukaryotic proteins and may function as adaptor/regulatory modules in signal transduction, pre-mRNA processing and cytoskeleton assembly. HMG boxes are found in many eukaryotic proteins involved in chromatin assembly, transcription and replication. Alternative splicing results in two transcript variants encoding different isoforms. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs374497720 p.Val210Ala | MIS, p.Val210Ala | Heterozygous | Unphased | Not reviewed | Tier1 | 25.8 | – | 24.5 (AlphaMissense) | 0.27 / 0.58 | chr14:55002157 AF: 0.0190% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000360586.8**
cDNA (HGVS): **c.629T>C**
Protein (HGVS): **p.Val210Ala**
VAF / Read Depth: **VAF 0.50 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0190%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.5** (Raw: 1.042 · Top 0.35%)
Primary Driver: **AlphaMissense**

42%
35%
19%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAAP24 FAAP24 chr19:32,974,193 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 25.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 FAAP24 is a component of the Fanconi anemia (FA) core complex (see MIM 227650), which plays a crucial role in DNA damage response (Ciccia et al., 2007 [PubMed 17289582]).[supplied by OMIM, Mar 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs36017455 p.Ser126Phe | MIS, p.Ser126Phe | Heterozygous | Unphased | Benign | Tier1 | 25.7 | — | 27.0 (AlphaMissense) | — / 0.69 | chr19:32974193 AF: 0.5937% |

### 1. INHERITANCE & OMIM PATHOLOGY

**FAAP24-related disorder**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

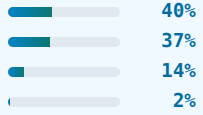
Transcript: ENST00000588258.6
cDNA (HGVS): c.377C>T
Protein (HGVS): p.Ser126Phe
VAF / Read Depth: VAF 0.44 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#784831 ↗
gnomAD v4 AF: 0.5937%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.0 (Raw: 1.385 · Top 0.20%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CTH CTH chr1:70,415,987 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 25.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a cytoplasmic enzyme in the trans-sulfuration pathway that converts cystathione derived from methionine into cysteine. Glutathione synthesis in the liver is dependent upon the availability of cysteine. Mutations in this gene cause cystathioninuria. Alternative splicing of this gene results in three transcript variants encoding different isoforms. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28941785 p.Thr67Ile | MIS, p.Thr67Ile | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 25.5 | — | 27.8 (AlphaMissense) | 0.93 / 0.81 | chr1:70415987 AF: 0.8026% |

### 1. INHERITANCE & OMIM PATHOLOGY

**CTH-related disorder**
Complex / Multifactorial OMIM #219500 ↗

**Cystathioninuria**
Complex / Multifactorial OMIM #219500 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

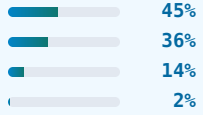
Transcript: ENST00000370938.8
cDNA (HGVS): c.200C>T
Protein (HGVS): p.Thr67Ile
VAF / Read Depth: VAF 0.46 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#2939 ↗
gnomAD v4 AF: 0.8026%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TEK TEK chr9:27,185,615 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 24.3 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a receptor that belongs to the protein tyrosine kinase Tie2 family. The encoded protein possesses a unique extracellular region that contains two immunoglobulin-like domains, three epidermal growth factor (EGF)-like domains and three fibronectin type III repeats. The ligand angiopoietin-1 binds to this receptor and mediates a signaling pathway that functions in embryonic vascular development. Mutations in this gene are associated with inherited venous malformations of the skin and mucous membranes. Alternative splicing results in multiple transcript variants. Additional alternatively spliced transcript variants of this gene have been described, but their full-length nature is not known. [provided by RefSeq, Feb 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs189543659 p.Asn438Ile | MIS, p.Asn438Ile | Heterozygous | Unphased | Not reviewed | Tier1 | 24.3 | – | 21.9 (Cactus) | 0.21 / 0.21 | chr9:27185615 AF: 0.0011% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

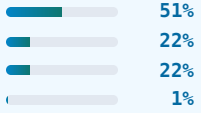
Transcript: ENST00000380036.10
cDNA (HGVS): c.1313A>T
Protein (HGVS): p.Asn438Ile
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0011%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 21.9 (Raw: 0.695 · Top 0.65%)
Primary Driver: Cactus




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs35030851 p.Val600Leu | MIS, p.Val600Leu | Heterozygous | Unphased | Benign | Tier3 | 17.9 | – | 22.9 (Cactus) | 0.06 / 0.16 | chr9:27197488 AF: 4.5993% |

## CHST14 CHST14 chr15:40,472,259 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 24.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the HNK-1 family of sulfotransferases. The encoded protein transfers sulfate to the C-4 hydroxyl of N-acetylgalactosamine residues in dermatan sulfate. Mutations in this gene have been associated with adducted thumb-clubfoot syndrome.[provided by RefSeq, Mar 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1463514561 p.Gln349Pro | MIS, p.Gln349Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 24.3 | – | 22.9 (Cactus) | 0.54 / 0.20 | chr15:40472259 AF: 0.0004% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Cardiovascular phenotype**
Complex / Multifactorial

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

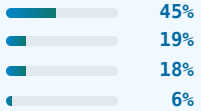
Transcript: ENST00000306243.7
cDNA (HGVS): c.1046A>C
Protein (HGVS): p.Gln349Pro
VAF / Read Depth: VAF 0.52 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#3229028 ↗
gnomAD v4 AF: 0.0004%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 22.9 (Raw: 0.798 · Top 0.51%)
Primary Driver: Cactus




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SUMF1 SUMF1 chr3:4,362,213 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 23.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an enzyme that catalyzes the hydrolysis of sulfate esters by oxidizing a cysteine residue in the substrate sulfatase to an active site 3-oxoalanine residue, which is also known as C-alpha-formylglycine. Mutations in this gene cause multiple sulfatase deficiency, a lysosomal storage disorder. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2009]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs376126385 p.Asn352Lys | MIS, p.Asn352Lys | Heterozygous | Unphased | Uncertain significance | Tier1 | 23.9 | — | 25.9 (AlphaMissense) | 0.64 / 1.00 | chr3:4362213 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Multiple sulfatase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000272902.10 |
| cDNA (HGVS): | c.1056C>G |
| Protein (HGVS): | p.Asn352Lys |
| VAF / Read Depth: | VAF 0.66 · DP — |
| Phased GT: | 0/1 (Unphased) |

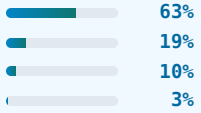
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#2161926 ↗ |
| gnomAD v4 AF: | 0.0009% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NPC2 NPC2 chr14:74,480,701 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 23.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein containing a lipid recognition domain. The encoded protein may function in regulating the transport of cholesterol through the late endosomal/lysosomal system. Mutations in this gene have been associated with Niemann-Pick disease, type C2 and frontal lobe atrophy. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs140130028 c.441+1G>A | SPL | Heterozygous | Maternal | Conflicting classifications of pathogenicity | Tier1 | 23.2 | 0.98 | 26.3 (Splicing) | 0.21 / — | chr14:74480701 AF: 0.6316% |

### 1. INHERITANCE & OMIM PATHOLOGY

**NPC2-related disorder**

Complex / Multifactorial OMIM #257220; 607625 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000555619.6 |
| cDNA (HGVS): | c.441+1G>A |
| Protein (HGVS): | — |
| VAF / Read Depth: | VAF 0.38 · DP — |
| Phased GT: | 0\|1 (Maternal) |

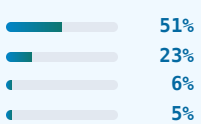
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.10 DL:0.98 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#100734 ↗ |
| gnomAD v4 AF: | 0.6316% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## APOB APOB chr2:21,016,593 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 22.8 Showing 3 of 8 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene product is the main apolipoprotein of chylomicrons and low density lipoproteins (LDL), and is the ligand for the LDL receptor. It occurs in plasma as two main isoforms, apoB-48 and apoB-100: the former is synthesized exclusively in the gut and the latter in the liver. The intestinal and the hepatic forms of apoB are encoded by a single gene from a single, very long mRNA. The two isoforms share a common N-terminal sequence. The shorter apoB-48 protein is produced after RNA editing of the apoB-100 transcript at residue 2180 (CAA->UAA), resulting in the creation of a stop codon, and early translation termination. Mutations in this gene or its regulatory region cause hypobetalipoproteinemia, normotriglyceridemic hypobetalipoproteinemia, and hypercholesterolemia due to ligand-defective apoB, diseases affecting plasma cholesterol and apoB levels. [provided by RefSeq, Dec 2019]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs72653073<br>p.Leu1060= | SYN, p.Leu1060= | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 9.1 | – | 11.5 (PhastCons 470) | – / – | chr2:21016593<br>AF: 0.1326% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Familial Hypercholesterolemia 2**

Autosomal Dominant OMIM #144010 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000233242.5 |
| cDNA (HGVS): | c.3178T>C |
| Protein (HGVS): | p.Leu1060= |
| VAF / Read Depth: | VAF 0.61 · DP – |
| Phased GT: | 0/1 (Unphased) |

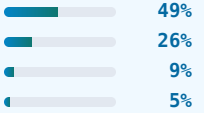
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#334158 ↗ |
| gnomAD v4 AF: | 0.1326% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 11.5 (Raw: 0.199 · Top 7.16%)

Primary Driver: PhastCons 470




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1367117<br>p.Thr98Ile | MIS, p.Thr98Ile | Homozygous | N/A | Benign/Likely benign | Tier2 | 22.8 | – | 18.4 (Cactus) | 0.09 / 0.14 | chr2:21041028<br>AF: 29.5755% |
| rs693<br>p.Thr2515= | SYN, p.Thr2515= | Heterozygous | Unphased | Benign | Tier2 | 5.4 | – | 10.5 (PhastCons 470) | – / – | chr2:21009323<br>AF: 45.9266% |

**PGK1** PGK1 chrX:78,114,140 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 22.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a glycolytic enzyme that catalyzes the conversion of 1,3-diphosphoglycerate to 3-phosphoglycerate. The encoded protein may also act as a cofactor for polymerase alpha. Additionally, this protein is secreted by tumor cells where it participates in angiogenesis by functioning to reduce disulfide bonds in the serine protease, plasmin, which consequently leads to the release of the tumor blood vessel inhibitor angiostatin. The encoded protein has been identified as a moonlighting protein based on its ability to perform mechanistically distinct functions. Deficiency of the enzyme is associated with a wide range of clinical phenotypes hemolytic anemia and neurological impairment. Pseudogenes of this gene have been defined on chromosomes 19, 21 and the X chromosome. [provided by RefSeq, Jan 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| **chrX:78114140**<br>p.Lys133Glu | MIS, p.Lys133Glu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.7 | – | 24.3 (AlphaMissense) | 0.54 / 0.57 | chrX:78114140<br>AF: 0.0008% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000373316.5 |
| cDNA (HGVS): | c.397A>G |
| Protein (HGVS): | p.Lys133Glu |
| VAF / Read Depth: | VAF 0.58 · DP – |
| Phased GT: | 0/1 (Unphased) |

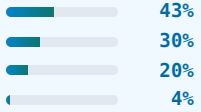
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0008% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.3 (Raw: 0.993 · Top 0.38%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-B HLA-B chr6:31,356,717 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 22.5 Showing 3 of 8 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-B belongs to the HLA class I heavy chain paralogues. This class I molecule is a heterodimer consisting of a heavy chain and a light chain (beta-2 microglobulin). The heavy chain is anchored in the membrane. Class I molecules play a central role in the immune system by presenting peptides derived from the endoplasmic reticulum lumen. They are expressed in nearly all cells. The heavy chain is approximately 45 kDa and its gene contains 8 exons. Exon 1 encodes the leader peptide, exon 2 and 3 encode the alpha1 and alpha2 domains, which both bind the peptide, exon 4 encodes the alpha3 domain, exon 5 encodes the transmembrane region and exons 6 and 7 encode the cytoplasmic tail. Polymorphisms within exon 2 and exon 3 are responsible for the peptide binding specificity of each class one molecule. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. Hundreds of HLA-B alleles have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9266161 p.Leu105Pro | MIS, p.Leu105Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 22.5 | — | 20.0 (AlphaMissense) | 0.20 / 0.79 | chr6:31356717 AF: 0.0006% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

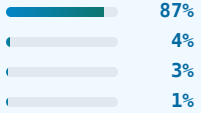
Transcript: **ENST00000412585.7**
cDNA (HGVS): **c.314T>C**
Protein (HGVS): **p.Leu105Pro**
VAF / Read Depth: **VAF 0.55 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0006%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **20.0** (Raw: 0.559 · Top 1.00%)
Primary Driver: **AlphaMissense**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1131213 p.Ala95Thr | MIS, p.Ala95Thr | Homozygous | N/A | Not reviewed | Tier2 | 13.9 | — | 6.4 (AlphaMissense) | 0.18 / 0.10 | chr6:31356748 AF: 0.0088% |
| rs1131212 p.Gln94His | MIS, p.Gln94His | Homozygous | N/A | Not reviewed | Tier2 | 7.1 | — | 11.3 (AlphaMissense) | 0.24 / 0.28 | chr6:31356749 AF: 0.2441% |

## VPS35L VPS35L chr16:19,629,820 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 22.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Involved in Golgi to plasma membrane transport and endocytic recycling. Located in endosome. Part of protein-containing complex. Implicated in Ritscher-Schinzel syndrome. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs144908771 p.Thr518= | SYN, p.Thr518= | Heterozygous | Unphased | Likely benign | Tier1 | 22.5 | 0.64 | 21.4 (Cactus) | — / — | chr16:19629820 AF: 0.3238% |

### 1. INHERITANCE & OMIM PATHOLOGY

**VPS35L-related disorder**
Complex / Multifactorial

**Ritscher-Schinzel syndrome**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

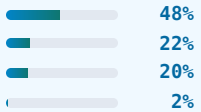
Transcript: **ENST00000417362.7**
cDNA (HGVS): **c.1554G>A**
Protein (HGVS): **p.Thr518=**
VAF / Read Depth: **VAF 0.45 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.64 DL:0.01**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#2672622 ↗
gnomAD v4 AF: **0.3238%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **21.4** (Raw: 0.657 · Top 0.73%)
Primary Driver: **Cactus**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## GUSB GUSB chr7:65,960,973 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 22.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a hydrolase that degrades glycosaminoglycans, including heparan sulfate, dermatan sulfate, and chondroitin-4,6-sulfate. The enzyme forms a homotetramer that is localized to the lysosome. Mutations in this gene result in mucopolysaccharidosis type VII. Alternative splicing results in multiple transcript variants. There are many pseudogenes of this locus in the human genome.[provided by RefSeq, May 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1236992554<br>p.Trp627Leu | MIS, p.Trp627Leu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.2 | – | 24.9 (Cactus) | 0.56 / 0.25 | chr7:65960973<br>AF: 0.0003% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000304895.9 |
| cDNA (HGVS): | c.1880G>T |
| Protein (HGVS): | p.Trp627Leu |
| VAF / Read Depth: | VAF 0.53 · DP – |
| Phased GT: | 0/1 (Unphased) |

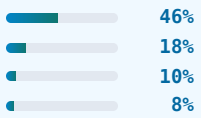
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0003% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 24.9 (Raw: 1.103 · Top 0.32%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FHIT FHIT chr3:60,547,291 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 16.4 Showing 3 of 7 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a P1-P3-bis(5'-adenosyl) triphosphate hydrolase involved in purine metabolism. This gene encompasses the common fragile site FRA3B on chromosome 3, where carcinogen-induced damage can lead to translocations and aberrant transcripts. In fact, aberrant transcripts from this gene have been found in about half of all esophageal, stomach, and colon carcinomas. The encoded protein is also a tumor suppressor, as loss of its activity results in replication stress and DNA damage. [provided by RefSeq, Aug 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114893898<br>c.-17-10312G>C | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 9.7 | 0.51 | 9.8 (Cactus) | – / – | chr3:60547291<br>AF: 4.4825% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000492590.6 |
| cDNA (HGVS): | c.-17-10312G>C |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.47 · DP – |
| Phased GT: | 0/1 (Unphased) |

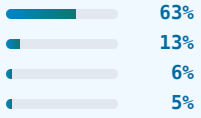
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.51 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 4.4825% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 9.8 (Raw: 0.147 · Top 10.59%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs115894613<br>c.104-75688A>G | INT | Heterozygous | Unphased | Not reviewed | Tier2 | 1.7 | 0.28 | 6.6 (Splicing) | – / – | chr3:60089840<br>AF: 3.4159% |
| chr3:60359998<br>c.103+176862del | SPL | Heterozygous | Unknown | Not reviewed | Tier2 | – | – | – | – / – | chr3:60359998<br>AF: 2.3997% |

## HEPACAM HEPACAM chr11:124,923,060 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 14.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a single-pass type I membrane protein that localizes to the cytoplasmic side of the cell membrane. The encoded protein acts as a homodimer and is involved in cell motility and cell-matrix interactions. The expression of this gene is downregulated or undetectable in many cancer cell lines, so this may be a tumor suppressor gene. [provided by RefSeq, Jul 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs11605766<br>c.804-242G>C | INT | Heterozygous | Unphased | Likely benign | Tier1 | 14.2 | 0.55 | 14.7 (Splicing) | — / — | chr11:124923060<br>AF: 1.0897% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

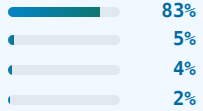
Transcript: **ENST00000298251.5**
cDNA (HGVS): **c.804-242G>C**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.57 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.55 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#1201816 ↗
gnomAD v4 AF: **1.0897%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 14.7 (Raw: 0.311 · Top 3.35%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLC5A4 SLC5A4 chr22:32,233,039 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 14.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable low-affinity D-glucose:sodium symporter activity and proton transmembrane transporter activity. Predicted to be involved in D-glucose import across plasma membrane and renal D-glucose absorption. Predicted to act upstream of or within proton transmembrane transport. Predicted to be located in membrane. Predicted to be active in plasma membrane. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9621415<br>c.886-5G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 14.0 | 0.97 | 21.1 (Splicing) | — / — | chr22:32233039<br>AF: 1.4374% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

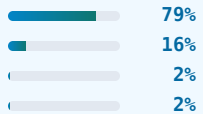
Transcript: **ENST00000266086.6**
cDNA (HGVS): **c.886-5G>A**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.41 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.97 AL:0.01 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4457848 ↗
gnomAD v4 AF: **1.4374%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.1 (Raw: 0.639 · Top 0.78%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## UCK2 UCK2 chr1:165,828,740 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 6.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a pyrimidine ribonucleoside kinase. The encoded protein (EC 2.7.1.48) catalyzes phosphorylation of uridine and cytidine to uridine monophosphate (UMP) and cytidine monophosphate (CMP), respectively.[provided by RefSeq, Oct 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs562989479 c.99+808G>T | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.9 | 0.57 | 19.2 (Splicing) | – / – | chr1:165828740 AF: 0.3704% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

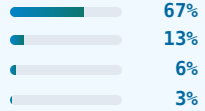
Transcript: **ENST00000367879.9**
cDNA (HGVS): **c.99+808G>T**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.32 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.57 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.3704%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **19.2** (Raw: 0.517 · Top 1.20%)
Primary Driver: **Splicing**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SYNPR SYNPR chr3:63,319,195 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 6.8 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to be involved in modulation of chemical synaptic transmission. Predicted to be located in membrane; neuron projection; and synaptic vesicle. Predicted to be active in hippocampal mossy fiber to CA3 synapse and synaptic vesicle membrane. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs73123134 c.84+40453T>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.8 | 0.54 | 6.0 (Cactus) | – / – | chr3:63319195 AF: 4.7697% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

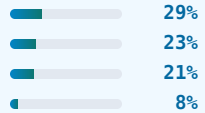
Transcript: **ENST00000478300.6**
cDNA (HGVS): **c.84+40453T>G**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.29 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.54 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **4.7697%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **6.0** (Raw: 0.048 · Top 25.19%)
Primary Driver: **Cactus**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs554940457 c.85-76281del | INT | Heterozygous | Unknown | Not reviewed | Tier3 | – | – | – | – / – | chr3:63404551 AF: 0.5713% |

## PLB1 PLB1 chr2:28,461,598 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 3.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a membrane-associated phospholipase that displays lysophospholipase and phospholipase A2 activities through removal of sn-1 and sn-2 fatty acids of glycerophospholipids. In addition, it displays lipase and retinyl ester hydrolase activities. The encoded protein is highly conserved and is composed of a large, glycosylated extracellular domain composed of four tandem homologous domains, followed by a hydrophobic segment that anchors the enzyme to the membrane and a short C-terminal cytoplasmic tail. This gene has been identified as a candidate rheumatoid arthritis risk gene. [provided by RefSeq, Jul 2016]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs56116810<br>c.-114+4401G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 3.5 | – | 3.6 (Cactus) | – / – | chr2:28461598<br>AF: 0.0012% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000416713.5 |
| cDNA (HGVS): | c.-114+4401G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.50 · DP – |
| Phased GT: | 0/1 (Unphased) |

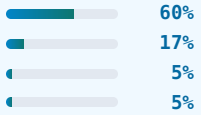
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0012% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 3.6 (Raw: 0.009 · Top 43.90%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ACOX3 ACOX3 chr4:8,440,048 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 3.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acyl-Coenzyme A oxidase 3 also know as pristanoyl -CoA oxidase (ACOX3)is involved in the desaturation of 2-methyl branched fatty acids in peroxisomes. Unlike the rat homolog, the human gene is expressed in very low amounts in liver such that its mRNA was undetectable by routine Northern-blot analysis or its product by immunoblotting or by enzyme activity measurements. However the human cDNA encoding a 700 amino acid protein with a peroxisomal targeting C-terminal tripeptide S-K-L was isolated and is thought to be expressed under special conditions such as specific developmental stages or in a tissue specific manner in tissues that have not yet been examined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs62287361<br>c.-15+600A>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 3.3 | 0.65 | 19.4 (Splicing) | – / – | chr4:8440048<br>AF: 2.7459% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000356406.10 |
| cDNA (HGVS): | c.-15+600A>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.51 · DP – |
| Phased GT: | 0/1 (Unphased) |

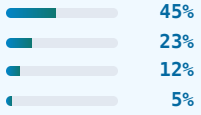
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.65 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 2.7459% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.4 (Raw: 0.526 · Top 1.15%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CD40 CD40 chr20:46,126,737 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 2.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the TNF-receptor superfamily. The encoded protein is a receptor on antigen-presenting cells of the immune system and is essential for mediating a broad variety of immune and inflammatory responses including T cell-dependent immunoglobulin class switching, memory B cell development, and germinal center formation. AT-hook transcription factor AKNA is reported to coordinately regulate the expression of this receptor and its ligand, which may be important for homotypic cell interactions. Adaptor protein TNFR2 interacts with this receptor and serves as a mediator of the signal transduction. The interaction of this receptor and its ligand is found to be necessary for amyloid-beta-induced microglial activation, and thus is thought to be an early event in Alzheimer disease pathogenesis. Mutations affecting this gene are the cause of autosomal recessive hyper-IgM immunodeficiency type 3 (HIGM3). Multiple alternatively spliced transcript variants of this gene encoding distinct isoforms have been reported. [provided by RefSeq, Nov 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs41282788 c.559+36G>C | INT | Heterozygous | Unphased | Benign | Tier1 | 2.0 | 0.51 | 19.4 (Splicing) | — / — | chr20:46126737 AF: 1.8172% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Hyper-IgM syndrome type 3**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

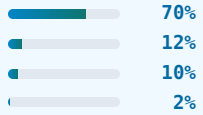
Transcript: **ENST00000372285.8**
cDNA (HGVS): **c.559+36G>C**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.58 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.51 DL:0.10**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#993383 ↗
gnomAD v4 AF: **1.8172%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **19.4** (Raw: 0.528 · Top 1.14%)
Primary Driver: **Splicing**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ADAMTS7 ADAMTS7 chr15:78,790,059 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 0.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ADAMTS (a disintegrin and metalloproteinase with thrombospondin motifs) family. Members of this family share several distinct protein modules, including a propeptide region, a metalloproteinase domain, a disintegrin-like domain, and a thrombospondin type 1 (TS) motif. Individual members of this family differ in the number of C-terminal TS motifs, and some have unique C-terminal domains. The encoded preproprotein is proteolytically processed to generate the mature enzyme. This enzyme contains two C-terminal TS motifs and may regulate vascular smooth muscle cell (VSMC) migration. Mutations in this gene may be associated with susceptibility to coronary artery disease. [provided by RefSeq, Feb 2016]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs142754252 c.1029-221G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 0.2 | 0.65 | 13.8 (Splicing) | — / — | chr15:78790059 AF: 0.1293% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

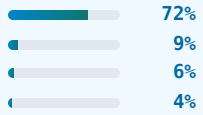
Transcript: **ENST00000388820.5**
cDNA (HGVS): **c.1029-221G>A**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.42 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.65 AL:0.00 DG:0.02 DL:0.00**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4406248 ↗
gnomAD v4 AF: **0.1293%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **13.8** (Raw: 0.277 · Top 4.16%)
Primary Driver: **Splicing**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ATM ATM chr11:108,250,699 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier1 · CADD 0.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PI3/PI4-kinase family. This protein is an important cell cycle checkpoint kinase that phosphorylates; thus, it functions as a regulator of a wide variety of downstream proteins, including tumor suppressor proteins p53 and BRCA1, checkpoint kinase CHK2, checkpoint proteins RAD17 and RAD9, and DNA repair protein NBS1. This protein and the closely related kinase ATR are thought to be master controllers of cell cycle checkpoint signaling pathways that are required for cell response to DNA damage and for genome stability. Mutations in this gene are associated with ataxia telangiectasia, an autosomal recessive disorder. [provided by RefSeq, Aug 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1565381646<br>c.1236-2del | SPL | Heterozygous | Unknown | Pathogenic/Likely pathogenic | Tier1 | – | – | – | – / – | chr11:108250699<br>AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Hereditary cancer-predisposing syndrome**
Complex / Multifactorial

**Familial cancer of breast**
Complex / Multifactorial

**Ataxia-telangiectasia syndrome**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000675843.1**
cDNA (HGVS): **c.1236-2del**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.17 · DP –**
Phased GT: **./. (Unknown)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#574589 ↗
gnomAD v4 AF: **0.0001%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

Evaluated via CADD / SpliceAI / REVEL ensemble.

LitVar2 Literature ↗

## VDR VDR chr12:47,906,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier3 · CADD 5.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes vitamin D3 receptor, which is a member of the nuclear hormone receptor superfamily of ligand-inducible transcription factors. This receptor also functions as a receptor for the secondary bile acid, lithocholic acid. Downstream targets of vitamin D3 receptor are principally involved in mineral metabolism, though this receptor regulates a variety of other metabolic pathways, such as those involved in immune response and cancer. Mutations in this gene are associated with type II vitamin D-resistant rickets. A single nucleotide polymorphism in the initiation codon results in an alternate translation start site three codons downstream. Alternatively spliced transcript variants encoding different isoforms have been described for this gene. A recent study provided evidence for translational readthrough in this gene, and expression of an additional C-terminally extended isoform via the use of an alternative in-frame translation termination codon. [provided by RefSeq, Jun 2018]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs4516035<br>c.-1172A>G | 2KU | Heterozygous | Unphased | Likely pathogenic | Tier3 | 5.4 | – | 2.1 (DNASE-seq) | – / – | chr12:47906043<br>AF: 30.5142% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Chronic obstructive pulmonary disease**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000549336.6**
cDNA (HGVS): **c.-1172A>G**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.19 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **no assertion criteria provided**
ClinVar VCV: VCV#3336650 ↗
gnomAD v4 AF: **30.5142%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **2.1** (Raw: -0.015 · Top 61.95%)
Primary Driver: **DNASE-seq**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CDKN2B CDKN2B chr9:22,003,368 Multisystem 🔥 Tier-1 / Pathogenic Gene

Tier3 · CADD 0.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene lies adjacent to the tumor suppressor gene CDKN2A in a region that is frequently mutated and deleted in a wide variety of tumors. This gene encodes a cyclin-dependent kinase inhibitor, which forms a complex with CDK4 or CDK6, and prevents the activation of the CDK kinases, thus the encoded protein functions as a cell growth regulator that controls cell cycle G1 progression. The expression of this gene was found to be dramatically induced by TGF beta, which suggested its role in the TGF beta induced growth inhibition. Two alternatively spliced transcript variants of this gene, which encode distinct proteins, have been reported. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1063192<br>c.*2619C>T | UT3 | Heterozygous | Maternal | Likely pathogenic\|protective | Tier3 | 0.2 | – | 1.3 (Cactus) | – / – | chr9:22003368<br>AF: 68.9409% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Coronary Artery Disease & Breast Carcinoma**

Complex / Polygenic Protective Allele OMIM #600431 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000276925.7 |
| cDNA (HGVS): | c.*2619C>T |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0\|1 (Maternal) |

### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | no assertion criteria provided |
| ClinVar VCV: | VCV#812639 ↗ |
| gnomAD v4 AF: | 68.9409% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.3 (Raw: -0.032 · Top 74.07%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## 2. ClinVar Pathogenicity & Clinical Curation — Tier-1 & Actionable Genes

Genes carrying ClinVar-curated Pathogenic, Likely Pathogenic, Risk Factor, Drug Response, or Protective alleles.

**Selected Scope:** 22 Genes (22 Tier-1/Pathogenic)
Filtered from 359 matching genes in view

**TAT** TAT chr16:71,576,247 Multisystem 🔥 Tier-1 / Pathogenic Gene — Pathogenic · CADD 36.0 — Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This nuclear gene encodes a mitochondrial protein tyrosine aminotransferase which is present in the liver and catalyzes the conversion of L-tyrosine into p-hydroxyphenylpyruvate. Mutations in this gene cause tyrosinemia (type II, Richner-Hanhart syndrome), a disorder accompanied by major skin and corneal lesions, with possible cognitive disability. A regulator gene for tyrosine aminotransferase is X-linked. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs118203914<br>p.Arg57Ter | STG, p.Arg57Ter | Heterozygous | Maternal | Pathogenic | Tier1 | 36.0 | – | 38.8 (Protein Termination) | – / – | chr16:71576247<br>AF: 0.0017% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Tyrosinemia type II**
Complex / Multifactorial OMIM #276600 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000355962.5 |
| cDNA (HGVS): | c.169C>T |
| Protein (HGVS): | p.Arg57Ter |
| VAF / Read Depth: | VAF 0.46 · DP – |
| Phased GT: | 0\|1 (Maternal) |

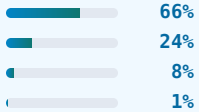
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#402 ↗ |
| gnomAD v4 AF: | 0.0017% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.8 (Raw: 2.285 · Top 0.01%)

Primary Driver: Protein Termination




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| rs572677025<br>c.*409G>A | UT3 | Heterozygous | Unphased | Uncertain significance | Tier2 | 0.8 | – | 5.8 (Splicing) | – / – | chr16:71567735<br>AF: 0.4009% |
|---|---|---|---|---|---|---|---|---|---|---|

## VDR VDR chr12:47,906,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

Likely pathogenic · CADD 5.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes vitamin D3 receptor, which is a member of the nuclear hormone receptor superfamily of ligand-inducible transcription factors. This receptor also functions as a receptor for the secondary bile acid, lithocholic acid. Downstream targets of vitamin D3 receptor are principally involved in mineral metabolism, though this receptor regulates a variety of other metabolic pathways, such as those involved in immune response and cancer. Mutations in this gene are associated with type II vitamin D-resistant rickets. A single nucleotide polymorphism in the initiation codon results in an alternate translation start site three codons downstream. Alternatively spliced transcript variants encoding different isoforms have been described for this gene. A recent study provided evidence for translational readthrough in this gene, and expression of an additional C-terminally extended isoform via the use of an alternative in-frame translation termination codon. [provided by RefSeq, Jun 2018]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs4516035<br>c.-1172A>G | 2KU | Heterozygous | Unphased | Likely pathogenic | Tier3 | 5.4 | – | 2.1 (DNASE-seq) | – / – | chr12:47906043<br>AF: 30.5142% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Chronic obstructive pulmonary disease**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000549336.6
cDNA (HGVS): c.-1172A>G
Protein (HGVS): –
VAF / Read Depth: VAF 0.19 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: no assertion criteria provided
ClinVar VCV: VCV#3336650 ↗
gnomAD v4 AF: 30.5142%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 2.1 (Raw: -0.015 · Top 61.95%)
Primary Driver: DNASE-seq




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## AMPD1 AMPD1 chr1:114,677,435 Multisystem 🔥 Tier-1 / Pathogenic Gene

Uncertain significance · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Adenosine monophosphate deaminase 1 catalyzes the deamination of AMP to IMP in skeletal muscle and plays an important role in the purine nucleotide cycle. Two other genes have been identified, AMPD2 and AMPD3, for the liver- and erythocyte-specific isoforms, respectively. Deficiency of the muscle-specific enzyme is apparently a common cause of exercise-induced myopathy and probably the most common cause of metabolic myopathy in the human. Alternatively spliced transcript variants encoding different isoforms have been identified in this gene.[provided by RefSeq, Feb 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs764950035<br>p.Glu435Gly | MIS, p.Glu435Gly | Heterozygous | Unphased | Uncertain significance | Tier1 | 32.0 | – | 28.2 (AlphaMissense) | 0.95 / 0.92 | chr1:114677435<br>AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Muscle AMP deaminase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

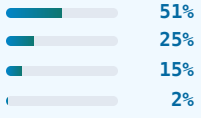
Transcript: ENST00000520113.7
cDNA (HGVS): c.1304A>G
Protein (HGVS): p.Glu435Gly
VAF / Read Depth: VAF 0.49 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#2173226 ↗
gnomAD v4 AF: 0.0075%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.2 (Raw: 1.454 · Top 0.15%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB1 HLA-DRB1 chr6:32,582,135 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

**Uncertain significance · CADD 26.8** Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB1 belongs to the HLA class II beta chain paralogs. The class II molecule is a heterodimer consisting of an alpha (DRA) and a beta chain (DRB), both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa. It is encoded by 6 exons. Exon one encodes the leader peptide; exons 2 and 3 encode the two extracellular domains; exon 4 encodes the transmembrane domain; and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Hundreds of DRB1 alleles have been described and some alleles have increased frequencies associated with certain diseases or conditions. For example, DRB1*1302 has been related to acute and chronic hepatitis B virus persistence. There are multiple pseudogenes of this gene. [provided by RefSeq, Jul 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs16822805<br>p.Ala103Pro | MIS, p.Ala103Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 0.0 | — | 20.2 (AlphaMissense) | 0.21 / 0.78 | chr6:32584172<br>AF: 0.1762% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.307G>C |
| Protein (HGVS): | p.Ala103Pro |
| VAF / Read Depth: | VAF 0.53 · DP — |
| Phased GT: | 0/1 (Unphased) |

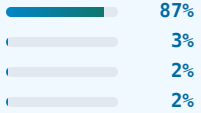
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#691505 ↗ |
| gnomAD v4 AF: | 0.1762% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.2 (Raw: 0.573 · Top 0.95%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9269957<br>p.Gln39Ter | STG, p.Gln39Ter | Heterozygous | Unphased | Uncertain significance | Tier2 | 26.8 | 0.13 | 28.2 (Protein Termination) | — / — | chr6:32584364<br>AF: 0.0015% |

## CHST14 CHST14 chr15:40,472,259 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

**Uncertain significance · CADD 24.3** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the HNK-1 family of sulfotransferases. The encoded protein transfers sulfate to the C-4 hydroxyl of N-acetylgalactosamine residues in dermatan sulfate. Mutations in this gene have been associated with adducted thumb-clubfoot syndrome.[provided by RefSeq, Mar 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1463514561<br>p.Gln349Pro | MIS, p.Gln349Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 24.3 | — | 22.9 (Cactus) | 0.54 / 0.20 | chr15:40472259<br>AF: 0.0004% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000306243.7 |
| cDNA (HGVS): | c.1046A>C |
| Protein (HGVS): | p.Gln349Pro |
| VAF / Read Depth: | VAF 0.52 · DP — |
| Phased GT: | 0/1 (Unphased) |

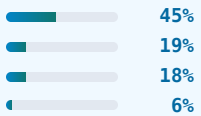
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#3229028 ↗ |
| gnomAD v4 AF: | 0.0004% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.9 (Raw: 0.798 · Top 0.51%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SUMF1 SUMF1 chr3:4,362,213 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Uncertain significance · CADD 23.9** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an enzyme that catalyzes the hydrolysis of sulfate esters by oxidizing a cysteine residue in the substrate sulfatase to an active site 3-oxoalanine residue, which is also known as C-alpha-formylglycine. Mutations in this gene cause multiple sulfatase deficiency, a lysosomal storage disorder. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2009]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs376126385 p.Asn352Lys | MIS, p.Asn352Lys | Heterozygous | Unphased | Uncertain significance | Tier1 | 23.9 | — | 25.9 (AlphaMissense) | 0.64 / 1.00 | chr3:4362213 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Multiple sulfatase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

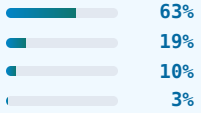
Transcript: **ENST00000272902.10**
cDNA (HGVS): **c.1056C>G**
Protein (HGVS): **p.Asn352Lys**
VAF / Read Depth: **VAF 0.66 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#2161926 ↗
gnomAD v4 AF: **0.0009%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.9** (Raw: 1.256 · Top 0.26%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## VWA3B VWA3B chr2:98,162,976 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Benign · CADD 46.0** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an intracellular protein that contains a von Willebrand factor type A domain. Intracellular proteins with VWA domains are thought to function in transcription, DNA repair, ribosomal and membrane transport and the proteasome. Mutations in this gene are associated with Spinocerebellar ataxia, autosomal recessive 22. Alternatively spliced transcript variants have been found for this gene. [provided by RefSeq, May 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs62154921 p.Glu372Ter | STG, p.Glu372Ter | Heterozygous | Unphased | Benign | Tier1 | 46.0 | 0.85 | 50.8 (Protein Termination) | — / — | chr2:98162976 AF: 4.2196% |

### 1. INHERITANCE & OMIM PATHOLOGY

**VWA3B-related disorder**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

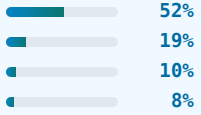
Transcript: **ENST00000477737.6**
cDNA (HGVS): **c.1114G>T**
Protein (HGVS): **p.Glu372Ter**
VAF / Read Depth: **VAF 0.44 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.05 DL:0.85**
ClinVar Review: **no assertion criteria provided**
ClinVar VCV: VCV#3056040 ↗
gnomAD v4 AF: **4.2196%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **50.8** (Raw: 2.870 · Top 0.00%)
Primary Driver: **Protein Termination**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CEP63 CEP63 chr3:134,537,268 Multisystem 🔥 Tier-1 / Pathogenic Gene

Benign · CADD 34.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with six coiled-coil domains. The protein is localized to the centrosome, a non-membraneous organelle that functions as the major microtubule-organizing center in animal cells. Several alternatively spliced transcript variants have been found, but their biological validity has not been determined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114108011 p.Gln185His | MIS, p.Gln185His | Heterozygous | Maternal | Benign | Tier1 | 34.0 | 0.93 | 38.6 (Splicing) | 0.28 / 0.54 | chr3:134537268 AF: 2.0823% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

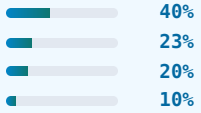
Transcript: ENST00000675561.1
cDNA (HGVS): c.555G>C
Protein (HGVS): p.Gln185His
VAF / Read Depth: VAF 0.74 · DP —
Phased GT: 0|1 (Maternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.93
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#128709 ↗
gnomAD v4 AF: 2.0823%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.6 (Raw: 2.279 · Top 0.01%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## IL17RC IL17RC chr3:9,932,859 Multisystem 🔥 Tier-1 / Pathogenic Gene

Benign/Likely benign · CADD 33.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a single-pass type I membrane protein that shares similarity with the interleukin-17 receptor (IL-17RA). Unlike IL-17RA, which is predominantly expressed in hemopoietic cells, and binds with high affinity to only IL-17A, this protein is expressed in nonhemopoietic tissues, and binds both IL-17A and IL-17F with similar affinities. The proinflammatory cytokines, IL-17A and IL-17F, have been implicated in the progression of inflammatory and autoimmune diseases. Multiple alternatively spliced transcript variants encoding different isoforms have been detected for this gene, and it has been proposed that soluble, secreted proteins lacking transmembrane and intracellular domains may function as extracellular antagonists to cytokine signaling. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs148575246 c.1522+1G>C | SPL | Heterozygous | Unphased | Benign/Likely benign | Tier1 | 33.0 | 0.97 | 25.9 (Splicing) | — / — | chr3:9932859 AF: 1.0744% |

### 1. INHERITANCE & OMIM PATHOLOGY

**IL17RC-related disorder**
Complex / Multifactorial OMIM #616445 ↗

**Candidiasis, familial, 9**
Complex / Multifactorial OMIM #616445 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

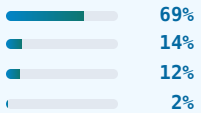
Transcript: ENST00000403601.8
cDNA (HGVS): c.1522+1G>C
Protein (HGVS): —
VAF / Read Depth: VAF 0.55 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.01 DL:0.97
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#542537 ↗
gnomAD v4 AF: 1.0744%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs115461448 p.Ala324Thr | MIS, p.Ala324Thr | Heterozygous | Unphased | Benign | Tier3 | 23.2 | — | 22.0 (Cactus) | 0.09 / 0.17 | chr3:9928397 AF: 1.4147% |

## TMEM43 TMEM43 chr3:14,139,244 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

**Conflicting classifications of pathogenicity · CADD 32.0** Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to the TMEM43 family. Defects in this gene are the cause of familial arrhythmogenic right ventricular dysplasia type 5 (ARVD5), also known as arrhythmogenic right ventricular cardiomyopathy type 5 (ARVC5). Arrhythmogenic right ventricular dysplasia is an inherited disorder, often involving both ventricles, and is characterized by ventricular tachycardia, heart failure, sudden cardiac death, and fibrofatty replacement of cardiomyocytes. This gene contains a response element for PPAR gamma (an adipogenic transcription factor), which may explain the fibrofatty replacement of the myocardium, a characteristic pathological finding in ARVC. [provided by RefSeq, Oct 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs199526104 p.Trp316Ser | MIS, p.Trp316Ser | Heterozygous | Paternal | Conflicting classifications of pathogenicity | Tier1 | 32.0 | — | 26.5 (AlphaMissense) | 0.53 / 0.76 | chr3:14139244 AF: 0.0126% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Cardiovascular phenotype**
Complex / Multifactorial OMIM #604400 ↗

**TMEM43-related disorder**
Complex / Multifactorial OMIM #604400 ↗

**Auditory neuropathy, autosomal dominant 3**
Autosomal Dominant OMIM #604400 ↗

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

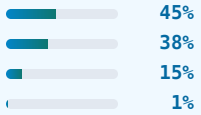
| | |
|---|---|
| Transcript: | ENST00000306077.5 |
| cDNA (HGVS): | c.947G>C |
| Protein (HGVS): | p.Trp316Ser |
| VAF / Read Depth: | VAF 0.44 · DP — |
| Phased GT: | 1\|0 (Paternal) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#46155 ↗ |
| gnomAD v4 AF: | 0.0126% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 26.5 (Raw: 1.335 · Top 0.22%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1043943 c.*115T>C | UT3 | Heterozygous | Paternal | Benign/Likely benign | Tier2 | 1.6 | — | 9.2 (ATAC-seq) | — / — | chr3:14141910 AF: 62.2376% |

## PLK4 PLK4 chr4:127,881,151 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Conflicting classifications of pathogenicity · CADD 28.5** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polo family of serine/threonine protein kinases. The protein localizes to centrioles, complex microtubule-based structures found in centrosomes, and regulates centriole duplication during the cell cycle. Three alternatively spliced transcript variants that encode different protein isoforms have been found for this gene. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs149003893 p.Gly6Glu | MIS, p.Gly6Glu | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 28.5 | — | 29.8 (AlphaMissense) | 0.36 / 0.99 | chr4:127881151 AF: 0.1341% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Inborn genetic diseases**
Complex / Multifactorial

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

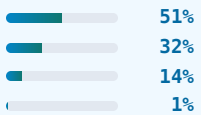
| | |
|---|---|
| Transcript: | ENST00000270861.10 |
| cDNA (HGVS): | c.17G>A |
| Protein (HGVS): | p.Gly6Glu |
| VAF / Read Depth: | VAF 0.45 · DP — |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#436338 ↗ |
| gnomAD v4 AF: | 0.1341% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 29.8 (Raw: 1.555 · Top 0.11%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## POLG POLG chr15:89,323,460 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

**Pathogenic/Likely pathogenic · CADD 26.3** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Mitochondrial DNA polymerase is heterotrimeric, consisting of a homodimer of accessory subunits plus a catalytic subunit. The protein encoded by this gene is the catalytic subunit of mitochondrial DNA polymerase. The encoded protein contains a polyglutamine tract near its N-terminus that may be polymorphic. Defects in this gene are a cause of progressive external ophthalmoplegia with mitochondrial DNA deletions 1 (PEOA1), sensory ataxic neuropathy dysarthria and ophthalmoparesis (SANDO), Alpers-Huttenlocher syndrome (AHS), and mitochondrial neurogastrointestinal encephalopathy syndrome (MNGIE). Two transcript variants encoding the same protein have been found for this gene. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs121918054 p.Gly737Arg | MIS, p.Gly737Arg | Heterozygous | Paternal | Pathogenic/Likely pathogenic | Tier1 | 26.3 | – | 30.3 (AlphaMissense) | 0.94 / 0.94 | chr15:89323460 AF: 0.1619% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Spinocerebellar ataxia with epilepsy**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Sensory ataxic neuropathy, dysarthria, and ophthalmoparesis**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 1**
Autosomal Dominant OMIM #203700; 607459; 613662; 157640; 258450 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000268124.11 |
| cDNA (HGVS): | c.2209G>C |
| Protein (HGVS): | p.Gly737Arg |
| VAF / Read Depth: | VAF 0.61 · DP – |
| Phased GT: | 1\|0 (Paternal) |

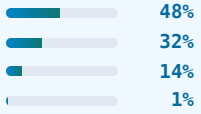
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#13513 ↗ |
| gnomAD v4 AF: | 0.1619% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.3 (Raw: 1.582 · Top 0.09%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## MID1IP1 MID1IP1 chrX:38,805,491 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Benign · CADD 26.1** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable enzyme activator activity. Predicted to be involved in several processes, including negative regulation of microtubule depolymerization; positive regulation of fatty acid biosynthetic process; and positive regulation of ligase activity. Predicted to be located in microtubule cytoskeleton. Predicted to be active in cytosol and nucleus. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs143008624 p.Gly182Asp | MIS, p.Gly182Asp | Heterozygous | Unphased | Benign | Tier1 | 26.1 | – | 27.1 (AlphaMissense) | 0.63 / 0.87 | chrX:38805491 AF: 0.3491% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000614558.3 |
| cDNA (HGVS): | c.545G>A |
| Protein (HGVS): | p.Gly182Asp |
| VAF / Read Depth: | VAF 0.48 · DP – |
| Phased GT: | 0/1 (Unphased) |

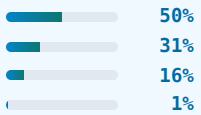
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#711083 ↗ |
| gnomAD v4 AF: | 0.3491% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.1 (Raw: 1.393 · Top 0.19%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAAP24 FAAP24 chr19:32,974,193 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Benign · CADD 25.7** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 FAAP24 is a component of the Fanconi anemia (FA) core complex (see MIM 227650), which plays a crucial role in DNA damage response (Ciccia et al., 2007 [PubMed 17289582]).[supplied by OMIM, Mar 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs36017455 p.Ser126Phe | MIS, p.Ser126Phe | Heterozygous | Unphased | Benign | Tier1 | 25.7 | — | 27.0 (AlphaMissense) | — / 0.69 | chr19:32974193 AF: 0.5937% |

### 1. INHERITANCE & OMIM PATHOLOGY

**FAAP24-related disorder**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

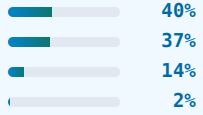
Transcript: **ENST00000588258.6**
cDNA (HGVS): **c.377C>T**
Protein (HGVS): **p.Ser126Phe**
VAF / Read Depth: **VAF 0.44 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#784831 ↗
gnomAD v4 AF: **0.5937%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **27.0** (Raw: 1.385 · Top 0.20%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CTH CTH chr1:70,415,987 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Conflicting classifications of pathogenicity · CADD 25.5** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a cytoplasmic enzyme in the trans-sulfuration pathway that converts cystathione derived from methionine into cysteine. Glutathione synthesis in the liver is dependent upon the availability of cysteine. Mutations in this gene cause cystathioninuria. Alternative splicing of this gene results in three transcript variants encoding different isoforms. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28941785 p.Thr67Ile | MIS, p.Thr67Ile | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 25.5 | — | 27.8 (AlphaMissense) | 0.93 / 0.81 | chr1:70415987 AF: 0.8026% |

### 1. INHERITANCE & OMIM PATHOLOGY

**CTH-related disorder**
Complex / Multifactorial OMIM #219500 ↗

**Cystathioninuria**
Complex / Multifactorial OMIM #219500 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

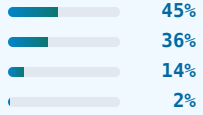
Transcript: **ENST00000370938.8**
cDNA (HGVS): **c.200C>T**
Protein (HGVS): **p.Thr67Ile**
VAF / Read Depth: **VAF 0.46 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, conflicting classifications**
ClinVar VCV: VCV#2939 ↗
gnomAD v4 AF: **0.8026%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **27.8** (Raw: 1.434 · Top 0.17%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NPC2 NPC2 chr14:74,480,701 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Conflicting classifications of pathogenicity · CADD 23.2** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein containing a lipid recognition domain. The encoded protein may function in regulating the transport of cholesterol through the late endosomal/lysosomal system. Mutations in this gene have been associated with Niemann-Pick disease, type C2 and frontal lobe atrophy. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs140130028<br>c.441+1G>A | SPL | Heterozygous | Maternal | Conflicting classifications of pathogenicity | Tier1 | 23.2 | 0.98 | 26.3 (Splicing) | 0.21 / — | chr14:74480701<br>AF: 0.6316% |

### 1. INHERITANCE & OMIM PATHOLOGY

**NPC2-related disorder**
Complex / Multifactorial OMIM #257220; 607625 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000555619.6 |
| cDNA (HGVS): | c.441+1G>A |
| Protein (HGVS): | — |
| VAF / Read Depth: | VAF 0.38 · DP — |
| Phased GT: | 0\|1 (Maternal) |

### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.10 DL:0.98 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#100734 ↗ |
| gnomAD v4 AF: | 0.6316% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)

Primary Driver: Splicing

51%
23%
6%
5%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## APOB APOB chr2:21,016,593 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

**Benign/Likely benign · CADD 22.8** Showing 3 of 8 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene product is the main apolipoprotein of chylomicrons and low density lipoproteins (LDL), and is the ligand for the LDL receptor. It occurs in plasma as two main isoforms, apoB-48 and apoB-100: the former is synthesized exclusively in the gut and the latter in the liver. The intestinal and the hepatic forms of apoB are encoded by a single gene from a single, very long mRNA. The two isoforms share a common N-terminal sequence. The shorter apoB-48 protein is produced after RNA editing of the apoB-100 transcript at residue 2180 (CAA->UAA), resulting in the creation of a stop codon, and early translation termination. Mutations in this gene or its regulatory region cause hypobetalipoproteinemia, normotriglyceridemic hypobetalipoproteinemia, and hypercholesterolemia due to ligand-defective apoB, diseases affecting plasma cholesterol and apoB levels. [provided by RefSeq, Dec 2019]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs72653073 p.Leu1060= | SYN, p.Leu1060= | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 9.1 | – | 11.5 (PhastCons 470) | – / – | chr2:21016593 AF: 0.1326% |
| rs1367117 p.Thr98Ile | MIS, p.Thr98Ile | Homozygous | N/A | Benign/Likely benign | Tier2 | 22.8 | – | 18.4 (Cactus) | 0.09 / 0.14 | chr2:21041028 AF: 29.5755% |
| rs488329 c.3843-77T>C | INT | Homozygous | N/A | Benign | Tier2 | 6.3 | – | 3.1 (Cactus) | – / – | chr2:21013610 AF: 99.7921% |

#### 1. INHERITANCE & OMIM PATHOLOGY

**Familial Hypercholesterolemia 2**
Autosomal Dominant OMIM #144010 ↗

Quadrant: **Curated**

#### 2. TRANSCRIPT, PHASING & DEPTH

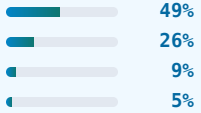
| | |
|---|---|
| Transcript: | ENST00000233242.5 |
| cDNA (HGVS): | c.3178T>C |
| Protein (HGVS): | p.Leu1060= |
| VAF / Read Depth: | VAF 0.61 · DP – |
| Phased GT: | 0/1 (Unphased) |

#### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#334158 ↗ |
| gnomAD v4 AF: | 0.1326% |

#### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 11.5 (Raw: 0.199 · Top 7.16%)
Primary Driver: PhastCons 470




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## VPS35L VPS35L chr16:19,629,820 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Likely benign · CADD 22.5** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Involved in Golgi to plasma membrane transport and endocytic recycling. Located in endosome. Part of protein-containing complex. Implicated in Ritscher-Schinzel syndrome. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs144908771 p.Thr518= | SYN, p.Thr518= | Heterozygous | Unphased | Likely benign | Tier1 | 22.5 | 0.64 | 21.4 (Cactus) | – / – | chr16:19629820 AF: 0.3238% |

#### 1. INHERITANCE & OMIM PATHOLOGY

**VPS35L-related disorder**
Complex / Multifactorial

**Ritscher-Schinzel syndrome**
Complex / Multifactorial

Quadrant: **Curated**

#### 2. TRANSCRIPT, PHASING & DEPTH

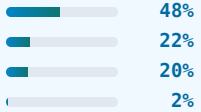
| | |
|---|---|
| Transcript: | ENST00000417362.7 |
| cDNA (HGVS): | c.1554G>A |
| Protein (HGVS): | p.Thr518= |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

#### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.64 DL:0.01 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#2672622 ↗ |
| gnomAD v4 AF: | 0.3238% |

#### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.4 (Raw: 0.657 · Top 0.73%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HEPACAM HEPACAM chr11:124,923,060 Multisystem 🔥 Tier-1 / Pathogenic Gene

Likely benign · CADD 14.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a single-pass type I membrane protein that localizes to the cytoplasmic side of the cell membrane. The encoded protein acts as a homodimer and is involved in cell motility and cell-matrix interactions. The expression of this gene is downregulated or undetectable in many cancer cell lines, so this may be a tumor suppressor gene. [provided by RefSeq, Jul 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs11605766<br>c.804-242G>C | INT | Heterozygous | Unphased | Likely benign | Tier1 | 14.2 | 0.55 | 14.7 (Splicing) | – / – | chr11:124923060<br>AF: 1.0897% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

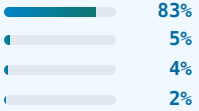
Transcript: ENST00000298251.5
cDNA (HGVS): c.804-242G>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.57 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.55 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#1201816 ↗
gnomAD v4 AF: 1.0897%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 14.7 (Raw: 0.311 · Top 3.35%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CD40 CD40 chr20:46,126,737 Multisystem 🔥 Tier-1 / Pathogenic Gene

Benign · CADD 2.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the TNF-receptor superfamily. The encoded protein is a receptor on antigen-presenting cells of the immune system and is essential for mediating a broad variety of immune and inflammatory responses including T cell-dependent immunoglobulin class switching, memory B cell development, and germinal center formation. AT-hook transcription factor AKNA is reported to coordinately regulate the expression of this receptor and its ligand, which may be important for homotypic cell interactions. Adaptor protein TNFR2 interacts with this receptor and serves as a mediator of the signal transduction. The interaction of this receptor and its ligand is found to be necessary for amyloid-beta-induced microglial activation, and thus is thought to be an early event in Alzheimer disease pathogenesis. Mutations affecting this gene are the cause of autosomal recessive hyper-IgM immunodeficiency type 3 (HIGM3). Multiple alternatively spliced transcript variants of this gene encoding distinct isoforms have been reported. [provided by RefSeq, Nov 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs41282788<br>c.559+36G>C | INT | Heterozygous | Unphased | Benign | Tier1 | 2.0 | 0.51 | 19.4 (Splicing) | – / – | chr20:46126737<br>AF: 1.8172% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Hyper-IgM syndrome type 3**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

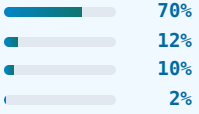
Transcript: ENST00000372285.8
cDNA (HGVS): c.559+36G>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.58 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.51 DL:0.10
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#993383 ↗
gnomAD v4 AF: 1.8172%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.4 (Raw: 0.528 · Top 1.14%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CDKN2B CDKN2B chr9:22,003,368 Multisystem 🔥 Tier-1 / Pathogenic Gene

Likely pathogenic|protective · CADD 0.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene lies adjacent to the tumor suppressor gene CDKN2A in a region that is frequently mutated and deleted in a wide variety of tumors. This gene encodes a cyclin-dependent kinase inhibitor, which forms a complex with CDK4 or CDK6, and prevents the activation of the CDK kinases, thus the encoded protein functions as a cell growth regulator that controls cell cycle G1 progression. The expression of this gene was found to be dramatically induced by TGF beta, which suggested its role in the TGF beta induced growth inhibition. Two alternatively spliced transcript variants of this gene, which encode distinct proteins, have been reported. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1063192<br>c.*2619C>T | UT3 | Heterozygous | Maternal | Likely pathogenic\|protective | Tier3 | 0.2 | – | 1.3 (Cactus) | – / – | chr9:22003368<br>AF: 68.9409% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Coronary Artery Disease & Breast Carcinoma**
Complex / Polygenic Protective Allele OMIM #600431 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000276925.7
cDNA (HGVS): c.*2619C>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.45 · DP –
Phased GT: 0|1 (Maternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: no assertion criteria provided
ClinVar VCV: VCV#812639 ↗
gnomAD v4 AF: 68.9409%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.3 (Raw: -0.032 · Top 74.07%)
Primary Driver: Cactus

22%
16%
14%
11%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ATM ATM chr11:108,250,699 Multisystem 🔥 Tier-1 / Pathogenic Gene

Pathogenic/Likely pathogenic · CADD 0.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PI3/PI4-kinase family. This protein is an important cell cycle checkpoint kinase that phosphorylates; thus, it functions as a regulator of a wide variety of downstream proteins, including tumor suppressor proteins p53 and BRCA1, checkpoint kinase CHK2, checkpoint proteins RAD17 and RAD9, and DNA repair protein NBS1. This protein and the closely related kinase ATR are thought to be master controllers of cell cycle checkpoint signaling pathways that are required for cell response to DNA damage and for genome stability. Mutations in this gene are associated with ataxia telangiectasia, an autosomal recessive disorder. [provided by RefSeq, Aug 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1565381646<br>c.1236-2del | SPL | Heterozygous | Unknown | Pathogenic/Likely pathogenic | Tier1 | – | – | – | – / – | chr11:108250699<br>AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Hereditary cancer-predisposing syndrome**
Complex / Multifactorial

**Familial cancer of breast**
Complex / Multifactorial

**Ataxia-telangiectasia syndrome**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000675843.1
cDNA (HGVS): c.1236-2del
Protein (HGVS): –
VAF / Read Depth: VAF 0.17 · DP –
Phased GT: ./. (Unknown)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#574589 ↗
gnomAD v4 AF: 0.0001%

### 4. ALPHAGENOME 18-MODALITY ATLAS

Evaluated via CADD / SpliceAI / REVEL ensemble.

LitVar2 Literature ↗

## 3. DeepMind AlphaGenome Atlas — Top Genes by AVI Phred & 18-Modality Drivers

Top genes ranked by DeepMind AlphaGenome Variant Impact (AVI) Phred score across 18 genomic and epigenomic modalities (RNA-seq, Splicing, ATAC/DNase, ChIP).

**Selected Scope:** 42 Genes (42 Tier-1/Pathogenic)
Filtered from 1004 matching genes in view

**VWA3B** VWA3B chr2:98,162,976 Multisystem 🔥 Tier-1 / Pathogenic Gene AlphaGenome AVI: 50.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an intracellular protein that contains a von Willebrand factor type A domain. Intracellular proteins with VWA domains are thought to function in transcription, DNA repair, ribosomal and membrane transport and the proteasome. Mutations in this gene are associated with Spinocerebellar ataxia, autosomal recessive 22. Alternatively spliced transcript variants have been found for this gene. [provided by RefSeq, May 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs62154921 p.Glu372Ter | STG, p.Glu372Ter | Heterozygous | Unphased | Benign | Tier1 | 46.0 | 0.85 | 50.8 (Protein Termination) | – / – | chr2:98162976 AF: 4.2196% |

### 1. INHERITANCE & OMIM PATHOLOGY

**VWA3B-related disorder**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000477737.6 |
| cDNA (HGVS): | c.1114G>T |
| Protein (HGVS): | p.Glu372Ter |
| VAF / Read Depth: | VAF 0.44 · DP – |
| Phased GT: | 0/1 (Unphased) |

### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.05 DL:0.85 |
| ClinVar Review: | no assertion criteria provided |
| ClinVar VCV: | VCV#3056040 ↗ |
| gnomAD v4 AF: | 4.2196% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 50.8 (Raw: 2.870 · Top 0.00%)
Primary Driver: Protein Termination

52%
19%
10%
8%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TAT TAT chr16:71,576,247 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 38.8 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This nuclear gene encodes a mitochondrial protein tyrosine aminotransferase which is present in the liver and catalyzes the conversion of L-tyrosine into p-hydroxyphenylpyruvate. Mutations in this gene cause tyrosinemia (type II, Richner-Hanhart syndrome), a disorder accompanied by major skin and corneal lesions, with possible cognitive disability. A regulator gene for tyrosine aminotransferase is X-linked. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs118203914 p.Arg57Ter | STG, p.Arg57Ter | Heterozygous | Maternal | Pathogenic | Tier1 | 36.0 | – | 38.8 (Protein Termination) | – / – | chr16:71576247 AF: 0.0017% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Tyrosinemia type II**

Complex / Multifactorial OMIM #276600 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000355962.5 |
| cDNA (HGVS): | c.169C>T |
| Protein (HGVS): | p.Arg57Ter |
| VAF / Read Depth: | VAF 0.46 · DP – |
| Phased GT: | 0\|1 (Maternal) |

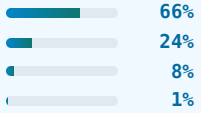
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#402 ↗ |
| gnomAD v4 AF: | 0.0017% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.8 (Raw: 2.285 · Top 0.01%)

Primary Driver: Protein Termination




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs572677025 c.*409G>A | UT3 | Heterozygous | Unphased | Uncertain significance | Tier2 | 0.8 | – | 5.8 (Splicing) | – / – | chr16:71567735 AF: 0.4009% |

## CEP63 CEP63 chr3:134,537,268 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 38.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with six coiled-coil domains. The protein is localized to the centrosome, a non-membraneous organelle that functions as the major microtubule-organizing center in animal cells. Several alternatively spliced transcript variants have been found, but their biological validity has not been determined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114108011 p.Gln185His | MIS, p.Gln185His | Heterozygous | Maternal | Benign | Tier1 | 34.0 | 0.93 | 38.6 (Splicing) | 0.28 / 0.54 | chr3:134537268 AF: 2.0823% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000675561.1 |
| cDNA (HGVS): | c.555G>C |
| Protein (HGVS): | p.Gln185His |
| VAF / Read Depth: | VAF 0.74 · DP – |
| Phased GT: | 0\|1 (Maternal) |

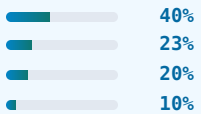
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.93 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#128709 ↗ |
| gnomAD v4 AF: | 2.0823% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.6 (Raw: 2.279 · Top 0.01%)

Primary Driver: Splicing




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## OGFOD3 OGFOD3 chr17:82,411,454 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 33.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable several functions, including L-ascorbic acid binding activity; dioxygenase activity; and iron ion binding activity. Located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs150938774<br>c.380+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 34.0 | 0.96 | 33.8 (Splicing) | – / – | chr17:82411454<br>AF: 0.6286% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000313056.10 |
| cDNA (HGVS): | c.380+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

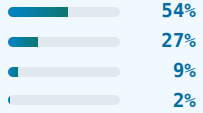
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.42 DL:0.96 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4429500 ↗ |
| gnomAD v4 AF: | 0.6286% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.882 · Top 0.04%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NIPSNAP2 NIPSNAP2 chr7:55,994,989 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 33.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the NipSnap family of proteins that may be involved in vesicular transport. The encoded protein is localized to mitochondria and plays a role in oxidative phosphorylation. A pseudogene of this gene is located on the long arm of chromosome 2. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs748801121<br>c.712+1G>T | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.97 | 33.8 (Splicing) | – / – | chr7:55994989<br>AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000322090.8 |
| cDNA (HGVS): | c.712+1G>T |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.50 · DP – |
| Phased GT: | 0/1 (Unphased) |

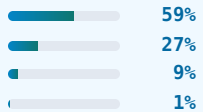
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.44 DL:0.97 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0075% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.881 · Top 0.04%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## POLG POLG chr15:89,323,460 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 30.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Mitochondrial DNA polymerase is heterotrimeric, consisting of a homodimer of accessory subunits plus a catalytic subunit. The protein encoded by this gene is the catalytic subunit of mitochondrial DNA polymerase. The encoded protein contains a polyglutamine tract near its N-terminus that may be polymorphic. Defects in this gene are a cause of progressive external ophthalmoplegia with mitochondrial DNA deletions 1 (PEOA1), sensory ataxic neuropathy dysarthria and ophthalmoparesis (SANDO), Alpers-Huttenlocher syndrome (AHS), and mitochondrial neurogastrointestinal encephalopathy syndrome (MNGIE). Two transcript variants encoding the same protein have been found for this gene. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs121918054 p.Gly737Arg | MIS, p.Gly737Arg | Heterozygous | Paternal | Pathogenic/Likely pathogenic | Tier1 | 26.3 | – | 30.3 (AlphaMissense) | 0.94 / 0.94 | chr15:89323460 AF: 0.1619% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Spinocerebellar ataxia with epilepsy**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Sensory ataxic neuropathy, dysarthria, and ophthalmoparesis**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 1**
Autosomal Dominant OMIM #203700; 607459; 613662; 157640; 258450 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

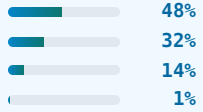
Transcript: ENST00000268124.11
cDNA (HGVS): c.2209G>C
Protein (HGVS): p.Gly737Arg
VAF / Read Depth: VAF 0.61 · DP –
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#13513 ↗
gnomAD v4 AF: 0.1619%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.3 (Raw: 1.582 · Top 0.09%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PLK4 PLK4 chr4:127,881,151 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 29.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polo family of serine/threonine protein kinases. The protein localizes to centrioles, complex microtubule-based structures found in centrosomes, and regulates centriole duplication during the cell cycle. Three alternatively spliced transcript variants that encode different protein isoforms have been found for this gene. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs149003893 p.Gly6Glu | MIS, p.Gly6Glu | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 28.5 | – | 29.8 (AlphaMissense) | 0.36 / 0.99 | chr4:127881151 AF: 0.1341% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Inborn genetic diseases**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

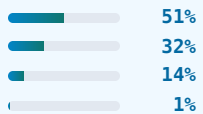
Transcript: ENST00000270861.10
cDNA (HGVS): c.17G>A
Protein (HGVS): p.Gly6Glu
VAF / Read Depth: VAF 0.45 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#436338 ↗
gnomAD v4 AF: 0.1341%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.8 (Raw: 1.555 · Top 0.11%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB1 HLA-DRB1 chr6:32,582,135 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene AlphaGenome AVI: 28.2 Showing 3 of 33 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB1 belongs to the HLA class II beta chain paralogs. The class II molecule is a heterodimer consisting of an alpha (DRA) and a beta chain (DRB), both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa. It is encoded by 6 exons. Exon one encodes the leader peptide; exons 2 and 3 encode the two extracellular domains; exon 4 encodes the transmembrane domain; and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Hundreds of DRB1 alleles have been described and some alleles have increased frequencies associated with certain diseases or conditions. For example, DRB1*1302 has been related to acute and chronic hepatitis B virus persistence. There are multiple pseudogenes of this gene. [provided by RefSeq, Jul 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs16822805<br>p.Ala103Pro | MIS, p.Ala103Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 0.0 | – | 20.2 (AlphaMissense) | 0.21 / 0.78 | chr6:32584172<br>AF: 0.1762% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.307G>C |
| Protein (HGVS): | p.Ala103Pro |
| VAF / Read Depth: | VAF 0.53 · DP – |
| Phased GT: | 0/1 (Unphased) |

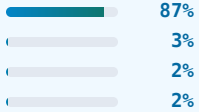
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#691505 ↗ |
| gnomAD v4 AF: | 0.1762% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 20.2 (Raw: 0.573 · Top 0.95%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28723976<br>c.371-297A>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 7.4 | 0.58 | 12.9 (Splicing) | – / – | chr6:32582135<br>AF: 0.7402% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.371-297A>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.41 · DP – |
| Phased GT: | 0/1 (Unphased) |

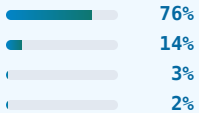
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.58 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.7402% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 12.9 (Raw: 0.248 · Top 5.08%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9269957<br>p.Gln39Ter | STG, p.Gln39Ter | Heterozygous | Unphased | Uncertain significance | Tier2 | 26.8 | 0.13 | 28.2 (Protein Termination) | – / – | chr6:32584364<br>AF: 0.0015% |

## AMPD1 AMPD1 chr1:114,677,435 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 28.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Adenosine monophosphate deaminase 1 catalyzes the deamination of AMP to IMP in skeletal muscle and plays an important role in the purine nucleotide cycle. Two other genes have been identified, AMPD2 and AMPD3, for the liver- and erythocyte-specific isoforms, respectively. Deficiency of the muscle-specific enzyme is apparently a common cause of exercise-induced myopathy and probably the most common cause of metabolic myopathy in the human. Alternatively spliced transcript variants encoding different isoforms have been identified in this gene.[provided by RefSeq, Feb 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs764950035 p.Glu435Gly | MIS, p.Glu435Gly | Heterozygous | Unphased | Uncertain significance | Tier1 | 32.0 | – | 28.2 (AlphaMissense) | 0.95 / 0.92 | chr1:114677435 AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Muscle AMP deaminase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000520113.7 |
| cDNA (HGVS): | c.1304A>G |
| Protein (HGVS): | p.Glu435Gly |
| VAF / Read Depth: | VAF 0.49 · DP – |
| Phased GT: | 0/1 (Unphased) |

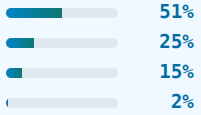
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#2173226 ↗ |
| gnomAD v4 AF: | 0.0075% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.2 (Raw: 1.454 · Top 0.15%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB5 HLA-DRB5 chr6:32,520,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 28.0 Showing 4 of 36 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB5 belongs to the HLA class II beta chain paralogues. This class II molecule is a heterodimer consisting of an alpha (DRA) and a beta (DRB) chain, both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa and its gene contains 6 exons. Exon one encodes the leader peptide, exons 2 and 3 encode the two extracellular domains, exon 4 encodes the transmembrane domain and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. There are multiple pseudogenes of this gene. [provided by RefSeq, Feb 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs201576424<br>c.101-1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.93 | 28.0 (Splicing) | – / – | chr6:32522175<br>AF: 0.0063% |

**1. INHERITANCE & OMIM PATHOLOGY**

Rheumatoid Arthritis Susceptibility
Complex / Polygenic OMIM #604305 ↗

Quadrant: Curated

**2. TRANSCRIPT, PHASING & DEPTH**

Transcript: ENST00000374975.4
cDNA (HGVS): c.101-1G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.65 AL:0.93 DG:0.00 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4326162 ↗
gnomAD v4 AF: 0.0063%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 28.0 (Raw: 1.444 · Top 0.16%)
Primary Driver: Splicing

77%
13%
4%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs796105724<br>c.371-392G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 8.4 | 1.00 | 25.7 (Splicing) | – / – | chr6:32520043<br>AF: 1.4415% |

**1. INHERITANCE & OMIM PATHOLOGY**

Rheumatoid Arthritis Susceptibility
Complex / Polygenic OMIM #604305 ↗

Quadrant: Curated

**2. TRANSCRIPT, PHASING & DEPTH**

Transcript: ENST00000374975.4
cDNA (HGVS): c.371-392G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.62 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:1.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 1.4415%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 25.7 (Raw: 1.223 · Top 0.27%)
Primary Driver: Splicing

85%
8%
2%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs774541765<br>c.100+17G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.7 | 0.68 | 18.9 (Splicing) | – / – | chr6:32530108<br>AF: 0.0027% |

**1. INHERITANCE & OMIM PATHOLOGY**

Rheumatoid Arthritis Susceptibility
Complex / Polygenic OMIM #604305 ↗

Quadrant: Curated

**2. TRANSCRIPT, PHASING & DEPTH**

Transcript: ENST00000374975.4
cDNA (HGVS): c.100+17G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.46 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.68 DL:0.01
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0027%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 18.9 (Raw: 0.503 · Top 1.28%)
Primary Driver: Splicing

80%
12%
1%
1%

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | 🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗ | | |
| rs796784269<br>c.371-393T>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 2.8 | 0.94 | 16.6 (Splicing) | – / – | chr6:32520044<br>AF: 0.1426% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000374975.4 |
| cDNA (HGVS): | c.371-393T>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.94 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.1426% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 16.6 (Raw: 0.386 · Top 2.21%)
Primary Driver: Splicing

87%
5%
3%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CTH CTH chr1:70,415,987 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 27.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a cytoplasmic enzyme in the trans-sulfuration pathway that converts cystathione derived from methionine into cysteine. Glutathione synthesis in the liver is dependent upon the availability of cysteine. Mutations in this gene cause cystathioninuria. Alternative splicing of this gene results in three transcript variants encoding different isoforms. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28941785<br>p.Thr67Ile | MIS, p.Thr67Ile | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 25.5 | – | 27.8 (AlphaMissense) | 0.93 / 0.81 | chr1:70415987<br>AF: 0.8026% |

**1. INHERITANCE & OMIM PATHOLOGY**

**CTH-related disorder**
Complex / Multifactorial OMIM #219500 ↗

**Cystathioninuria**
Complex / Multifactorial OMIM #219500 ↗

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000370938.8 |
| cDNA (HGVS): | c.200C>T |
| Protein (HGVS): | p.Thr67Ile |
| VAF / Read Depth: | VAF 0.46 · DP – |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#2939 ↗ |
| gnomAD v4 AF: | 0.8026% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense

45%
36%
14%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## MID1IP1 MID1IP1 chrX:38,805,491 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 27.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable enzyme activator activity. Predicted to be involved in several processes, including negative regulation of microtubule depolymerization; positive regulation of fatty acid biosynthetic process; and positive regulation of ligase activity. Predicted to be located in microtubule cytoskeleton. Predicted to be active in cytosol and nucleus. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs143008624 p.Gly182Asp | MIS, p.Gly182Asp | Heterozygous | Unphased | Benign | Tier1 | 26.1 | – | 27.1 (AlphaMissense) | 0.63 / 0.87 | chrX:38805491 AF: 0.3491% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000614558.3 |
| cDNA (HGVS): | c.545G>A |
| Protein (HGVS): | p.Gly182Asp |
| VAF / Read Depth: | VAF 0.48 · DP – |
| Phased GT: | 0/1 (Unphased) |

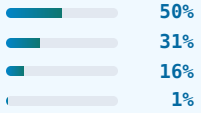
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#711083 ↗ |
| gnomAD v4 AF: | 0.3491% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.1 (Raw: 1.393 · Top 0.19%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAAP24 FAAP24 chr19:32,974,193 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 27.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 FAAP24 is a component of the Fanconi anemia (FA) core complex (see MIM 227650), which plays a crucial role in DNA damage response (Ciccia et al., 2007 [PubMed 17289582]).[supplied by OMIM, Mar 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs36017455 p.Ser126Phe | MIS, p.Ser126Phe | Heterozygous | Unphased | Benign | Tier1 | 25.7 | – | 27.0 (AlphaMissense) | – / 0.69 | chr19:32974193 AF: 0.5937% |

### 1. INHERITANCE & OMIM PATHOLOGY

**FAAP24-related disorder**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000588258.6 |
| cDNA (HGVS): | c.377C>T |
| Protein (HGVS): | p.Ser126Phe |
| VAF / Read Depth: | VAF 0.44 · DP – |
| Phased GT: | 0/1 (Unphased) |

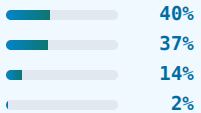
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#784831 ↗ |
| gnomAD v4 AF: | 0.5937% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.0 (Raw: 1.385 · Top 0.20%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLC27A3 SLC27A3 chr1:153,776,015 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 26.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to a family of integral membrane proteins and encodes a protein that is involved in lipid metabolism. The increased expression of this gene in human neural stem cells derived from induced pluripotent stem cells suggests that it plays an important role in early brain development. Naturally occurring mutations in this gene are associated with autism spectrum disorders. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Feb 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs756978807 p.Leu173Pro | MIS, p.Leu173Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 26.9 (AlphaMissense) | 0.41 / 0.94 | chr1:153776015 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

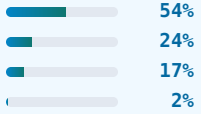
Transcript: ENST00000624995.4
cDNA (HGVS): c.518T>C
Protein (HGVS): p.Leu173Pro
VAF / Read Depth: VAF 0.23 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0001%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.9 (Raw: 1.371 · Top 0.21%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TMEM43 TMEM43 chr3:14,139,244 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 26.5 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to the TMEM43 family. Defects in this gene are the cause of familial arrhythmogenic right ventricular dysplasia type 5 (ARVD5), also known as arrhythmogenic right ventricular cardiomyopathy type 5 (ARVC5). Arrhythmogenic right ventricular dysplasia is an inherited disorder, often involving both ventricles, and is characterized by ventricular tachycardia, heart failure, sudden cardiac death, and fibrofatty replacement of cardiomyocytes. This gene contains a response element for PPAR gamma (an adipogenic transcription factor), which may explain the fibrofatty replacement of the myocardium, a characteristic pathological finding in ARVC. [provided by RefSeq, Oct 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs199526104 p.Trp316Ser | MIS, p.Trp316Ser | Heterozygous | Paternal | Conflicting classifications of pathogenicity | Tier1 | 32.0 | – | 26.5 (AlphaMissense) | 0.53 / 0.76 | chr3:14139244 AF: 0.0126% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**
Complex / Multifactorial OMIM #604400 ↗

**TMEM43-related disorder**
Complex / Multifactorial OMIM #604400 ↗

**Auditory neuropathy, autosomal dominant 3**
Autosomal Dominant OMIM #604400 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

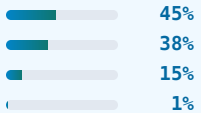
Transcript: ENST00000306077.5
cDNA (HGVS): c.947G>C
Protein (HGVS): p.Trp316Ser
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#46155 ↗
gnomAD v4 AF: 0.0126%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.5 (Raw: 1.335 · Top 0.22%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1043943 c.*115T>C | UT3 | Heterozygous | Paternal | Benign/Likely benign | Tier2 | 1.6 | – | 9.2 (ATAC-seq) | – / – | chr3:14141910 AF: 62.2376% |

## NPC2 NPC2 chr14:74,480,701 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 26.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein containing a lipid recognition domain. The encoded protein may function in regulating the transport of cholesterol through the late endosomal/lysosomal system. Mutations in this gene have been associated with Niemann-Pick disease, type C2 and frontal lobe atrophy. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs140130028<br>c.441+1G>A | SPL | Heterozygous | Maternal | Conflicting classifications of pathogenicity | Tier1 | 23.2 | 0.98 | 26.3 (Splicing) | 0.21 / – | chr14:74480701<br>AF: 0.6316% |

### 1. INHERITANCE & OMIM PATHOLOGY

**NPC2-related disorder**

Complex / Multifactorial OMIM #257220; 607625 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000555619.6 |
| cDNA (HGVS): | c.441+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0\|1 (Maternal) |

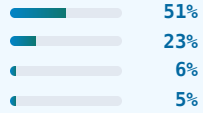
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.10 DL:0.98 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#100734 ↗ |
| gnomAD v4 AF: | 0.6316% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SUMF1 SUMF1 chr3:4,362,213 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 25.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an enzyme that catalyzes the hydrolysis of sulfate esters by oxidizing a cysteine residue in the substrate sulfatase to an active site 3-oxoalanine residue, which is also known as C-alpha-formylglycine. Mutations in this gene cause multiple sulfatase deficiency, a lysosomal storage disorder. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2009]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs376126385<br>p.Asn352Lys | MIS, p.Asn352Lys | Heterozygous | Unphased | Uncertain significance | Tier1 | 23.9 | – | 25.9 (AlphaMissense) | 0.64 / 1.00 | chr3:4362213<br>AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Multiple sulfatase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000272902.10 |
| cDNA (HGVS): | c.1056C>G |
| Protein (HGVS): | p.Asn352Lys |
| VAF / Read Depth: | VAF 0.66 · DP – |
| Phased GT: | 0/1 (Unphased) |

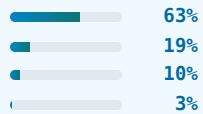
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#2161926 ↗ |
| gnomAD v4 AF: | 0.0009% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## IL17RC IL17RC chr3:9,932,859 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 25.9 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a single-pass type I membrane protein that shares similarity with the interleukin-17 receptor (IL-17RA). Unlike IL-17RA, which is predominantly expressed in hemopoietic cells, and binds with high affinity to only IL-17A, this protein is expressed in nonhemopoietic tissues, and binds both IL-17A and IL-17F with similar affinities. The proinflammatory cytokines, IL-17A and IL-17F, have been implicated in the progression of inflammatory and autoimmune diseases. Multiple alternatively spliced transcript variants encoding different isoforms have been detected for this gene, and it has been proposed that soluble, secreted proteins lacking transmembrane and intracellular domains may function as extracellular antagonists to cytokine signaling. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs148575246<br>c.1522+1G>C | SPL | Heterozygous | Unphased | Benign/Likely benign | Tier1 | 33.0 | 0.97 | 25.9 (Splicing) | – / – | chr3:9932859<br>AF: 1.0744% |

### 1. INHERITANCE & OMIM PATHOLOGY

**IL17RC-related disorder**
Complex / Multifactorial OMIM #616445 ↗

**Candidiasis, familial, 9**
Complex / Multifactorial OMIM #616445 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

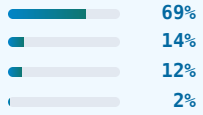
Transcript: ENST00000403601.8
cDNA (HGVS): c.1522+1G>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.01 DL:0.97
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#542537 ↗
gnomAD v4 AF: 1.0744%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs115461448<br>p.Ala324Thr | MIS, p.Ala324Thr | Heterozygous | Unphased | Benign | Tier3 | 23.2 | – | 22.0 (Cactus) | 0.09 / 0.17 | chr3:9928397<br>AF: 1.4147% |

## GUSB GUSB chr7:65,960,973 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a hydrolase that degrades glycosaminoglycans, including heparan sulfate, dermatan sulfate, and chondroitin-4,6-sulfate. The enzyme forms a homotetramer that is localized to the lysosome. Mutations in this gene result in mucopolysaccharidosis type VII. Alternative splicing results in multiple transcript variants. There are many pseudogenes of this locus in the human genome.[provided by RefSeq, May 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1236992554<br>p.Trp627Leu | MIS, p.Trp627Leu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.2 | – | 24.9 (Cactus) | 0.56 / 0.25 | chr7:65960973<br>AF: 0.0003% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

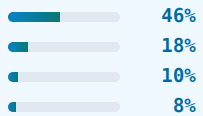
Transcript: ENST00000304895.9
cDNA (HGVS): c.1880G>T
Protein (HGVS): p.Trp627Leu
VAF / Read Depth: VAF 0.53 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0003%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.9 (Raw: 1.103 · Top 0.32%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## WDHD1 WDHD1 chr14:55,002,157 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene contains multiple N-terminal WD40 domains and a C-terminal high mobility group (HMG) box. WD40 domains are found in a variety of eukaryotic proteins and may function as adaptor/regulatory modules in signal transduction, pre-mRNA processing and cytoskeleton assembly. HMG boxes are found in many eukaryotic proteins involved in chromatin assembly, transcription and replication. Alternative splicing results in two transcript variants encoding different isoforms. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs374497720 p.Val210Ala | MIS, p.Val210Ala | Heterozygous | Unphased | Not reviewed | Tier1 | 25.8 | – | 24.5 (AlphaMissense) | 0.27 / 0.58 | chr14:55002157 AF: 0.0190% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

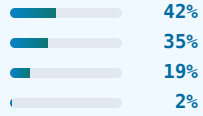
Transcript: **ENST00000360586.8**
cDNA (HGVS): **c.629T>C**
Protein (HGVS): **p.Val210Ala**
VAF / Read Depth: **VAF 0.50 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0190%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.5** (Raw: 1.042 · Top 0.35%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ACSM2B ACSM2B chr16:20,540,665 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables benzoate-CoA ligase activity. Predicted to be involved in acyl-CoA metabolic process and fatty acid biosynthetic process. Located in mitochondrion. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs139253186 p.Tyr540Asn | MIS, p.Tyr540Asn | Heterozygous | Unphased | Not reviewed | Tier1 | 27.6 | – | 24.4 (AlphaMissense) | 0.40 / 0.58 | chr16:20540665 AF: 0.4705% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

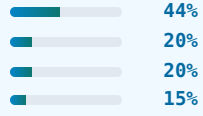
Transcript: **ENST00000329697.10**
cDNA (HGVS): **c.1618T>A**
Protein (HGVS): **p.Tyr540Asn**
VAF / Read Depth: **VAF 0.54 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.4705%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.4** (Raw: 1.012 · Top 0.37%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PGK1 PGK1 chrX:78,114,140 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 24.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a glycolytic enzyme that catalyzes the conversion of 1,3-diphosphoglycerate to 3-phosphoglycerate. The encoded protein may also act as a cofactor for polymerase alpha. Additionally, this protein is secreted by tumor cells where it participates in angiogenesis by functioning to reduce disulfide bonds in the serine protease, plasmin, which consequently leads to the release of the tumor blood vessel inhibitor angiostatin. The encoded protein has been identified as a moonlighting protein based on its ability to perform mechanistically distinct functions. Deficiency of the enzyme is associated with a wide range of clinical phenotypes hemolytic anemia and neurological impairment. Pseudogenes of this gene have been defined on chromosomes 19, 21 and the X chromosome. [provided by RefSeq, Jan 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| **chrX:78114140** p.Lys133Glu | MIS, p.Lys133Glu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.7 | – | 24.3 (AlphaMissense) | 0.54 / 0.57 | chrX:78114140 AF: 0.0008% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

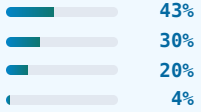
Transcript: **ENST00000373316.5**
cDNA (HGVS): **c.397A>G**
Protein (HGVS): **p.Lys133Glu**
VAF / Read Depth: **VAF 0.58 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0008%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.3** (Raw: 0.993 · Top 0.38%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLIT2 SLIT2 chr4:20,567,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 23.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the slit family of secreted glycoproteins, which are ligands for the Robo family of immunoglobulin receptors. Slit proteins play highly conserved roles in axon guidance and neuronal migration and may also have functions during other cell migration processes including leukocyte migration. Members of the slit family are characterized by an N-terminal signal peptide, four leucine-rich repeats, nine epidermal growth factor repeats, and a C-terminal cysteine knot. Proteolytic processing of this protein gives rise to an N-terminal fragment that contains the four leucine-rich repeats and five epidermal growth factor repeats and a C-terminal fragment that contains four epidermal growth factor repeats and the cysteine knot. Both full length and cleaved proteins are secreted extracellularly and can function in axon repulsion as well as other specific processes. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2015]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1020555095 p.Gly909Asp | MIS, p.Gly909Asp | Heterozygous | Unphased | Not reviewed | Tier1 | 32.0 | – | 23.4 (Cactus) | 0.74 / 0.35 | chr4:20567262 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

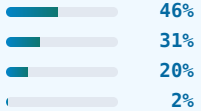
Transcript: **ENST00000504154.6**
cDNA (HGVS): **c.2726G>A**
Protein (HGVS): **p.Gly909Asp**
VAF / Read Depth: **VAF 0.54 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0009%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **23.4** (Raw: 0.859 · Top 0.46%)
Primary Driver: **Cactus**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## DMAP1 DMAP1 chr1:44,218,751 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 23.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a subunit of several, distinct complexes involved in the repression or activation of transcription. The encoded protein can independently repress transcription and is targeted to replication foci throughout S phase by interacting directly with the N-terminus of DNA methyltransferase 1. During late S phase, histone deacetylase 2 is added to this complex, providing a means to deacetylate histones in transcriptionally inactive heterochromatin following replication. The encoded protein is also a component of the nucleosome acetyltransferase of H4 complex and interacts with the transcriptional corepressor tumor susceptibility gene 101 and the pro-apoptotic death-associated protein 6, among others. Alternatively spliced transcript variants encoding the same protein have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| **chr1:44218751** p.Glu239Gly | MIS, p.Glu239Gly | **Heterozygous** | Unphased | Not reviewed | Tier1 | **29.7** | – | **23.4 (Cactus)** | 0.38 / 0.38 | chr1:44218751 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

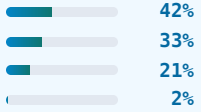
Transcript: **ENST00000372289.7**
cDNA (HGVS): **c.716A>G**
Protein (HGVS): **p.Glu239Gly**
VAF / Read Depth: **VAF 0.38 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0001%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **23.4** (Raw: 0.854 · Top 0.46%)
Primary Driver: **Cactus**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## LDHAL6A LDHAL6A chr11:18,475,640 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 23.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables L-lactate dehydrogenase (NAD+) activity. Predicted to be involved in lactate metabolic process and pyruvate catabolic process. Located in cytoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs145833201 c.592+1G>A | SPL | **Heterozygous** | Unphased | Not reviewed | Tier1 | **31.0** | 0.69 | **23.3 (Splicing)** | – / – | chr11:18475640 AF: 0.1631% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

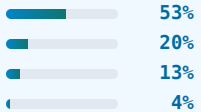
Transcript: **ENST00000280706.3**
cDNA (HGVS): **c.592+1G>A**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.36 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.69**
ClinVar Review: **criteria provided**
ClinVar VCV: VCV#4370113 ↗
gnomAD v4 AF: **0.1631%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **23.3** (Raw: 0.845 · Top 0.47%)
Primary Driver: **Splicing**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CHST14 CHST14 chr15:40,472,259 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 22.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the HNK-1 family of sulfotransferases. The encoded protein transfers sulfate to the C-4 hydroxyl of N-acetylgalactosamine residues in dermatan sulfate. Mutations in this gene have been associated with adducted thumb-clubfoot syndrome.[provided by RefSeq, Mar 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1463514561 p.Gln349Pro | MIS, p.Gln349Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 24.3 | — | 22.9 (Cactus) | 0.54 / 0.20 | chr15:40472259 AF: 0.0004% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

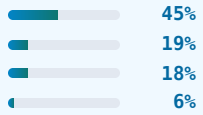
Transcript: **ENST00000306243.7**
cDNA (HGVS): **c.1046A>C**
Protein (HGVS): **p.Gln349Pro**
VAF / Read Depth: **VAF 0.52 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#3229028 ↗
gnomAD v4 AF: **0.0004%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **22.9** (Raw: 0.798 · Top 0.51%)
Primary Driver: **Cactus**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TEK TEK chr9:27,185,615 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 22.9 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a receptor that belongs to the protein tyrosine kinase Tie2 family. The encoded protein possesses a unique extracellular region that contains two immunoglobulin-like domains, three epidermal growth factor (EGF)-like domains and three fibronectin type III repeats. The ligand angiopoietin-1 binds to this receptor and mediates a signaling pathway that functions in embryonic vascular development. Mutations in this gene are associated with inherited venous malformations of the skin and mucous membranes. Alternative splicing results in multiple transcript variants. Additional alternatively spliced transcript variants of this gene have been described, but their full-length nature is not known. [provided by RefSeq, Feb 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs189543659 p.Asn438Ile | MIS, p.Asn438Ile | Heterozygous | Unphased | Not reviewed | Tier1 | 24.3 | — | 21.9 (Cactus) | 0.21 / 0.21 | chr9:27185615 AF: 0.0011% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

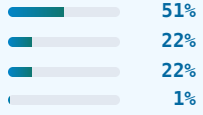
Transcript: **ENST00000380036.10**
cDNA (HGVS): **c.1313A>T**
Protein (HGVS): **p.Asn438Ile**
VAF / Read Depth: **VAF 0.50 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0011%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **21.9** (Raw: 0.695 · Top 0.65%)
Primary Driver: **Cactus**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs35030851 p.Val600Leu | MIS, p.Val600Leu | Heterozygous | Unphased | Benign | Tier3 | 17.9 | — | 22.9 (Cactus) | 0.06 / 0.16 | chr9:27197488 AF: 4.5993% |

## FAHD1 FAHD1 chr16:1,827,542 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 22.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables hydrolase activity, acting on carbon-carbon bonds, in ketonic substances; oxaloacetate decarboxylase activity; and oxaloacetate tautomerase activity. Involved in oxaloacetate metabolic process. Located in cytosol; mitochondrion; and nucleoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs200357596 p.Ala102Ser | MIS, p.Ala102Ser | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | — | 22.1 (Cactus) | 0.71 / 0.24 | chr16:1827542 AF: 0.0037% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000427358.5
cDNA (HGVS): c.304G>T
Protein (HGVS): p.Ala102Ser
VAF / Read Depth: VAF 0.49 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0037%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.1 (Raw: 0.710 · Top 0.62%)
Primary Driver: Cactus

45%
25%
22%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## VPS35L VPS35L chr16:19,629,820 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 21.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Involved in Golgi to plasma membrane transport and endocytic recycling. Located in endosome. Part of protein-containing complex. Implicated in Ritscher-Schinzel syndrome. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs144908771 p.Thr518= | SYN, p.Thr518= | Heterozygous | Unphased | Likely benign | Tier1 | 22.5 | 0.64 | 21.4 (Cactus) | — / — | chr16:19629820 AF: 0.3238% |

### 1. INHERITANCE & OMIM PATHOLOGY

**VPS35L-related disorder**
Complex / Multifactorial

**Ritscher-Schinzel syndrome**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000417362.7
cDNA (HGVS): c.1554G>A
Protein (HGVS): p.Thr518=
VAF / Read Depth: VAF 0.45 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.64 DL:0.01
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#2672622 ↗
gnomAD v4 AF: 0.3238%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.4 (Raw: 0.657 · Top 0.73%)
Primary Driver: Cactus

48%
22%
20%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLC5A4 SLC5A4 chr22:32,233,039 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 21.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable low-affinity D-glucose:sodium symporter activity and proton transmembrane transporter activity. Predicted to be involved in D-glucose import across plasma membrane and renal D-glucose absorption. Predicted to act upstream of or within proton transmembrane transport. Predicted to be located in membrane. Predicted to be active in plasma membrane. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9621415 c.886-5G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 14.0 | 0.97 | 21.1 (Splicing) | – / – | chr22:32233039 AF: 1.4374% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000266086.6 |
| cDNA (HGVS): | c.886-5G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.41 · DP – |
| Phased GT: | 0/1 (Unphased) |

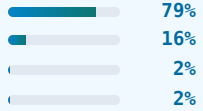
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.97 AL:0.01 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4457848 ↗ |
| gnomAD v4 AF: | 1.4374% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 21.1 (Raw: 0.639 · Top 0.78%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-B HLA-B chr6:31,356,717 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 20.0 Showing 3 of 8 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-B belongs to the HLA class I heavy chain paralogues. This class I molecule is a heterodimer consisting of a heavy chain and a light chain (beta-2 microglobulin). The heavy chain is anchored in the membrane. Class I molecules play a central role in the immune system by presenting peptides derived from the endoplasmic reticulum lumen. They are expressed in nearly all cells. The heavy chain is approximately 45 kDa and its gene contains 8 exons. Exon 1 encodes the leader peptide, exon 2 and 3 encode the alpha1 and alpha2 domains, which both bind the peptide, exon 4 encodes the alpha3 domain, exon 5 encodes the transmembrane region and exons 6 and 7 encode the cytoplasmic tail. Polymorphisms within exon 2 and exon 3 are responsible for the peptide binding specificity of each class one molecule. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. Hundreds of HLA-B alleles have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9266161 p.Leu105Pro | MIS, p.Leu105Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 22.5 | – | 20.0 (AlphaMissense) | 0.20 / 0.79 | chr6:31356717 AF: 0.0006% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000412585.7 |
| cDNA (HGVS): | c.314T>C |
| Protein (HGVS): | p.Leu105Pro |
| VAF / Read Depth: | VAF 0.55 · DP – |
| Phased GT: | 0/1 (Unphased) |

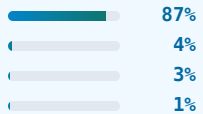
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0006% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 20.0 (Raw: 0.559 · Top 1.00%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1131212 p.Gln94His | MIS, p.Gln94His | Homozygous | N/A | Not reviewed | Tier2 | 7.1 | – | 11.3 (AlphaMissense) | 0.24 / 0.28 | chr6:31356749 AF: 0.2441% |
| rs9266144 p.Arg180Gly | MIS, p.Arg180Gly | Homozygous | N/A | Not reviewed | Tier2 | 0.0 | – | 10.6 (AlphaMissense) | 0.25 / 0.22 | chr6:31356248 AF: 0.0002% |

## CD40 CD40 chr20:46,126,737 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 19.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the TNF-receptor superfamily. The encoded protein is a receptor on antigen-presenting cells of the immune system and is essential for mediating a broad variety of immune and inflammatory responses including T cell-dependent immunoglobulin class switching, memory B cell development, and germinal center formation. AT-hook transcription factor AKNA is reported to coordinately regulate the expression of this receptor and its ligand, which may be important for homotypic cell interactions. Adaptor protein TNFR2 interacts with this receptor and serves as a mediator of the signal transduction. The interaction of this receptor and its ligand is found to be necessary for amyloid-beta-induced microglial activation, and thus is thought to be an early event in Alzheimer disease pathogenesis. Mutations affecting this gene are the cause of autosomal recessive hyper-IgM immunodeficiency type 3 (HIGM3). Multiple alternatively spliced transcript variants of this gene encoding distinct isoforms have been reported. [provided by RefSeq, Nov 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs41282788<br>c.559+36G>C | INT | Heterozygous | Unphased | Benign | Tier1 | 2.0 | 0.51 | 19.4 (Splicing) | — / — | chr20:46126737<br>AF: 1.8172% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Hyper-IgM syndrome type 3**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

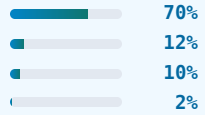
Transcript: ENST00000372285.8
cDNA (HGVS): c.559+36G>C
Protein (HGVS): —
VAF / Read Depth: VAF 0.58 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.51 DL:0.10
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#993383 ↗
gnomAD v4 AF: 1.8172%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.4 (Raw: 0.528 · Top 1.14%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ACOX3 ACOX3 chr4:8,440,048 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 19.4 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acyl-Coenzyme A oxidase 3 also know as pristanoyl -CoA oxidase (ACOX3)is involved in the desaturation of 2-methyl branched fatty acids in peroxisomes. Unlike the rat homolog, the human gene is expressed in very low amounts in liver such that its mRNA was undetectable by routine Northern-blot analysis or its product by immunoblotting or by enzyme activity measurements. However the human cDNA encoding a 700 amino acid protein with a peroxisomal targeting C-terminal tripeptide S-K-L was isolated and is thought to be expressed under special conditions such as specific developmental stages or in a tissue specific manner in tissues that have not yet been examined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs62287361<br>c.-15+600A>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 3.3 | 0.65 | 19.4 (Splicing) | — / — | chr4:8440048<br>AF: 2.7459% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

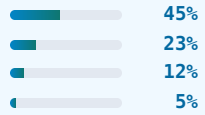
Transcript: ENST00000356406.10
cDNA (HGVS): c.-15+600A>G
Protein (HGVS): —
VAF / Read Depth: VAF 0.51 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.65 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 2.7459%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.4 (Raw: 0.526 · Top 1.15%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## UCK2 UCK2 chr1:165,828,740 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 19.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a pyrimidine ribonucleoside kinase. The encoded protein (EC 2.7.1.48) catalyzes phosphorylation of uridine and cytidine to uridine monophosphate (UMP) and cytidine monophosphate (CMP), respectively.[provided by RefSeq, Oct 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs562989479<br>c.99+808G>T | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.9 | 0.57 | 19.2 (Splicing) | – / – | chr1:165828740<br>AF: 0.3704% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

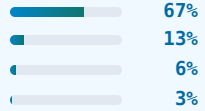
Transcript: ENST00000367879.9
cDNA (HGVS): c.99+808G>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.32 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.57 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.3704%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.2 (Raw: 0.517 · Top 1.20%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## APOB APOB chr2:21,016,593 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 18.4 Showing 3 of 7 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene product is the main apolipoprotein of chylomicrons and low density lipoproteins (LDL), and is the ligand for the LDL receptor. It occurs in plasma as two main isoforms, apoB-48 and apoB-100: the former is synthesized exclusively in the gut and the latter in the liver. The intestinal and the hepatic forms of apoB are encoded by a single gene from a single, very long mRNA. The two isoforms share a common N-terminal sequence. The shorter apoB-48 protein is produced after RNA editing of the apoB-100 transcript at residue 2180 (CAA->UAA), resulting in the creation of a stop codon, and early translation termination. Mutations in this gene or its regulatory region cause hypobetalipoproteinemia, normotriglyceridemic hypobetalipoproteinemia, and hypercholesterolemia due to ligand-defective apoB, diseases affecting plasma cholesterol and apoB levels. [provided by RefSeq, Dec 2019]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs72653073<br>p.Leu1060= | SYN, p.Leu1060= | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 9.1 | – | 11.5 (PhastCons 470) | – / – | chr2:21016593<br>AF: 0.1326% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Familial Hypercholesterolemia 2**
Autosomal Dominant OMIM #144010 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

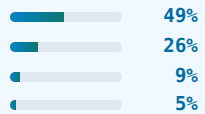
Transcript: ENST00000233242.5
cDNA (HGVS): c.3178T>C
Protein (HGVS): p.Leu1060=
VAF / Read Depth: VAF 0.61 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#334158 ↗
gnomAD v4 AF: 0.1326%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 11.5 (Raw: 0.199 · Top 7.16%)
Primary Driver: PhastCons 470




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1367117<br>p.Thr98Ile | MIS, p.Thr98Ile | Homozygous | N/A | Benign/Likely benign | Tier2 | 22.8 | – | 18.4 (Cactus) | 0.09 / 0.14 | chr2:21041028<br>AF: 29.5755% |
| rs693<br>p.Thr2515= | SYN, p.Thr2515= | Heterozygous | Unphased | Benign | Tier2 | 5.4 | – | 10.5 (PhastCons 470) | – / – | chr2:21009323<br>AF: 45.9266% |

## HEPACAM HEPACAM chr11:124,923,060 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 14.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a single-pass type I membrane protein that localizes to the cytoplasmic side of the cell membrane. The encoded protein acts as a homodimer and is involved in cell motility and cell-matrix interactions. The expression of this gene is downregulated or undetectable in many cancer cell lines, so this may be a tumor suppressor gene. [provided by RefSeq, Jul 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs11605766<br>c.804-242G>C | INT | Heterozygous | Unphased | Likely benign | Tier1 | 14.2 | 0.55 | 14.7 (Splicing) | – / – | chr11:124923060<br>AF: 1.0897% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000298251.5 |
| cDNA (HGVS): | c.804-242G>C |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.57 · DP – |
| Phased GT: | 0/1 (Unphased) |

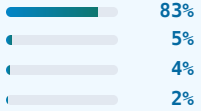
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.55 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#1201816 ↗ |
| gnomAD v4 AF: | 1.0897% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 14.7 (Raw: 0.311 · Top 3.35%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ADAMTS7 ADAMTS7 chr15:78,790,059 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 13.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ADAMTS (a disintegrin and metalloproteinase with thrombospondin motifs) family. Members of this family share several distinct protein modules, including a propeptide region, a metalloproteinase domain, a disintegrin-like domain, and a thrombospondin type 1 (TS) motif. Individual members of this family differ in the number of C-terminal TS motifs, and some have unique C-terminal domains. The encoded preproprotein is proteolytically processed to generate the mature enzyme. This enzyme contains two C-terminal TS motifs and may regulate vascular smooth muscle cell (VSMC) migration. Mutations in this gene may be associated with susceptibility to coronary artery disease. [provided by RefSeq, Feb 2016]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs142754252<br>c.1029-221G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 0.2 | 0.65 | 13.8 (Splicing) | – / – | chr15:78790059<br>AF: 0.1293% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000388820.5 |
| cDNA (HGVS): | c.1029-221G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.42 · DP – |
| Phased GT: | 0/1 (Unphased) |

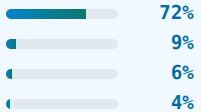
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.65 AL:0.00 DG:0.02 DL:0.00 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4406248 ↗ |
| gnomAD v4 AF: | 0.1293% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.8 (Raw: 0.277 · Top 4.16%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FHIT FHIT chr3:60,547,291 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 12.9 Showing 3 of 5 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a P1-P3-bis(5'-adenosyl) triphosphate hydrolase involved in purine metabolism. This gene encompasses the common fragile site FRA3B on chromosome 3, where carcinogen-induced damage can lead to translocations and aberrant transcripts. In fact, aberrant transcripts from this gene have been found in about half of all esophageal, stomach, and colon carcinomas. The encoded protein is also a tumor suppressor, as loss of its activity results in replication stress and DNA damage. [provided by RefSeq, Aug 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114893898<br>c.-17-10312G>C | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 9.7 | 0.51 | 9.8 (Cactus) | – / – | chr3:60547291<br>AF: 4.4825% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000492590.6 |
| cDNA (HGVS): | c.-17-10312G>C |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.47 · DP – |
| Phased GT: | 0/1 (Unphased) |

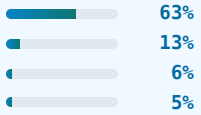
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.51 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 4.4825% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 9.8 (Raw: 0.147 · Top 10.59%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1029925290<br>c.103+75925A>G | INT | Heterozygous | Unphased | Not reviewed | Tier3 | 16.4 | – | 12.9 (PhastCons 470) | – / – | chr3:60460935<br>AF: 0.0092% |
| rs956035979<br>c.-110-5170T>A | INT | Heterozygous | Unphased | Not reviewed | Tier3 | 9.3 | – | 10.7 (Splicing) | – / – | chr3:60827181<br>AF: 0.0191% |

## SYNPR SYNPR chr3:63,319,195 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 6.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to be involved in modulation of chemical synaptic transmission. Predicted to be located in membrane; neuron projection; and synaptic vesicle. Predicted to be active in hippocampal mossy fiber to CA3 synapse and synaptic vesicle membrane. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs73123134<br>c.84+40453T>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.8 | 0.54 | 6.0 (Cactus) | – / – | chr3:63319195<br>AF: 4.7697% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000478300.6 |
| cDNA (HGVS): | c.84+40453T>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.29 · DP – |
| Phased GT: | 0/1 (Unphased) |

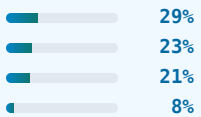
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.54 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 4.7697% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 6.0 (Raw: 0.048 · Top 25.19%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PLB1 PLB1 chr2:28,461,598 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 3.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a membrane-associated phospholipase that displays lysophospholipase and phospholipase A2 activities through removal of sn-1 and sn-2 fatty acids of glycerophospholipids. In addition, it displays lipase and retinyl ester hydrolase activities. The encoded protein is highly conserved and is composed of a large, glycosylated extracellular domain composed of four tandem homologous domains, followed by a hydrophobic segment that anchors the enzyme to the membrane and a short C-terminal cytoplasmic tail. This gene has been identified as a candidate rheumatoid arthritis risk gene. [provided by RefSeq, Jul 2016]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs56116810<br>c.-114+4401G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 3.5 | – | 3.6 (Cactus) | – / – | chr2:28461598<br>AF: 0.0012% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

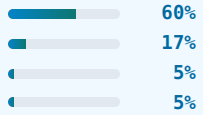
Transcript: ENST00000416713.5
cDNA (HGVS): c.-114+4401G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0012%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 3.6 (Raw: 0.009 · Top 43.90%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## VDR VDR chr12:47,906,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 2.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes vitamin D3 receptor, which is a member of the nuclear hormone receptor superfamily of ligand-inducible transcription factors. This receptor also functions as a receptor for the secondary bile acid, lithocholic acid. Downstream targets of vitamin D3 receptor are principally involved in mineral metabolism, though this receptor regulates a variety of other metabolic pathways, such as those involved in immune response and cancer. Mutations in this gene are associated with type II vitamin D-resistant rickets. A single nucleotide polymorphism in the initiation codon results in an alternate translation start site three codons downstream. Alternatively spliced transcript variants encoding different isoforms have been described for this gene. A recent study provided evidence for translational readthrough in this gene, and expression of an additional C-terminally extended isoform via the use of an alternative in-frame translation termination codon. [provided by RefSeq, Jun 2018]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs4516035<br>c.-1172A>G | 2KU | Heterozygous | Unphased | Likely pathogenic | Tier3 | 5.4 | – | 2.1 (DNASE-seq) | – / – | chr12:47906043<br>AF: 30.5142% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Chronic obstructive pulmonary disease**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000549336.6
cDNA (HGVS): c.-1172A>G
Protein (HGVS): –
VAF / Read Depth: VAF 0.19 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: no assertion criteria provided
ClinVar VCV: VCV#3336650 ↗
gnomAD v4 AF: 30.5142%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 2.1 (Raw: -0.015 · Top 61.95%)
Primary Driver: DNASE-seq




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CDKN2B CDKN2B chr9:22,003,368 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaGenome AVI: 1.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene lies adjacent to the tumor suppressor gene CDKN2A in a region that is frequently mutated and deleted in a wide variety of tumors. This gene encodes a cyclin-dependent kinase inhibitor, which forms a complex with CDK4 or CDK6, and prevents the activation of the CDK kinases, thus the encoded protein functions as a cell growth regulator that controls cell cycle G1 progression. The expression of this gene was found to be dramatically induced by TGF beta, which suggested its role in the TGF beta induced growth inhibition. Two alternatively spliced transcript variants of this gene, which encode distinct proteins, have been reported. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1063192<br>c.*2619C>T | UT3 | Heterozygous | Maternal | Likely pathogenic\|protective | Tier3 | 0.2 | – | 1.3 (Cactus) | – / – | chr9:22003368<br>AF: 68.9409% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Coronary Artery Disease & Breast Carcinoma**

Complex / Polygenic Protective Allele OMIM #600431 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000276925.7 |
| cDNA (HGVS): | c.*2619C>T |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0\|1 (Maternal) |

### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | no assertion criteria provided |
| ClinVar VCV: | VCV#812639 ↗ |
| gnomAD v4 AF: | 68.9409% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 1.3 (Raw: -0.032 · Top 74.07%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## 4. CADD Deleteriousness Ranking — Tier-1 & Top Genes (Phred ≥ 15)

Top deleterious genes ranked by Combined Annotation Dependent Depletion (CADD v1.7) Phred-scaled score.

**Selected Scope:** 29 Genes (29 Tier-1/Pathogenic)
Filtered from 530 matching genes in view

### VWA3B VWA3B chr2:98,162,976 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 46.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an intracellular protein that contains a von Willebrand factor type A domain. Intracellular proteins with VWA domains are thought to function in transcription, DNA repair, ribosomal and membrane transport and the proteasome. Mutations in this gene are associated with Spinocerebellar ataxia, autosomal recessive 22. Alternatively spliced transcript variants have been found for this gene. [provided by RefSeq, May 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs62154921 p.Glu372Ter | STG, p.Glu372Ter | Heterozygous | Unphased | Benign | Tier1 | 46.0 | 0.85 | 50.8 (Protein Termination) | – / – | chr2:98162976 AF: 4.2196% |

**1. INHERITANCE & OMIM PATHOLOGY**

VWA3B-related disorder
Complex / Multifactorial

Quadrant: Curated

**2. TRANSCRIPT, PHASING & DEPTH**

Transcript: ENST00000477737.6
cDNA (HGVS): c.1114G>T
Protein (HGVS): p.Glu372Ter
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.85
ClinVar Review: no assertion criteria provided
ClinVar VCV: VCV#3056040 ↗
gnomAD v4 AF: 4.2196%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 50.8 (Raw: 2.870 · Top 0.00%)
Primary Driver: Protein Termination

52%
19%
10%
8%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

### TAT TAT chr16:71,576,247 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 36.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This nuclear gene encodes a mitochondrial protein tyrosine aminotransferase which is present in the liver and catalyzes the conversion of L-tyrosine into p-hydroxyphenylpyruvate. Mutations in this gene cause tyrosinemia (type II, Richner-Hanhart syndrome), a disorder accompanied by major skin and corneal lesions, with possible cognitive disability. A regulator gene for tyrosine aminotransferase is X-linked. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs118203914 p.Arg57Ter | STG, p.Arg57Ter | Heterozygous | Maternal | Pathogenic | Tier1 | 36.0 | – | 38.8 (Protein Termination) | – / – | chr16:71576247 AF: 0.0017% |

**1. INHERITANCE & OMIM PATHOLOGY**

Tyrosinemia type II
Complex / Multifactorial OMIM #276600 ↗

Quadrant: Curated

**2. TRANSCRIPT, PHASING & DEPTH**

Transcript: ENST00000355962.5
cDNA (HGVS): c.169C>T
Protein (HGVS): p.Arg57Ter
VAF / Read Depth: VAF 0.46 · DP –
Phased GT: 0|1 (Maternal)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#402 ↗
gnomAD v4 AF: 0.0017%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 38.8 (Raw: 2.285 · Top 0.01%)
Primary Driver: Protein Termination

66%
24%
8%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CEP63 CEP63 chr3:134,537,268 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 34.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with six coiled-coil domains. The protein is localized to the centrosome, a non-membraneous organelle that functions as the major microtubule-organizing center in animal cells. Several alternatively spliced transcript variants have been found, but their biological validity has not been determined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114108011 p.Gln185His | MIS, p.Gln185His | Heterozygous | Maternal | Benign | Tier1 | 34.0 | 0.93 | 38.6 (Splicing) | 0.28 / 0.54 | chr3:134537268 AF: 2.0823% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000675561.1 |
| cDNA (HGVS): | c.555G>C |
| Protein (HGVS): | p.Gln185His |
| VAF / Read Depth: | VAF 0.74 · DP — |
| Phased GT: | 0\|1 (Maternal) |

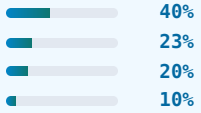
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.93 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#128709 ↗ |
| gnomAD v4 AF: | 2.0823% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.6 (Raw: 2.279 · Top 0.01%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## OGFOD3 OGFOD3 chr17:82,411,454 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 34.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable several functions, including L-ascorbic acid binding activity; dioxygenase activity; and iron ion binding activity. Located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs150938774 c.380+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 34.0 | 0.96 | 33.8 (Splicing) | — / — | chr17:82411454 AF: 0.6286% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000313056.10 |
| cDNA (HGVS): | c.380+1G>A |
| Protein (HGVS): | — |
| VAF / Read Depth: | VAF 0.45 · DP — |
| Phased GT: | 0/1 (Unphased) |

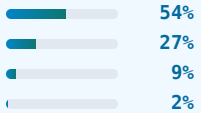
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.42 DL:0.96 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4429500 ↗ |
| gnomAD v4 AF: | 0.6286% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.882 · Top 0.04%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB5 HLA-DRB5 chr6:32,520,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 33.0 Showing 3 of 6 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB5 belongs to the HLA class II beta chain paralogues. This class II molecule is a heterodimer consisting of an alpha (DRA) and a beta (DRB) chain, both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa and its gene contains 6 exons. Exon one encodes the leader peptide, exons 2 and 3 encode the two extracellular domains, exon 4 encodes the transmembrane domain and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. There are multiple pseudogenes of this gene. [provided by RefSeq, Feb 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs201576424<br>c.101-1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.93 | 28.0 (Splicing) | – / – | chr6:32522175<br>AF: 0.0063% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**

Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000374975.4 |
| cDNA (HGVS): | c.101-1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.44 · DP – |
| Phased GT: | 0/1 (Unphased) |

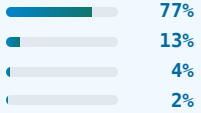
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.65 AL:0.93 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4326162 ↗ |
| gnomAD v4 AF: | 0.0063% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.0 (Raw: 1.444 · Top 0.16%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs140580884<br>p.Asp70Asn | MIS, p.Asp70Asn | Heterozygous | Unphased | Not reviewed | Tier3 | 23.5 | – | 20.3 (AlphaMissense) | 0.12 / 0.57 | chr6:32522067<br>AF: 3.5002% |
| rs796616593<br>c.-152T>C | 2KU | Heterozygous | Unphased | Not reviewed | Tier3 | 16.2 | – | 24.1 (PRO-cap) | – / – | chr6:32530376<br>AF: 0.0006% |

## IL17RC IL17RC chr3:9,932,859 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 33.0 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a single-pass type I membrane protein that shares similarity with the interleukin-17 receptor (IL-17RA). Unlike IL-17RA, which is predominantly expressed in hemopoietic cells, and binds with high affinity to only IL-17A, this protein is expressed in nonhemopoietic tissues, and binds both IL-17A and IL-17F with similar affinities. The proinflammatory cytokines, IL-17A and IL-17F, have been implicated in the progression of inflammatory and autoimmune diseases. Multiple alternatively spliced transcript variants encoding different isoforms have been detected for this gene, and it has been proposed that soluble, secreted proteins lacking transmembrane and intracellular domains may function as extracellular antagonists to cytokine signaling. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs148575246<br>c.1522+1G>C | SPL | Heterozygous | Unphased | Benign/Likely benign | Tier1 | 33.0 | 0.97 | 25.9 (Splicing) | – / – | chr3:9932859<br>AF: 1.0744% |

### 1. INHERITANCE & OMIM PATHOLOGY

**IL17RC-related disorder**
Complex / Multifactorial OMIM #616445 ↗

**Candidiasis, familial, 9**
Complex / Multifactorial OMIM #616445 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

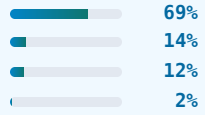
Transcript: ENST00000403601.8
cDNA (HGVS): c.1522+1G>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.01 DL:0.97
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#542537 ↗
gnomAD v4 AF: 1.0744%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs115461448<br>p.Ala324Thr | MIS, p.Ala324Thr | Heterozygous | Unphased | Benign | Tier3 | 23.2 | – | 22.0 (Cactus) | 0.09 / 0.17 | chr3:9928397<br>AF: 1.4147% |

## NIPSNAP2 NIPSNAP2 chr7:55,994,989 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 33.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the NipSnap family of proteins that may be involved in vesicular transport. The encoded protein is localized to mitochondria and plays a role in oxidative phosphorylation. A pseudogene of this gene is located on the long arm of chromosome 2. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs748801121<br>c.712+1G>T | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.97 | 33.8 (Splicing) | – / – | chr7:55994989<br>AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

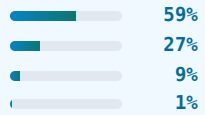
Transcript: ENST00000322090.8
cDNA (HGVS): c.712+1G>T
Protein (HGVS): –
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.44 DL:0.97
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0075%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.881 · Top 0.04%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLIT2 SLIT2 chr4:20,567,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the slit family of secreted glycoproteins, which are ligands for the Robo family of immunoglobulin receptors. Slit proteins play highly conserved roles in axon guidance and neuronal migration and may also have functions during other cell migration processes including leukocyte migration. Members of the slit family are characterized by an N-terminal signal peptide, four leucine-rich repeats, nine epidermal growth factor repeats, and a C-terminal cysteine knot. Proteolytic processing of this protein gives rise to an N-terminal fragment that contains the four leucine-rich repeats and five epidermal growth factor repeats and a C-terminal fragment that contains four epidermal growth factor repeats and the cysteine knot. Both full length and cleaved proteins are secreted extracellularly and can function in axon repulsion as well as other specific processes. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2015]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1020555095 p.Gly909Asp | MIS, p.Gly909Asp | Heterozygous | Unphased | Not reviewed | Tier1 | 32.0 | – | 23.4 (Cactus) | 0.74 / 0.35 | chr4:20567262 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

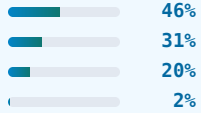
Transcript: ENST00000504154.6
cDNA (HGVS): c.2726G>A
Protein (HGVS): p.Gly909Asp
VAF / Read Depth: VAF 0.54 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0009%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.4 (Raw: 0.859 · Top 0.46%)
Primary Driver: Cactus




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TMEM43 TMEM43 chr3:14,139,244 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to the TMEM43 family. Defects in this gene are the cause of familial arrhythmogenic right ventricular dysplasia type 5 (ARVD5), also known as arrhythmogenic right ventricular cardiomyopathy type 5 (ARVC5). Arrhythmogenic right ventricular dysplasia is an inherited disorder, often involving both ventricles, and is characterized by ventricular tachycardia, heart failure, sudden cardiac death, and fibrofatty replacement of cardiomyocytes. This gene contains a response element for PPAR gamma (an adipogenic transcription factor), which may explain the fibrofatty replacement of the myocardium, a characteristic pathological finding in ARVC. [provided by RefSeq, Oct 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs199526104 p.Trp316Ser | MIS, p.Trp316Ser | Heterozygous | Paternal | Conflicting classifications of pathogenicity | Tier1 | 32.0 | – | 26.5 (AlphaMissense) | 0.53 / 0.76 | chr3:14139244 AF: 0.0126% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**
Complex / Multifactorial OMIM #604400 ↗

**TMEM43-related disorder**
Complex / Multifactorial OMIM #604400 ↗

**Auditory neuropathy, autosomal dominant 3**
Autosomal Dominant OMIM #604400 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000306077.5
cDNA (HGVS): c.947G>C
Protein (HGVS): p.Trp316Ser
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#46155 ↗
gnomAD v4 AF: 0.0126%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.5 (Raw: 1.335 · Top 0.22%)
Primary Driver: AlphaMissense




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## AMPD1 AMPD1 chr1:114,677,435 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Adenosine monophosphate deaminase 1 catalyzes the deamination of AMP to IMP in skeletal muscle and plays an important role in the purine nucleotide cycle. Two other genes have been identified, AMPD2 and AMPD3, for the liver- and erythocyte-specific isoforms, respectively. Deficiency of the muscle-specific enzyme is apparently a common cause of exercise-induced myopathy and probably the most common cause of metabolic myopathy in the human. Alternatively spliced transcript variants encoding different isoforms have been identified in this gene.[provided by RefSeq, Feb 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs764950035 p.Glu435Gly | MIS, p.Glu435Gly | Heterozygous | Unphased | Uncertain significance | Tier1 | 32.0 | — | 28.2 (AlphaMissense) | 0.95 / 0.92 | chr1:114677435 AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Muscle AMP deaminase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000520113.7 |
| cDNA (HGVS): | c.1304A>G |
| Protein (HGVS): | p.Glu435Gly |
| VAF / Read Depth: | VAF 0.49 · DP — |
| Phased GT: | 0/1 (Unphased) |

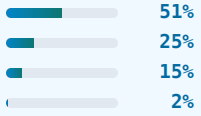
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#2173226 ↗ |
| gnomAD v4 AF: | 0.0075% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.2 (Raw: 1.454 · Top 0.15%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## LDHAL6A LDHAL6A chr11:18,475,640 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 31.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables L-lactate dehydrogenase (NAD+) activity. Predicted to be involved in lactate metabolic process and pyruvate catabolic process. Located in cytoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs145833201 c.592+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 31.0 | 0.69 | 23.3 (Splicing) | — / — | chr11:18475640 AF: 0.1631% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000280706.3 |
| cDNA (HGVS): | c.592+1G>A |
| Protein (HGVS): | — |
| VAF / Read Depth: | VAF 0.36 · DP — |
| Phased GT: | 0/1 (Unphased) |

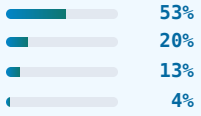
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.69 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4370113 ↗ |
| gnomAD v4 AF: | 0.1631% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.3 (Raw: 0.845 · Top 0.47%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## DMAP1 DMAP1 chr1:44,218,751 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 29.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a subunit of several, distinct complexes involved in the repression or activation of transcription. The encoded protein can independently repress transcription and is targeted to replication foci throughout S phase by interacting directly with the N-terminus of DNA methyltransferase 1. During late S phase, histone deacetylase 2 is added to this complex, providing a means to deacetylate histones in transcriptionally inactive heterochromatin following replication. The encoded protein is also a component of the nucleosome acetyltransferase of H4 complex and interacts with the transcriptional corepressor tumor susceptibility gene 101 and the pro-apoptotic death-associated protein 6, among others. Alternatively spliced transcript variants encoding the same protein have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| **chr1:44218751** p.Glu239Gly | MIS, p.Glu239Gly | Heterozygous | Unphased | Not reviewed | Tier1 | 29.7 | – | 23.4 (Cactus) | 0.38 / 0.38 | chr1:44218751 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000372289.7 |
| cDNA (HGVS): | c.716A>G |
| Protein (HGVS): | p.Glu239Gly |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0/1 (Unphased) |

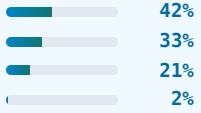
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0001% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.4 (Raw: 0.854 · Top 0.46%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PLK4 PLK4 chr4:127,881,151 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 28.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polo family of serine/threonine protein kinases. The protein localizes to centrioles, complex microtubule-based structures found in centrosomes, and regulates centriole duplication during the cell cycle. Three alternatively spliced transcript variants that encode different protein isoforms have been found for this gene. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs149003893 p.Gly6Glu | MIS, p.Gly6Glu | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 28.5 | – | 29.8 (AlphaMissense) | 0.36 / 0.99 | chr4:127881151 AF: 0.1341% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Inborn genetic diseases**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000270861.10 |
| cDNA (HGVS): | c.17G>A |
| Protein (HGVS): | p.Gly6Glu |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

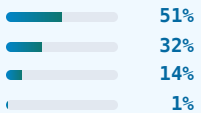
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#436338 ↗ |
| gnomAD v4 AF: | 0.1341% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.8 (Raw: 1.555 · Top 0.11%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ACSM2B ACSM2B chr16:20,540,665 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 27.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables benzoate-CoA ligase activity. Predicted to be involved in acyl-CoA metabolic process and fatty acid biosynthetic process. Located in mitochondrion. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs139253186 p.Tyr540Asn | MIS, p.Tyr540Asn | Heterozygous | Unphased | Not reviewed | Tier1 | 27.6 | – | 24.4 (AlphaMissense) | 0.40 / 0.58 | chr16:20540665 AF: 0.4705% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000329697.10 |
| cDNA (HGVS): | c.1618T>A |
| Protein (HGVS): | p.Tyr540Asn |
| VAF / Read Depth: | VAF 0.54 · DP – |
| Phased GT: | 0/1 (Unphased) |

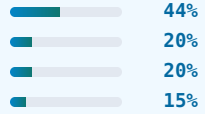
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.4705% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.4 (Raw: 1.012 · Top 0.37%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAHD1 FAHD1 chr16:1,827,542 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 27.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables hydrolase activity, acting on carbon-carbon bonds, in ketonic substances; oxaloacetate decarboxylase activity; and oxaloacetate tautomerase activity. Involved in oxaloacetate metabolic process. Located in cytosol; mitochondrion; and nucleoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs200357596 p.Ala102Ser | MIS, p.Ala102Ser | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 22.1 (Cactus) | 0.71 / 0.24 | chr16:1827542 AF: 0.0037% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000427358.5 |
| cDNA (HGVS): | c.304G>T |
| Protein (HGVS): | p.Ala102Ser |
| VAF / Read Depth: | VAF 0.49 · DP – |
| Phased GT: | 0/1 (Unphased) |

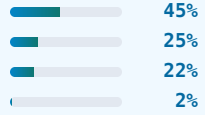
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0037% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.1 (Raw: 0.710 · Top 0.62%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLC27A3 SLC27A3 chr1:153,776,015 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 27.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to a family of integral membrane proteins and encodes a protein that is involved in lipid metabolism. The increased expression of this gene in human neural stem cells derived from induced pluripotent stem cells suggests that it plays an important role in early brain development. Naturally occurring mutations in this gene are associated with autism spectrum disorders. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Feb 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs756978807 p.Leu173Pro | MIS, p.Leu173Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 26.9 (AlphaMissense) | 0.41 / 0.94 | chr1:153776015 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

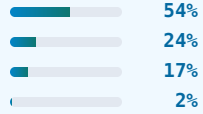
Transcript: ENST00000624995.4
cDNA (HGVS): c.518T>C
Protein (HGVS): p.Leu173Pro
VAF / Read Depth: VAF 0.23 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0001%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.9 (Raw: 1.371 · Top 0.21%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## POLG POLG chr15:89,323,460 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 26.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Mitochondrial DNA polymerase is heterotrimeric, consisting of a homodimer of accessory subunits plus a catalytic subunit. The protein encoded by this gene is the catalytic subunit of mitochondrial DNA polymerase. The encoded protein contains a polyglutamine tract near its N-terminus that may be polymorphic. Defects in this gene are a cause of progressive external ophthalmoplegia with mitochondrial DNA deletions 1 (PEOA1), sensory ataxic neuropathy dysarthria and ophthalmoparesis (SANDO), Alpers-Huttenlocher syndrome (AHS), and mitochondrial neurogastrointestinal encephalopathy syndrome (MNGIE). Two transcript variants encoding the same protein have been found for this gene. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs121918054 p.Gly737Arg | MIS, p.Gly737Arg | Heterozygous | Paternal | Pathogenic/Likely pathogenic | Tier1 | 26.3 | – | 30.3 (AlphaMissense) | 0.94 / 0.94 | chr15:89323460 AF: 0.1619% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Spinocerebellar ataxia with epilepsy**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Sensory ataxic neuropathy, dysarthria, and ophthalmoparesis**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 1**
Autosomal Dominant OMIM #203700; 607459; 613662; 157640; 258450 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

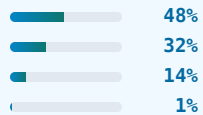
Transcript: ENST00000268124.11
cDNA (HGVS): c.2209G>C
Protein (HGVS): p.Gly737Arg
VAF / Read Depth: VAF 0.61 · DP –
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#13513 ↗
gnomAD v4 AF: 0.1619%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.3 (Raw: 1.582 · Top 0.09%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## MID1IP1 MID1IP1 chrX:38,805,491 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 26.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable enzyme activator activity. Predicted to be involved in several processes, including negative regulation of microtubule depolymerization; positive regulation of fatty acid biosynthetic process; and positive regulation of ligase activity. Predicted to be located in microtubule cytoskeleton. Predicted to be active in cytosol and nucleus. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs143008624 p.Gly182Asp | MIS, p.Gly182Asp | Heterozygous | Unphased | Benign | Tier1 | 26.1 | – | 27.1 (AlphaMissense) | 0.63 / 0.87 | chrX:38805491 AF: 0.3491% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000614558.3
cDNA (HGVS): c.545G>A
Protein (HGVS): p.Gly182Asp
VAF / Read Depth: VAF 0.48 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#711083 ↗
gnomAD v4 AF: 0.3491%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.1 (Raw: 1.393 · Top 0.19%)
Primary Driver: AlphaMissense

50%
31%
16%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## WDHD1 WDHD1 chr14:55,002,157 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 25.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene contains multiple N-terminal WD40 domains and a C-terminal high mobility group (HMG) box. WD40 domains are found in a variety of eukaryotic proteins and may function as adaptor/regulatory modules in signal transduction, pre-mRNA processing and cytoskeleton assembly. HMG boxes are found in many eukaryotic proteins involved in chromatin assembly, transcription and replication. Alternative splicing results in two transcript variants encoding different isoforms. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs374497720 p.Val210Ala | MIS, p.Val210Ala | Heterozygous | Unphased | Not reviewed | Tier1 | 25.8 | – | 24.5 (AlphaMissense) | 0.27 / 0.58 | chr14:55002157 AF: 0.0190% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000360586.8
cDNA (HGVS): c.629T>C
Protein (HGVS): p.Val210Ala
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0190%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.5 (Raw: 1.042 · Top 0.35%)
Primary Driver: AlphaMissense

42%
35%
19%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAAP24 FAAP24 chr19:32,974,193 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 25.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 FAAP24 is a component of the Fanconi anemia (FA) core complex (see MIM 227650), which plays a crucial role in DNA damage response (Ciccia et al., 2007 [PubMed 17289582]).[supplied by OMIM, Mar 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs36017455 p.Ser126Phe | MIS, p.Ser126Phe | Heterozygous | Unphased | Benign | Tier1 | 25.7 | — | 27.0 (AlphaMissense) | — / 0.69 | chr19:32974193 AF: 0.5937% |

### 1. INHERITANCE & OMIM PATHOLOGY

**FAAP24-related disorder**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000588258.6
cDNA (HGVS): c.377C>T
Protein (HGVS): p.Ser126Phe
VAF / Read Depth: VAF 0.44 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#784831 ↗
gnomAD v4 AF: 0.5937%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.0 (Raw: 1.385 · Top 0.20%)
Primary Driver: AlphaMissense

40%
37%
14%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CTH CTH chr1:70,415,987 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 25.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a cytoplasmic enzyme in the trans-sulfuration pathway that converts cystathione derived from methionine into cysteine. Glutathione synthesis in the liver is dependent upon the availability of cysteine. Mutations in this gene cause cystathioninuria. Alternative splicing of this gene results in three transcript variants encoding different isoforms. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28941785 p.Thr67Ile | MIS, p.Thr67Ile | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 25.5 | — | 27.8 (AlphaMissense) | 0.93 / 0.81 | chr1:70415987 AF: 0.8026% |

### 1. INHERITANCE & OMIM PATHOLOGY

**CTH-related disorder**
Complex / Multifactorial OMIM #219500 ↗

**Cystathioninuria**
Complex / Multifactorial OMIM #219500 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000370938.8
cDNA (HGVS): c.200C>T
Protein (HGVS): p.Thr67Ile
VAF / Read Depth: VAF 0.46 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#2939 ↗
gnomAD v4 AF: 0.8026%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense

45%
36%
14%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TEK TEK chr9:27,185,615 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 24.3 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a receptor that belongs to the protein tyrosine kinase Tie2 family. The encoded protein possesses a unique extracellular region that contains two immunoglobulin-like domains, three epidermal growth factor (EGF)-like domains and three fibronectin type III repeats. The ligand angiopoietin-1 binds to this receptor and mediates a signaling pathway that functions in embryonic vascular development. Mutations in this gene are associated with inherited venous malformations of the skin and mucous membranes. Alternative splicing results in multiple transcript variants. Additional alternatively spliced transcript variants of this gene have been described, but their full-length nature is not known. [provided by RefSeq, Feb 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs189543659 p.Asn438Ile | MIS, p.Asn438Ile | Heterozygous | Unphased | Not reviewed | Tier1 | 24.3 | – | 21.9 (Cactus) | 0.21 / 0.21 | chr9:27185615 AF: 0.0011% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

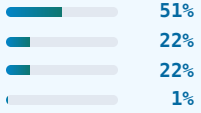
Transcript: ENST00000380036.10
cDNA (HGVS): c.1313A>T
Protein (HGVS): p.Asn438Ile
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0011%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 21.9 (Raw: 0.695 · Top 0.65%)
Primary Driver: Cactus




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs35030851 p.Val600Leu | MIS, p.Val600Leu | Heterozygous | Unphased | Benign | Tier3 | 17.9 | – | 22.9 (Cactus) | 0.06 / 0.16 | chr9:27197488 AF: 4.5993% |

## CHST14 CHST14 chr15:40,472,259 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 24.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the HNK-1 family of sulfotransferases. The encoded protein transfers sulfate to the C-4 hydroxyl of N-acetylgalactosamine residues in dermatan sulfate. Mutations in this gene have been associated with adducted thumb-clubfoot syndrome.[provided by RefSeq, Mar 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1463514561 p.Gln349Pro | MIS, p.Gln349Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 24.3 | – | 22.9 (Cactus) | 0.54 / 0.20 | chr15:40472259 AF: 0.0004% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Cardiovascular phenotype**
Complex / Multifactorial

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

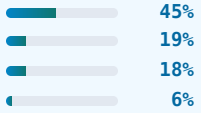
Transcript: ENST00000306243.7
cDNA (HGVS): c.1046A>C
Protein (HGVS): p.Gln349Pro
VAF / Read Depth: VAF 0.52 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#3229028 ↗
gnomAD v4 AF: 0.0004%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 22.9 (Raw: 0.798 · Top 0.51%)
Primary Driver: Cactus




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SUMF1 SUMF1 chr3:4,362,213 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 23.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an enzyme that catalyzes the hydrolysis of sulfate esters by oxidizing a cysteine residue in the substrate sulfatase to an active site 3-oxoalanine residue, which is also known as C-alpha-formylglycine. Mutations in this gene cause multiple sulfatase deficiency, a lysosomal storage disorder. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2009]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs376126385 p.Asn352Lys | MIS, p.Asn352Lys | Heterozygous | Unphased | Uncertain significance | Tier1 | 23.9 | — | 25.9 (AlphaMissense) | 0.64 / 1.00 | chr3:4362213 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Multiple sulfatase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000272902.10**
cDNA (HGVS): **c.1056C>G**
Protein (HGVS): **p.Asn352Lys**
VAF / Read Depth: **VAF 0.66 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#2161926 ↗
gnomAD v4 AF: **0.0009%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.9** (Raw: 1.256 · Top 0.26%)
Primary Driver: **AlphaMissense**

63%
19%
10%
3%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NPC2 NPC2 chr14:74,480,701 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 23.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein containing a lipid recognition domain. The encoded protein may function in regulating the transport of cholesterol through the late endosomal/lysosomal system. Mutations in this gene have been associated with Niemann-Pick disease, type C2 and frontal lobe atrophy. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs140130028 c.441+1G>A | SPL | Heterozygous | Maternal | Conflicting classifications of pathogenicity | Tier1 | 23.2 | 0.98 | 26.3 (Splicing) | 0.21 / — | chr14:74480701 AF: 0.6316% |

### 1. INHERITANCE & OMIM PATHOLOGY

**NPC2-related disorder**

Complex / Multifactorial OMIM #257220; 607625 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000555619.6**
cDNA (HGVS): **c.441+1G>A**
Protein (HGVS): **—**
VAF / Read Depth: **VAF 0.38 · DP —**
Phased GT: **0|1 (Maternal)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.10 DL:0.98**
ClinVar Review: **criteria provided, conflicting classifications**
ClinVar VCV: VCV#100734 ↗
gnomAD v4 AF: **0.6316%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **26.3** (Raw: 1.312 · Top 0.23%)
Primary Driver: **Splicing**

51%
23%
6%
5%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PGK1 PGK1 chrX:78,114,140 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 22.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a glycolytic enzyme that catalyzes the conversion of 1,3-diphosphoglycerate to 3-phosphoglycerate. The encoded protein may also act as a cofactor for polymerase alpha. Additionally, this protein is secreted by tumor cells where it participates in angiogenesis by functioning to reduce disulfide bonds in the serine protease, plasmin, which consequently leads to the release of the tumor blood vessel inhibitor angiostatin. The encoded protein has been identified as a moonlighting protein based on its ability to perform mechanistically distinct functions. Deficiency of the enzyme is associated with a wide range of clinical phenotypes hemolytic anemia and neurological impairment. Pseudogenes of this gene have been defined on chromosomes 19, 21 and the X chromosome. [provided by RefSeq, Jan 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| **chrX:78114140** p.Lys133Glu | MIS, p.Lys133Glu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.7 | – | 24.3 (AlphaMissense) | 0.54 / 0.57 | chrX:78114140 AF: 0.0008% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

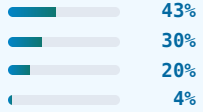
Transcript: **ENST00000373316.5**
cDNA (HGVS): **c.397A>G**
Protein (HGVS): **p.Lys133Glu**
VAF / Read Depth: **VAF 0.58 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0008%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.3** (Raw: 0.993 · Top 0.38%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-B HLA-B chr6:31,356,717 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 22.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-B belongs to the HLA class I heavy chain paralogues. This class I molecule is a heterodimer consisting of a heavy chain and a light chain (beta-2 microglobulin). The heavy chain is anchored in the membrane. Class I molecules play a central role in the immune system by presenting peptides derived from the endoplasmic reticulum lumen. They are expressed in nearly all cells. The heavy chain is approximately 45 kDa and its gene contains 8 exons. Exon 1 encodes the leader peptide, exon 2 and 3 encode the alpha1 and alpha2 domains, which both bind the peptide, exon 4 encodes the alpha3 domain, exon 5 encodes the transmembrane region and exons 6 and 7 encode the cytoplasmic tail. Polymorphisms within exon 2 and exon 3 are responsible for the peptide binding specificity of each class one molecule. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. Hundreds of HLA-B alleles have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9266161 p.Leu105Pro | MIS, p.Leu105Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 22.5 | – | 20.0 (AlphaMissense) | 0.20 / 0.79 | chr6:31356717 AF: 0.0006% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

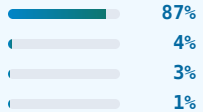
Transcript: **ENST00000412585.7**
cDNA (HGVS): **c.314T>C**
Protein (HGVS): **p.Leu105Pro**
VAF / Read Depth: **VAF 0.55 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0006%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **20.0** (Raw: 0.559 · Top 1.00%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## VPS35L VPS35L chr16:19,629,820 Multisystem 🔥 Tier-1 / Pathogenic Gene

Max CADD: 22.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Involved in Golgi to plasma membrane transport and endocytic recycling. Located in endosome. Part of protein-containing complex. Implicated in Ritscher-Schinzel syndrome. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs144908771 p.Thr518= | SYN, p.Thr518= | Heterozygous | Unphased | Likely benign | Tier1 | 22.5 | 0.64 | 21.4 (Cactus) | – / – | chr16:19629820 AF: 0.3238% |

### 1. INHERITANCE & OMIM PATHOLOGY

**VPS35L-related disorder**
Complex / Multifactorial

**Ritscher-Schinzel syndrome**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

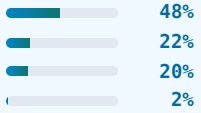
Transcript: ENST00000417362.7
cDNA (HGVS): c.1554G>A
Protein (HGVS): p.Thr518=
VAF / Read Depth: VAF 0.45 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.64 DL:0.01
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#2672622 ↗
gnomAD v4 AF: 0.3238%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.4 (Raw: 0.657 · Top 0.73%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## GUSB GUSB chr7:65,960,973 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Max CADD: 22.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a hydrolase that degrades glycosaminoglycans, including heparan sulfate, dermatan sulfate, and chondroitin-4,6-sulfate. The enzyme forms a homotetramer that is localized to the lysosome. Mutations in this gene result in mucopolysaccharidosis type VII. Alternative splicing results in multiple transcript variants. There are many pseudogenes of this locus in the human genome.[provided by RefSeq, May 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1236992554 p.Trp627Leu | MIS, p.Trp627Leu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.2 | – | 24.9 (Cactus) | 0.56 / 0.25 | chr7:65960973 AF: 0.0003% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

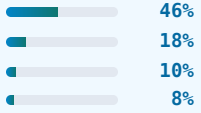
Transcript: ENST00000304895.9
cDNA (HGVS): c.1880G>T
Protein (HGVS): p.Trp627Leu
VAF / Read Depth: VAF 0.53 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0003%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.9 (Raw: 1.103 · Top 0.32%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## 5. SpliceAI Deep Learning Splicing Disruption — Top Genes (Δ ≥ 0.10)

Genes exhibiting cryptic splice-site activation or canonical donor/acceptor loss (Δ score ≥ 0.10).

**Selected Scope:** 18 Genes (18 Tier-1/Pathogenic)
Filtered from 95 matching genes in view

## HLA-DRB5 HLA-DRB5 chr6:32,520,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 1.00 Showing 4 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB5 belongs to the HLA class II beta chain paralogues. This class II molecule is a heterodimer consisting of an alpha (DRA) and a beta (DRB) chain, both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa and its gene contains 6 exons. Exon one encodes the leader peptide, exons 2 and 3 encode the two extracellular domains, exon 4 encodes the transmembrane domain and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. There are multiple pseudogenes of this gene. [provided by RefSeq, Feb 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs796105724<br>c.371-392G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 8.4 | 1.00 | 25.7 (Splicing) | – / – | chr6:32520043<br>AF: 1.4415% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

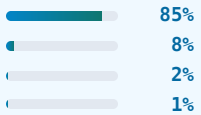
Transcript: **ENST00000374975.4**
cDNA (HGVS): **c.371-392G>A**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.62 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:1.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **1.4415%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.7** (Raw: 1.223 · Top 0.27%)
Primary Driver: **Splicing**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs796784269<br>c.371-393T>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 2.8 | 0.94 | 16.6 (Splicing) | – / – | chr6:32520044<br>AF: 0.1426% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

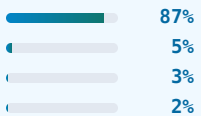
Transcript: **ENST00000374975.4**
cDNA (HGVS): **c.371-393T>G**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.38 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.94 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.1426%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **16.6** (Raw: 0.386 · Top 2.21%)
Primary Driver: **Splicing**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs201576424<br>c.101-1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.93 | 28.0 (Splicing) | – / – | chr6:32522175<br>AF: 0.0063% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

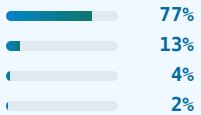
Transcript: **ENST00000374975.4**
cDNA (HGVS): **c.101-1G>A**
Protein (HGVS): **–**
VAF / Read Depth: **VAF 0.44 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.65 AL:0.93 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
ClinVar VCV: **VCV#4326162 ↗**
gnomAD v4 AF: **0.0063%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **28.0** (Raw: 1.444 · Top 0.16%)
Primary Driver: **Splicing**



| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | 🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗ | | |
| rs774541765<br>c.100+17G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.7 | 0.68 | 18.9 (Splicing) | – / – | chr6:32530108<br>AF: 0.0027% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

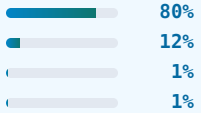
| | |
|---|---|
| Transcript: | ENST00000374975.4 |
| cDNA (HGVS): | c.100+17G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.46 · DP – |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.68 DL:0.01 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0027% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 18.9 (Raw: 0.503 · Top 1.28%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NPC2 NPC2 chr14:74,480,701 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.98 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein containing a lipid recognition domain. The encoded protein may function in regulating the transport of cholesterol through the late endosomal/lysosomal system. Mutations in this gene have been associated with Niemann-Pick disease, type C2 and frontal lobe atrophy. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs140130028<br>c.441+1G>A | SPL | Heterozygous | Maternal | Conflicting classifications of pathogenicity | Tier1 | 23.2 | 0.98 | 26.3 (Splicing) | 0.21 / – | chr14:74480701<br>AF: 0.6316% |

**1. INHERITANCE & OMIM PATHOLOGY**

**NPC2-related disorder**
Complex / Multifactorial OMIM #257220; 607625 ↗

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

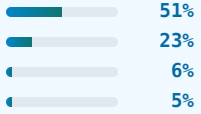
| | |
|---|---|
| Transcript: | ENST00000555619.6 |
| cDNA (HGVS): | c.441+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0\|1 (Maternal) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.10 DL:0.98 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#100734 ↗ |
| gnomAD v4 AF: | 0.6316% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## IL17RC IL17RC chr3:9,932,859 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.97 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a single-pass type I membrane protein that shares similarity with the interleukin-17 receptor (IL-17RA). Unlike IL-17RA, which is predominantly expressed in hemopoietic cells, and binds with high affinity to only IL-17A, this protein is expressed in nonhemopoietic tissues, and binds both IL-17A and IL-17F with similar affinities. The proinflammatory cytokines, IL-17A and IL-17F, have been implicated in the progression of inflammatory and autoimmune diseases. Multiple alternatively spliced transcript variants encoding different isoforms have been detected for this gene, and it has been proposed that soluble, secreted proteins lacking transmembrane and intracellular domains may function as extracellular antagonists to cytokine signaling. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs148575246<br>c.1522+1G>C | SPL | Heterozygous | Unphased | Benign/Likely benign | Tier1 | 33.0 | 0.97 | 25.9 (Splicing) | – / – | chr3:9932859<br>AF: 1.0744% |

### 1. INHERITANCE & OMIM PATHOLOGY

**IL17RC-related disorder**
Complex / Multifactorial OMIM #616445 ↗

**Candidiasis, familial, 9**
Complex / Multifactorial OMIM #616445 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

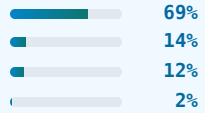
Transcript: ENST00000403601.8
cDNA (HGVS): c.1522+1G>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.01 DL:0.97
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#542537 ↗
gnomAD v4 AF: 1.0744%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLC5A4 SLC5A4 chr22:32,233,039 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.97 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable low-affinity D-glucose:sodium symporter activity and proton transmembrane transporter activity. Predicted to be involved in D-glucose import across plasma membrane and renal D-glucose absorption. Predicted to act upstream of or within proton transmembrane transport. Predicted to be located in membrane. Predicted to be active in plasma membrane. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9621415<br>c.886-5G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 14.0 | 0.97 | 21.1 (Splicing) | – / – | chr22:32233039<br>AF: 1.4374% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

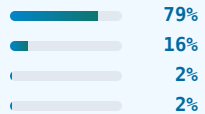
Transcript: ENST00000266086.6
cDNA (HGVS): c.886-5G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.41 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.97 AL:0.01 DG:0.00 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4457848 ↗
gnomAD v4 AF: 1.4374%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.1 (Raw: 0.639 · Top 0.78%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NIPSNAP2 NIPSNAP2 chr7:55,994,989 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.97 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the NipSnap family of proteins that may be involved in vesicular transport. The encoded protein is localized to mitochondria and plays a role in oxidative phosphorylation. A pseudogene of this gene is located on the long arm of chromosome 2. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs748801121 c.712+1G>T | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.97 | 33.8 (Splicing) | — / — | chr7:55994989 AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000322090.8 |
| cDNA (HGVS): | c.712+1G>T |
| Protein (HGVS): | — |
| VAF / Read Depth: | VAF 0.50 · DP — |
| Phased GT: | 0/1 (Unphased) |

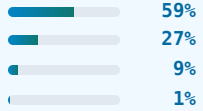
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.44 DL:0.97 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0075% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.881 · Top 0.04%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## OGFOD3 OGFOD3 chr17:82,411,454 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.96 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable several functions, including L-ascorbic acid binding activity; dioxygenase activity; and iron ion binding activity. Located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs150938774 c.380+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 34.0 | 0.96 | 33.8 (Splicing) | — / — | chr17:82411454 AF: 0.6286% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000313056.10 |
| cDNA (HGVS): | c.380+1G>A |
| Protein (HGVS): | — |
| VAF / Read Depth: | VAF 0.45 · DP — |
| Phased GT: | 0/1 (Unphased) |

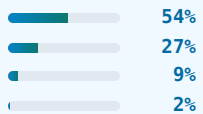
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.42 DL:0.96 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4429500 ↗ |
| gnomAD v4 AF: | 0.6286% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.882 · Top 0.04%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CEP63 CEP63 chr3:134,537,268 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.93 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with six coiled-coil domains. The protein is localized to the centrosome, a non-membraneous organelle that functions as the major microtubule-organizing center in animal cells. Several alternatively spliced transcript variants have been found, but their biological validity has not been determined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114108011 p.Gln185His | MIS, p.Gln185His | Heterozygous | Maternal | Benign | Tier1 | 34.0 | 0.93 | 38.6 (Splicing) | 0.28 / 0.54 | chr3:134537268 AF: 2.0823% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

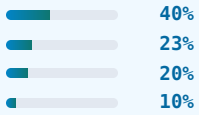
Transcript: ENST00000675561.1
cDNA (HGVS): c.555G>C
Protein (HGVS): p.Gln185His
VAF / Read Depth: VAF 0.74 · DP –
Phased GT: 0|1 (Maternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.93
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#128709 ↗
gnomAD v4 AF: 2.0823%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.6 (Raw: 2.279 · Top 0.01%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## VWA3B VWA3B chr2:98,162,976 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.85 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an intracellular protein that contains a von Willebrand factor type A domain. Intracellular proteins with VWA domains are thought to function in transcription, DNA repair, ribosomal and membrane transport and the proteasome. Mutations in this gene are associated with Spinocerebellar ataxia, autosomal recessive 22. Alternatively spliced transcript variants have been found for this gene. [provided by RefSeq, May 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs62154921 p.Glu372Ter | STG, p.Glu372Ter | Heterozygous | Unphased | Benign | Tier1 | 46.0 | 0.85 | 50.8 (Protein Termination) | – / – | chr2:98162976 AF: 4.2196% |

### 1. INHERITANCE & OMIM PATHOLOGY

**VWA3B-related disorder**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

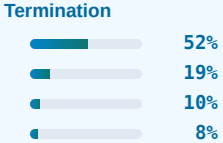
Transcript: ENST00000477737.6
cDNA (HGVS): c.1114G>T
Protein (HGVS): p.Glu372Ter
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.05 DL:0.85
ClinVar Review: no assertion criteria provided
ClinVar VCV: VCV#3056040 ↗
gnomAD v4 AF: 4.2196%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 50.8 (Raw: 2.870 · Top 0.00%)
Primary Driver: Protein Termination




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## LDHAL6A LDHAL6A chr11:18,475,640 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.69 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables L-lactate dehydrogenase (NAD+) activity. Predicted to be involved in lactate metabolic process and pyruvate catabolic process. Located in cytoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs145833201<br>c.592+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 31.0 | 0.69 | 23.3 (Splicing) | – / – | chr11:18475640<br>AF: 0.1631% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000280706.3 |
| cDNA (HGVS): | c.592+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.36 · DP – |
| Phased GT: | 0/1 (Unphased) |

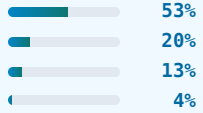
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.69 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4370113 ↗ |
| gnomAD v4 AF: | 0.1631% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.3 (Raw: 0.845 · Top 0.47%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ADAMTS7 ADAMTS7 chr15:78,790,059 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.65 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ADAMTS (a disintegrin and metalloproteinase with thrombospondin motifs) family. Members of this family share several distinct protein modules, including a propeptide region, a metalloproteinase domain, a disintegrin-like domain, and a thrombospondin type 1 (TS) motif. Individual members of this family differ in the number of C-terminal TS motifs, and some have unique C-terminal domains. The encoded preproprotein is proteolytically processed to generate the mature enzyme. This enzyme contains two C-terminal TS motifs and may regulate vascular smooth muscle cell (VSMC) migration. Mutations in this gene may be associated with susceptibility to coronary artery disease. [provided by RefSeq, Feb 2016]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs142754252<br>c.1029-221G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 0.2 | 0.65 | 13.8 (Splicing) | – / – | chr15:78790059<br>AF: 0.1293% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000388820.5 |
| cDNA (HGVS): | c.1029-221G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.42 · DP – |
| Phased GT: | 0/1 (Unphased) |

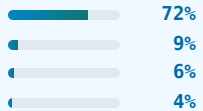
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.65 AL:0.00 DG:0.02 DL:0.00 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4406248 ↗ |
| gnomAD v4 AF: | 0.1293% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.8 (Raw: 0.277 · Top 4.16%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ACOX3 ACOX3 chr4:8,440,048 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.65 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Acyl-Coenzyme A oxidase 3 also know as pristanoyl -CoA oxidase (ACOX3)is involved in the desaturation of 2-methyl branched fatty acids in peroxisomes. Unlike the rat homolog, the human gene is expressed in very low amounts in liver such that its mRNA was undetectable by routine Northern-blot analysis or its product by immunoblotting or by enzyme activity measurements. However the human cDNA encoding a 700 amino acid protein with a peroxisomal targeting C-terminal tripeptide S-K-L was isolated and is thought to be expressed under special conditions such as specific developmental stages or in a tissue specific manner in tissues that have not yet been examined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs62287361<br>c.-15+600A>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 3.3 | 0.65 | 19.4 (Splicing) | – / – | chr4:8440048<br>AF: 2.7459% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000356406.10 |
| cDNA (HGVS): | c.-15+600A>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.51 · DP – |
| Phased GT: | 0/1 (Unphased) |

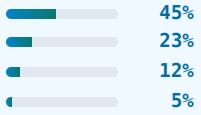
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.65 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 2.7459% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.4 (Raw: 0.526 · Top 1.15%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## VPS35L VPS35L chr16:19,629,820 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.64 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Involved in Golgi to plasma membrane transport and endocytic recycling. Located in endosome. Part of protein-containing complex. Implicated in Ritscher-Schinzel syndrome. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs144908771<br>p.Thr518= | SYN, p.Thr518= | Heterozygous | Unphased | Likely benign | Tier1 | 22.5 | 0.64 | 21.4 (Cactus) | – / – | chr16:19629820<br>AF: 0.3238% |

### 1. INHERITANCE & OMIM PATHOLOGY

**VPS35L-related disorder**
Complex / Multifactorial

**Ritscher-Schinzel syndrome**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000417362.7 |
| cDNA (HGVS): | c.1554G>A |
| Protein (HGVS): | p.Thr518= |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

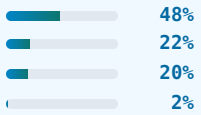
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.64 DL:0.01 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#2672622 ↗ |
| gnomAD v4 AF: | 0.3238% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.4 (Raw: 0.657 · Top 0.73%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB1 HLA-DRB1 chr6:32,582,135 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.58 Showing 3 of 7 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB1 belongs to the HLA class II beta chain paralogs. The class II molecule is a heterodimer consisting of an alpha (DRA) and a beta chain (DRB), both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa. It is encoded by 6 exons. Exon one encodes the leader peptide; exons 2 and 3 encode the two extracellular domains; exon 4 encodes the transmembrane domain; and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Hundreds of DRB1 alleles have been described and some alleles have increased frequencies associated with certain diseases or conditions. For example, DRB1*1302 has been related to acute and chronic hepatitis B virus persistence. There are multiple pseudogenes of this gene. [provided by RefSeq, Jul 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28723976<br>c.371-297A>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 7.4 | 0.58 | 12.9 (Splicing) | – / – | chr6:32582135<br>AF: 0.7402% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.371-297A>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.41 · DP – |
| Phased GT: | 0/1 (Unphased) |

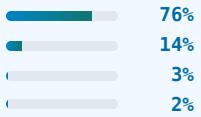
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.58 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.7402% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 12.9 (Raw: 0.248 · Top 5.08%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28723975<br>c.371-270T>G | INT | Heterozygous | Unphased | Not reviewed | Tier2 | 6.4 | 0.46 | 11.3 (Splicing) | – / – | chr6:32582108<br>AF: 0.0331% |
| rs28724109<br>c.101-43C>T | INT | Heterozygous | Unphased | Not reviewed | Tier2 | 6.7 | 0.41 | 19.2 (Splicing) | – / – | chr6:32584421<br>AF: 0.0888% |

## UCK2 UCK2 chr1:165,828,740 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.57 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a pyrimidine ribonucleoside kinase. The encoded protein (EC 2.7.1.48) catalyzes phosphorylation of uridine and cytidine to uridine monophosphate (UMP) and cytidine monophosphate (CMP), respectively.[provided by RefSeq, Oct 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs562989479<br>c.99+808G>T | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.9 | 0.57 | 19.2 (Splicing) | – / – | chr1:165828740<br>AF: 0.3704% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000367879.9 |
| cDNA (HGVS): | c.99+808G>T |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.32 · DP – |
| Phased GT: | 0/1 (Unphased) |

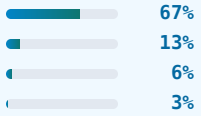
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.57 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.3704% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 19.2 (Raw: 0.517 · Top 1.20%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HEPACAM HEPACAM chr11:124,923,060 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.55 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a single-pass type I membrane protein that localizes to the cytoplasmic side of the cell membrane. The encoded protein acts as a homodimer and is involved in cell motility and cell-matrix interactions. The expression of this gene is downregulated or undetectable in many cancer cell lines, so this may be a tumor suppressor gene. [provided by RefSeq, Jul 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs11605766<br>c.804-242G>C | INT | Heterozygous | Unphased | Likely benign | Tier1 | 14.2 | 0.55 | 14.7 (Splicing) | – / – | chr11:124923060<br>AF: 1.0897% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000298251.5 |
| cDNA (HGVS): | c.804-242G>C |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.57 · DP – |
| Phased GT: | 0/1 (Unphased) |

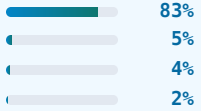
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.55 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#1201816 ↗ |
| gnomAD v4 AF: | 1.0897% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 14.7 (Raw: 0.311 · Top 3.35%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SYNPR SYNPR chr3:63,319,195 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.54 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to be involved in modulation of chemical synaptic transmission. Predicted to be located in membrane; neuron projection; and synaptic vesicle. Predicted to be active in hippocampal mossy fiber to CA3 synapse and synaptic vesicle membrane. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs73123134<br>c.84+40453T>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.8 | 0.54 | 6.0 (Cactus) | – / – | chr3:63319195<br>AF: 4.7697% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000478300.6 |
| cDNA (HGVS): | c.84+40453T>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.29 · DP – |
| Phased GT: | 0/1 (Unphased) |

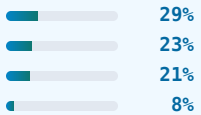
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.54 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 4.7697% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 6.0 (Raw: 0.048 · Top 25.19%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CD40 CD40 chr20:46,126,737 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.51 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene is a member of the TNF-receptor superfamily. The encoded protein is a receptor on antigen-presenting cells of the immune system and is essential for mediating a broad variety of immune and inflammatory responses including T cell-dependent immunoglobulin class switching, memory B cell development, and germinal center formation. AT-hook transcription factor AKNA is reported to coordinately regulate the expression of this receptor and its ligand, which may be important for homotypic cell interactions. Adaptor protein TNFR2 interacts with this receptor and serves as a mediator of the signal transduction. The interaction of this receptor and its ligand is found to be necessary for amyloid-beta-induced microglial activation, and thus is thought to be an early event in Alzheimer disease pathogenesis. Mutations affecting this gene are the cause of autosomal recessive hyper-IgM immunodeficiency type 3 (HIGM3). Multiple alternatively spliced transcript variants of this gene encoding distinct isoforms have been reported. [provided by RefSeq, Nov 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs41282788<br>c.559+36G>C | INT | Heterozygous | Unphased | Benign | Tier1 | 2.0 | 0.51 | 19.4 (Splicing) | – / – | chr20:46126737<br>AF: 1.8172% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Hyper-IgM syndrome type 3**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

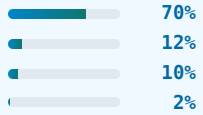
Transcript: ENST00000372285.8
cDNA (HGVS): c.559+36G>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.58 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.51 DL:0.10
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#993383 ↗
gnomAD v4 AF: 1.8172%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.4 (Raw: 0.528 · Top 1.14%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FHIT FHIT chr3:60,547,291 Multisystem 🔥 Tier-1 / Pathogenic Gene

SpliceAI Δ: 0.51 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a P1-P3-bis(5'-adenosyl) triphosphate hydrolase involved in purine metabolism. This gene encompasses the common fragile site FRA3B on chromosome 3, where carcinogen-induced damage can lead to translocations and aberrant transcripts. In fact, aberrant transcripts from this gene have been found in about half of all esophageal, stomach, and colon carcinomas. The encoded protein is also a tumor suppressor, as loss of its activity results in replication stress and DNA damage. [provided by RefSeq, Aug 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114893898<br>c.-17-10312G>C | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 9.7 | 0.51 | 9.8 (Cactus) | – / – | chr3:60547291<br>AF: 4.4825% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

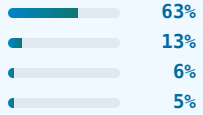
Transcript: ENST00000492590.6
cDNA (HGVS): c.-17-10312G>C
Protein (HGVS): –
VAF / Read Depth: VAF 0.47 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.51 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 4.4825%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 9.8 (Raw: 0.147 · Top 10.59%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs115894613<br>c.104-75688A>G | INT | Heterozygous | Unphased | Not reviewed | Tier2 | 1.7 | 0.28 | 6.6 (Splicing) | – / – | chr3:60089840<br>AF: 3.4159% |

## 6. AlphaMissense 3D Structural Pathogenicity — Top Missense Genes

Top protein-coding missense variants ranked by DeepMind AlphaFold-derived AlphaMissense structural pathogenicity probability.

**Selected Scope:** 21 Genes (21 Tier-1/Pathogenic)
Filtered from 378 matching genes in view

### SUMF1 SUMF1 chr3:4,362,213 Multisystem 🔥 Tier-1 / Pathogenic Gene

**AlphaMissense: 0.995** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an enzyme that catalyzes the hydrolysis of sulfate esters by oxidizing a cysteine residue in the substrate sulfatase to an active site 3-oxoalanine residue, which is also known as C-alpha-formylglycine. Mutations in this gene cause multiple sulfatase deficiency, a lysosomal storage disorder. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2009]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs376126385 p.Asn352Lys | MIS, p.Asn352Lys | Heterozygous | Unphased | Uncertain significance | Tier1 | 23.9 | – | 25.9 (AlphaMissense) | 0.64 / 1.00 | chr3:4362213 AF: 0.0009% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Multiple sulfatase deficiency**
Complex / Multifactorial

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

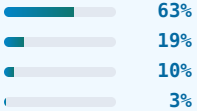
| | |
|---|---|
| Transcript: | ENST00000272902.10 |
| cDNA (HGVS): | c.1056C>G |
| Protein (HGVS): | p.Asn352Lys |
| VAF / Read Depth: | VAF 0.66 · DP – |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#2161926 ↗ |
| gnomAD v4 AF: | 0.0009% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

### PLK4 PLK4 chr4:127,881,151 Multisystem 🔥 Tier-1 / Pathogenic Gene

**AlphaMissense: 0.989** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polo family of serine/threonine protein kinases. The protein localizes to centrioles, complex microtubule-based structures found in centrosomes, and regulates centriole duplication during the cell cycle. Three alternatively spliced transcript variants that encode different protein isoforms have been found for this gene. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs149003893 p.Gly6Glu | MIS, p.Gly6Glu | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 28.5 | – | 29.8 (AlphaMissense) | 0.36 / 0.99 | chr4:127881151 AF: 0.1341% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Inborn genetic diseases**
Complex / Multifactorial

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

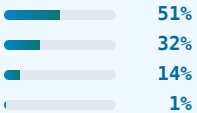
| | |
|---|---|
| Transcript: | ENST00000270861.10 |
| cDNA (HGVS): | c.17G>A |
| Protein (HGVS): | p.Gly6Glu |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#436338 ↗ |
| gnomAD v4 AF: | 0.1341% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 29.8 (Raw: 1.555 · Top 0.11%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## POLG POLG chr15:89,323,460 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.936 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Mitochondrial DNA polymerase is heterotrimeric, consisting of a homodimer of accessory subunits plus a catalytic subunit. The protein encoded by this gene is the catalytic subunit of mitochondrial DNA polymerase. The encoded protein contains a polyglutamine tract near its N-terminus that may be polymorphic. Defects in this gene are a cause of progressive external ophthalmoplegia with mitochondrial DNA deletions 1 (PEOA1), sensory ataxic neuropathy dysarthria and ophthalmoparesis (SANDO), Alpers-Huttenlocher syndrome (AHS), and mitochondrial neurogastrointestinal encephalopathy syndrome (MNGIE). Two transcript variants encoding the same protein have been found for this gene. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs121918054 p.Gly737Arg | MIS, p.Gly737Arg | Heterozygous | Paternal | Pathogenic/Likely pathogenic | Tier1 | 26.3 | – | 30.3 (AlphaMissense) | 0.94 / 0.94 | chr15:89323460 AF: 0.1619% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Spinocerebellar ataxia with epilepsy**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Sensory ataxic neuropathy, dysarthria, and ophthalmoparesis**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 1**
Autosomal Dominant OMIM #203700; 607459; 613662; 157640; 258450 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000268124.11 |
| cDNA (HGVS): | c.2209G>C |
| Protein (HGVS): | p.Gly737Arg |
| VAF / Read Depth: | VAF 0.61 · DP – |
| Phased GT: | 1\|0 (Paternal) |

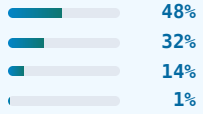
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#13513 ↗ |
| gnomAD v4 AF: | 0.1619% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.3 (Raw: 1.582 · Top 0.09%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLC27A3 SLC27A3 chr1:153,776,015 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.935 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to a family of integral membrane proteins and encodes a protein that is involved in lipid metabolism. The increased expression of this gene in human neural stem cells derived from induced pluripotent stem cells suggests that it plays an important role in early brain development. Naturally occurring mutations in this gene are associated with autism spectrum disorders. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Feb 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs756978807 p.Leu173Pro | MIS, p.Leu173Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 26.9 (AlphaMissense) | 0.41 / 0.94 | chr1:153776015 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000624995.4 |
| cDNA (HGVS): | c.518T>C |
| Protein (HGVS): | p.Leu173Pro |
| VAF / Read Depth: | VAF 0.23 · DP – |
| Phased GT: | 0/1 (Unphased) |

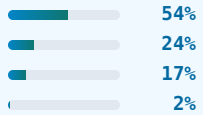
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0001% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.9 (Raw: 1.371 · Top 0.21%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## AMPD1 AMPD1 chr1:114,677,435 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.919 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Adenosine monophosphate deaminase 1 catalyzes the deamination of AMP to IMP in skeletal muscle and plays an important role in the purine nucleotide cycle. Two other genes have been identified, AMPD2 and AMPD3, for the liver- and erythocyte-specific isoforms, respectively. Deficiency of the muscle-specific enzyme is apparently a common cause of exercise-induced myopathy and probably the most common cause of metabolic myopathy in the human. Alternatively spliced transcript variants encoding different isoforms have been identified in this gene.[provided by RefSeq, Feb 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs764950035 p.Glu435Gly | MIS, p.Glu435Gly | Heterozygous | Unphased | Uncertain significance | Tier1 | 32.0 | – | 28.2 (AlphaMissense) | 0.95 / 0.92 | chr1:114677435 AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Muscle AMP deaminase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

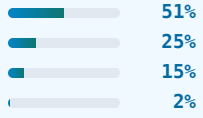
Transcript: ENST00000520113.7
cDNA (HGVS): c.1304A>G
Protein (HGVS): p.Glu435Gly
VAF / Read Depth: VAF 0.49 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#2173226 ↗
gnomAD v4 AF: 0.0075%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.2 (Raw: 1.454 · Top 0.15%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## MID1IP1 MID1IP1 chrX:38,805,491 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.874 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable enzyme activator activity. Predicted to be involved in several processes, including negative regulation of microtubule depolymerization; positive regulation of fatty acid biosynthetic process; and positive regulation of ligase activity. Predicted to be located in microtubule cytoskeleton. Predicted to be active in cytosol and nucleus. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs143008624 p.Gly182Asp | MIS, p.Gly182Asp | Heterozygous | Unphased | Benign | Tier1 | 26.1 | – | 27.1 (AlphaMissense) | 0.63 / 0.87 | chrX:38805491 AF: 0.3491% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

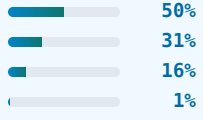
Transcript: ENST00000614558.3
cDNA (HGVS): c.545G>A
Protein (HGVS): p.Gly182Asp
VAF / Read Depth: VAF 0.48 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#711083 ↗
gnomAD v4 AF: 0.3491%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.1 (Raw: 1.393 · Top 0.19%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CTH CTH chr1:70,415,987 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.814 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a cytoplasmic enzyme in the trans-sulfuration pathway that converts cystathione derived from methionine into cysteine. Glutathione synthesis in the liver is dependent upon the availability of cysteine. Mutations in this gene cause cystathioninuria. Alternative splicing of this gene results in three transcript variants encoding different isoforms. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28941785<br>p.Thr67Ile | MIS, p.Thr67Ile | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 25.5 | – | 27.8 (AlphaMissense) | 0.93 / 0.81 | chr1:70415987<br>AF: 0.8026% |

### 1. INHERITANCE & OMIM PATHOLOGY

**CTH-related disorder**
Complex / Multifactorial OMIM #219500 ↗

**Cystathioninuria**
Complex / Multifactorial OMIM #219500 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

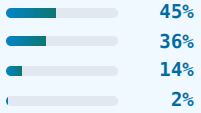
Transcript: ENST00000370938.8
cDNA (HGVS): c.200C>T
Protein (HGVS): p.Thr67Ile
VAF / Read Depth: VAF 0.46 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#2939 ↗
gnomAD v4 AF: 0.8026%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-B HLA-B chr6:31,356,717 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.788 Showing 3 of 7 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-B belongs to the HLA class I heavy chain paralogues. This class I molecule is a heterodimer consisting of a heavy chain and a light chain (beta-2 microglobulin). The heavy chain is anchored in the membrane. Class I molecules play a central role in the immune system by presenting peptides derived from the endoplasmic reticulum lumen. They are expressed in nearly all cells. The heavy chain is approximately 45 kDa and its gene contains 8 exons. Exon 1 encodes the leader peptide, exon 2 and 3 encode the alpha1 and alpha2 domains, which both bind the peptide, exon 4 encodes the alpha3 domain, exon 5 encodes the transmembrane region and exons 6 and 7 encode the cytoplasmic tail. Polymorphisms within exon 2 and exon 3 are responsible for the peptide binding specificity of each class one molecule. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. Hundreds of HLA-B alleles have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9266161<br>p.Leu105Pro | MIS, p.Leu105Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 22.5 | – | 20.0 (AlphaMissense) | 0.20 / 0.79 | chr6:31356717<br>AF: 0.0006% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

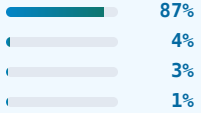
Transcript: ENST00000412585.7
cDNA (HGVS): c.314T>C
Protein (HGVS): p.Leu105Pro
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0006%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.0 (Raw: 0.559 · Top 1.00%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1131212<br>p.Gln94His | MIS, p.Gln94His | Homozygous | N/A | Not reviewed | Tier2 | 7.1 | – | 11.3 (AlphaMissense) | 0.24 / 0.28 | chr6:31356749<br>AF: 0.2441% |
| rs9266144<br>p.Arg180Gly | MIS, p.Arg180Gly | Homozygous | N/A | Not reviewed | Tier2 | 0.0 | – | 10.6 (AlphaMissense) | 0.25 / 0.22 | chr6:31356248<br>AF: 0.0002% |

## HLA-DRB1 HLA-DRB1 chr6:32,582,135 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene AlphaMissense: 0.776 Showing 3 of 13 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB1 belongs to the HLA class II beta chain paralogs. The class II molecule is a heterodimer consisting of an alpha (DRA) and a beta chain (DRB), both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa. It is encoded by 6 exons. Exon one encodes the leader peptide; exons 2 and 3 encode the two extracellular domains; exon 4 encodes the transmembrane domain; and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Hundreds of DRB1 alleles have been described and some alleles have increased frequencies associated with certain diseases or conditions. For example, DRB1*1302 has been related to acute and chronic hepatitis B virus persistence. There are multiple pseudogenes of this gene. [provided by RefSeq, Jul 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs16822805<br>p.Ala103Pro | MIS, p.Ala103Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 0.0 | – | 20.2 (AlphaMissense) | 0.21 / 0.78 | chr6:32584172<br>AF: 0.1762% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.307G>C |
| Protein (HGVS): | p.Ala103Pro |
| VAF / Read Depth: | VAF 0.53 · DP – |
| Phased GT: | 0/1 (Unphased) |

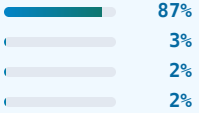
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#691505 ↗ |
| gnomAD v4 AF: | 0.1762% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.2 (Raw: 0.573 · Top 0.95%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs17884043<br>p.Tyr107Asp | MIS, p.Tyr107Asp | Heterozygous | Unphased | Not reviewed | Tier2 | 16.5 | – | 8.6 (AlphaMissense) | 0.13 / 0.45 | chr6:32584160<br>AF: 0.0001% |
| rs17879599<br>p.Gln99His | MIS, p.Gln99His | Heterozygous | Unphased | Not reviewed | Tier2 | 0.6 | – | 11.3 (AlphaMissense) | 0.22 / 0.28 | chr6:32584182<br>AF: 0.3123% |

## TMEM43 TMEM43 chr3:14,139,244 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.761 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to the TMEM43 family. Defects in this gene are the cause of familial arrhythmogenic right ventricular dysplasia type 5 (ARVD5), also known as arrhythmogenic right ventricular cardiomyopathy type 5 (ARVC5). Arrhythmogenic right ventricular dysplasia is an inherited disorder, often involving both ventricles, and is characterized by ventricular tachycardia, heart failure, sudden cardiac death, and fibrofatty replacement of cardiomyocytes. This gene contains a response element for PPAR gamma (an adipogenic transcription factor), which may explain the fibrofatty replacement of the myocardium, a characteristic pathological finding in ARVC. [provided by RefSeq, Oct 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs199526104 p.Trp316Ser | MIS, p.Trp316Ser | Heterozygous | Paternal | Conflicting classifications of pathogenicity | Tier1 | 32.0 | – | 26.5 (AlphaMissense) | 0.53 / 0.76 | chr3:14139244 AF: 0.0126% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**
Complex / Multifactorial OMIM #604400 ↗

**TMEM43-related disorder**
Complex / Multifactorial OMIM #604400 ↗

**Auditory neuropathy, autosomal dominant 3**
Autosomal Dominant OMIM #604400 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

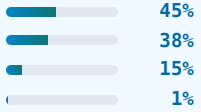
Transcript: ENST00000306077.5
cDNA (HGVS): c.947G>C
Protein (HGVS): p.Trp316Ser
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#46155 ↗
gnomAD v4 AF: 0.0126%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.5 (Raw: 1.335 · Top 0.22%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAAP24 FAAP24 chr19:32,974,193 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.690 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 FAAP24 is a component of the Fanconi anemia (FA) core complex (see MIM 227650), which plays a crucial role in DNA damage response (Ciccia et al., 2007 [PubMed 17289582]).[supplied by OMIM, Mar 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs36017455 p.Ser126Phe | MIS, p.Ser126Phe | Heterozygous | Unphased | Benign | Tier1 | 25.7 | – | 27.0 (AlphaMissense) | – / 0.69 | chr19:32974193 AF: 0.5937% |

### 1. INHERITANCE & OMIM PATHOLOGY

**FAAP24-related disorder**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

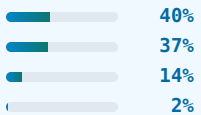
Transcript: ENST00000588258.6
cDNA (HGVS): c.377C>T
Protein (HGVS): p.Ser126Phe
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#784831 ↗
gnomAD v4 AF: 0.5937%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.0 (Raw: 1.385 · Top 0.20%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ACSM2B ACSM2B chr16:20,540,665 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.580 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables benzoate-CoA ligase activity. Predicted to be involved in acyl-CoA metabolic process and fatty acid biosynthetic process. Located in mitochondrion. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs139253186 p.Tyr540Asn | MIS, p.Tyr540Asn | Heterozygous | Unphased | Not reviewed | Tier1 | 27.6 | – | 24.4 (AlphaMissense) | 0.40 / 0.58 | chr16:20540665 AF: 0.4705% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000329697.10 |
| cDNA (HGVS): | c.1618T>A |
| Protein (HGVS): | p.Tyr540Asn |
| VAF / Read Depth: | VAF 0.54 · DP – |
| Phased GT: | 0/1 (Unphased) |

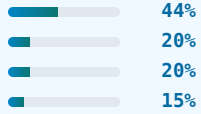
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.4705% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.4 (Raw: 1.012 · Top 0.37%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## WDHD1 WDHD1 chr14:55,002,157 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.577 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene contains multiple N-terminal WD40 domains and a C-terminal high mobility group (HMG) box. WD40 domains are found in a variety of eukaryotic proteins and may function as adaptor/regulatory modules in signal transduction, pre-mRNA processing and cytoskeleton assembly. HMG boxes are found in many eukaryotic proteins involved in chromatin assembly, transcription and replication. Alternative splicing results in two transcript variants encoding different isoforms. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs374497720 p.Val210Ala | MIS, p.Val210Ala | Heterozygous | Unphased | Not reviewed | Tier1 | 25.8 | – | 24.5 (AlphaMissense) | 0.27 / 0.58 | chr14:55002157 AF: 0.0190% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000360586.8 |
| cDNA (HGVS): | c.629T>C |
| Protein (HGVS): | p.Val210Ala |
| VAF / Read Depth: | VAF 0.50 · DP – |
| Phased GT: | 0/1 (Unphased) |

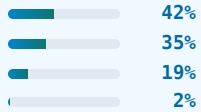
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0190% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.5 (Raw: 1.042 · Top 0.35%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PGK1 PGK1 chrX:78,114,140 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.569 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a glycolytic enzyme that catalyzes the conversion of 1,3-diphosphoglycerate to 3-phosphoglycerate. The encoded protein may also act as a cofactor for polymerase alpha. Additionally, this protein is secreted by tumor cells where it participates in angiogenesis by functioning to reduce disulfide bonds in the serine protease, plasmin, which consequently leads to the release of the tumor blood vessel inhibitor angiostatin. The encoded protein has been identified as a moonlighting protein based on its ability to perform mechanistically distinct functions. Deficiency of the enzyme is associated with a wide range of clinical phenotypes hemolytic anemia and neurological impairment. Pseudogenes of this gene have been defined on chromosomes 19, 21 and the X chromosome. [provided by RefSeq, Jan 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| **chrX:78114140** p.Lys133Glu | MIS, p.Lys133Glu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.7 | – | 24.3 (AlphaMissense) | 0.54 / 0.57 | chrX:78114140 AF: 0.0008% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

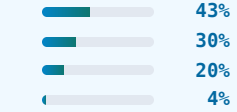
Transcript: **ENST00000373316.5**
cDNA (HGVS): **c.397A>G**
Protein (HGVS): **p.Lys133Glu**
VAF / Read Depth: **VAF 0.58 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0008%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.3** (Raw: 0.993 · Top 0.38%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CEP63 CEP63 chr3:134,537,268 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.542 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with six coiled-coil domains. The protein is localized to the centrosome, a non-membraneous organelle that functions as the major microtubule-organizing center in animal cells. Several alternatively spliced transcript variants have been found, but their biological validity has not been determined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114108011 p.Gln185His | MIS, p.Gln185His | Heterozygous | Maternal | Benign | Tier1 | 34.0 | 0.93 | 38.6 (Splicing) | 0.28 / 0.54 | chr3:134537268 AF: 2.0823% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

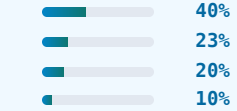
Transcript: **ENST00000675561.1**
cDNA (HGVS): **c.555G>C**
Protein (HGVS): **p.Gln185His**
VAF / Read Depth: **VAF 0.74 · DP –**
Phased GT: **0|1 (Maternal)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.93**
ClinVar Review: **criteria provided, multiple submitters, no conflicts**
ClinVar VCV: VCV#128709 ↗
gnomAD v4 AF: **2.0823%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **38.6** (Raw: 2.279 · Top 0.01%)
Primary Driver: **Splicing**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## DMAP1 DMAP1 chr1:44,218,751 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.379 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a subunit of several, distinct complexes involved in the repression or activation of transcription. The encoded protein can independently repress transcription and is targeted to replication foci throughout S phase by interacting directly with the N-terminus of DNA methyltransferase 1. During late S phase, histone deacetylase 2 is added to this complex, providing a means to deacetylate histones in transcriptionally inactive heterochromatin following replication. The encoded protein is also a component of the nucleosome acetyltransferase of H4 complex and interacts with the transcriptional corepressor tumor susceptibility gene 101 and the pro-apoptotic death-associated protein 6, among others. Alternatively spliced transcript variants encoding the same protein have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| **chr1:44218751** p.Glu239Gly | MIS, p.Glu239Gly | Heterozygous | Unphased | Not reviewed | Tier1 | 29.7 | – | 23.4 (Cactus) | 0.38 / 0.38 | chr1:44218751 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

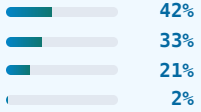
Transcript: **ENST00000372289.7**
cDNA (HGVS): **c.716A>G**
Protein (HGVS): **p.Glu239Gly**
VAF / Read Depth: **VAF 0.38 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0001%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **23.4** (Raw: 0.854 · Top 0.46%)
Primary Driver: **Cactus**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLIT2 SLIT2 chr4:20,567,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.352 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the slit family of secreted glycoproteins, which are ligands for the Robo family of immunoglobulin receptors. Slit proteins play highly conserved roles in axon guidance and neuronal migration and may also have functions during other cell migration processes including leukocyte migration. Members of the slit family are characterized by an N-terminal signal peptide, four leucine-rich repeats, nine epidermal growth factor repeats, and a C-terminal cysteine knot. Proteolytic processing of this protein gives rise to an N-terminal fragment that contains the four leucine-rich repeats and five epidermal growth factor repeats and a C-terminal fragment that contains four epidermal growth factor repeats and the cysteine knot. Both full length and cleaved proteins are secreted extracellularly and can function in axon repulsion as well as other specific processes. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2015]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1020555095 p.Gly909Asp | MIS, p.Gly909Asp | Heterozygous | Unphased | Not reviewed | Tier1 | 32.0 | – | 23.4 (Cactus) | 0.74 / 0.35 | chr4:20567262 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

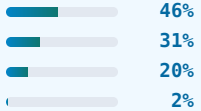
Transcript: **ENST00000504154.6**
cDNA (HGVS): **c.2726G>A**
Protein (HGVS): **p.Gly909Asp**
VAF / Read Depth: **VAF 0.54 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0009%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **23.4** (Raw: 0.859 · Top 0.46%)
Primary Driver: **Cactus**




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## GUSB GUSB chr7:65,960,973 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.246 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a hydrolase that degrades glycosaminoglycans, including heparan sulfate, dermatan sulfate, and chondroitin-4,6-sulfate. The enzyme forms a homotetramer that is localized to the lysosome. Mutations in this gene result in mucopolysaccharidosis type VII. Alternative splicing results in multiple transcript variants. There are many pseudogenes of this locus in the human genome.[provided by RefSeq, May 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1236992554 p.Trp627Leu | MIS, p.Trp627Leu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.2 | – | 24.9 (Cactus) | 0.56 / 0.25 | chr7:65960973 AF: 0.0003% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

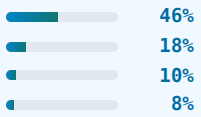
Transcript: ENST00000304895.9
cDNA (HGVS): c.1880G>T
Protein (HGVS): p.Trp627Leu
VAF / Read Depth: VAF 0.53 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0003%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.9 (Raw: 1.103 · Top 0.32%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAHD1 FAHD1 chr16:1,827,542 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.240 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables hydrolase activity, acting on carbon-carbon bonds, in ketonic substances; oxaloacetate decarboxylase activity; and oxaloacetate tautomerase activity. Involved in oxaloacetate metabolic process. Located in cytosol; mitochondrion; and nucleoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs200357596 p.Ala102Ser | MIS, p.Ala102Ser | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 22.1 (Cactus) | 0.71 / 0.24 | chr16:1827542 AF: 0.0037% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

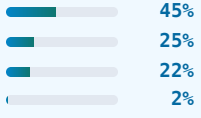
Transcript: ENST00000427358.5
cDNA (HGVS): c.304G>T
Protein (HGVS): p.Ala102Ser
VAF / Read Depth: VAF 0.49 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0037%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.1 (Raw: 0.710 · Top 0.62%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TEK TEK chr9:27,185,615 Multisystem 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.207 Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a receptor that belongs to the protein tyrosine kinase Tie2 family. The encoded protein possesses a unique extracellular region that contains two immunoglobulin-like domains, three epidermal growth factor (EGF)-like domains and three fibronectin type III repeats. The ligand angiopoietin-1 binds to this receptor and mediates a signaling pathway that functions in embryonic vascular development. Mutations in this gene are associated with inherited venous malformations of the skin and mucous membranes. Alternative splicing results in multiple transcript variants. Additional alternatively spliced transcript variants of this gene have been described, but their full-length nature is not known. [provided by RefSeq, Feb 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs189543659 p.Asn438Ile | MIS, p.Asn438Ile | Heterozygous | Unphased | Not reviewed | Tier1 | 24.3 | – | 21.9 (Cactus) | 0.21 / 0.21 | chr9:27185615 AF: 0.0011% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

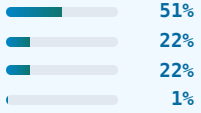
Transcript: ENST00000380036.10
cDNA (HGVS): c.1313A>T
Protein (HGVS): p.Asn438Ile
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0011%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 21.9 (Raw: 0.695 · Top 0.65%)
Primary Driver: Cactus




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs35030851 p.Val600Leu | MIS, p.Val600Leu | Heterozygous | Unphased | Benign | Tier3 | 17.9 | – | 22.9 (Cactus) | 0.06 / 0.16 | chr9:27197488 AF: 4.5993% |

## CHST14 CHST14 chr15:40,472,259 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

AlphaMissense: 0.201 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the HNK-1 family of sulfotransferases. The encoded protein transfers sulfate to the C-4 hydroxyl of N-acetylgalactosamine residues in dermatan sulfate. Mutations in this gene have been associated with adducted thumb-clubfoot syndrome.[provided by RefSeq, Mar 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1463514561 p.Gln349Pro | MIS, p.Gln349Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 24.3 | – | 22.9 (Cactus) | 0.54 / 0.20 | chr15:40472259 AF: 0.0004% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Cardiovascular phenotype**
Complex / Multifactorial

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

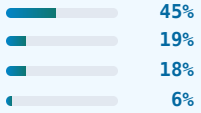
Transcript: ENST00000306243.7
cDNA (HGVS): c.1046A>C
Protein (HGVS): p.Gln349Pro
VAF / Read Depth: VAF 0.52 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#3229028 ↗
gnomAD v4 AF: 0.0004%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 22.9 (Raw: 0.798 · Top 0.51%)
Primary Driver: Cactus




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## 7. REVEL Ensembled Missense Pathogenicity — Top Genes (Score ≥ 0.20)

Top missense variants ranked by Rare Exome Variant Ensemble Learner (REVEL) pathogenicity score.

**Selected Scope:** 21 Genes (21 Tier-1/Pathogenic)
Filtered from 137 matching genes in view

### AMPD1 AMPD1 chr1:114,677,435 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.949 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Adenosine monophosphate deaminase 1 catalyzes the deamination of AMP to IMP in skeletal muscle and plays an important role in the purine nucleotide cycle. Two other genes have been identified, AMPD2 and AMPD3, for the liver- and erythocyte-specific isoforms, respectively. Deficiency of the muscle-specific enzyme is apparently a common cause of exercise-induced myopathy and probably the most common cause of metabolic myopathy in the human. Alternatively spliced transcript variants encoding different isoforms have been identified in this gene.[provided by RefSeq, Feb 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs764950035<br>p.Glu435Gly | MIS, p.Glu435Gly | Heterozygous | Unphased | Uncertain significance | Tier1 | 32.0 | – | 28.2 (AlphaMissense) | 0.95 / 0.92 | chr1:114677435<br>AF: 0.0075% |

#### 1. INHERITANCE & OMIM PATHOLOGY

**Muscle AMP deaminase deficiency**
Complex / Multifactorial

Quadrant: **Curated**

#### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000520113.7 |
| cDNA (HGVS): | c.1304A>G |
| Protein (HGVS): | p.Glu435Gly |
| VAF / Read Depth: | VAF 0.49 · DP – |
| Phased GT: | 0/1 (Unphased) |

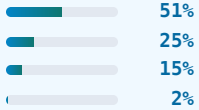
#### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#2173226 ↗ |
| gnomAD v4 AF: | 0.0075% |

#### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.2 (Raw: 1.454 · Top 0.15%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## POLG POLG chr15:89,323,460 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.936 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Mitochondrial DNA polymerase is heterotrimeric, consisting of a homodimer of accessory subunits plus a catalytic subunit. The protein encoded by this gene is the catalytic subunit of mitochondrial DNA polymerase. The encoded protein contains a polyglutamine tract near its N-terminus that may be polymorphic. Defects in this gene are a cause of progressive external ophthalmoplegia with mitochondrial DNA deletions 1 (PEOA1), sensory ataxic neuropathy dysarthria and ophthalmoparesis (SANDO), Alpers-Huttenlocher syndrome (AHS), and mitochondrial neurogastrointestinal encephalopathy syndrome (MNGIE). Two transcript variants encoding the same protein have been found for this gene. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs121918054 p.Gly737Arg | MIS, p.Gly737Arg | Heterozygous | Paternal | Pathogenic/Likely pathogenic | Tier1 | 26.3 | – | 30.3 (AlphaMissense) | 0.94 / 0.94 | chr15:89323460 AF: 0.1619% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Spinocerebellar ataxia with epilepsy**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Sensory ataxic neuropathy, dysarthria, and ophthalmoparesis**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 1**
Autosomal Dominant OMIM #203700; 607459; 613662; 157640; 258450 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

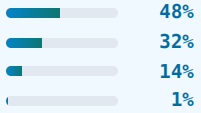
Transcript: ENST00000268124.11
cDNA (HGVS): c.2209G>C
Protein (HGVS): p.Gly737Arg
VAF / Read Depth: VAF 0.61 · DP –
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#13513 ↗
gnomAD v4 AF: 0.1619%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.3 (Raw: 1.582 · Top 0.09%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CTH CTH chr1:70,415,987 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.931 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a cytoplasmic enzyme in the trans-sulfuration pathway that converts cystathione derived from methionine into cysteine. Glutathione synthesis in the liver is dependent upon the availability of cysteine. Mutations in this gene cause cystathioninuria. Alternative splicing of this gene results in three transcript variants encoding different isoforms. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28941785 p.Thr67Ile | MIS, p.Thr67Ile | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 25.5 | – | 27.8 (AlphaMissense) | 0.93 / 0.81 | chr1:70415987 AF: 0.8026% |

### 1. INHERITANCE & OMIM PATHOLOGY

**CTH-related disorder**
Complex / Multifactorial OMIM #219500 ↗

**Cystathioninuria**
Complex / Multifactorial OMIM #219500 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

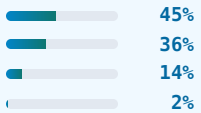
Transcript: ENST00000370938.8
cDNA (HGVS): c.200C>T
Protein (HGVS): p.Thr67Ile
VAF / Read Depth: VAF 0.46 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#2939 ↗
gnomAD v4 AF: 0.8026%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLIT2 SLIT2 chr4:20,567,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.743 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the slit family of secreted glycoproteins, which are ligands for the Robo family of immunoglobulin receptors. Slit proteins play highly conserved roles in axon guidance and neuronal migration and may also have functions during other cell migration processes including leukocyte migration. Members of the slit family are characterized by an N-terminal signal peptide, four leucine-rich repeats, nine epidermal growth factor repeats, and a C-terminal cysteine knot. Proteolytic processing of this protein gives rise to an N-terminal fragment that contains the four leucine-rich repeats and five epidermal growth factor repeats and a C-terminal fragment that contains four epidermal growth factor repeats and the cysteine knot. Both full length and cleaved proteins are secreted extracellularly and can function in axon repulsion as well as other specific processes. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2015]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1020555095 p.Gly909Asp | MIS, p.Gly909Asp | Heterozygous | Unphased | Not reviewed | Tier1 | 32.0 | – | 23.4 (Cactus) | 0.74 / 0.35 | chr4:20567262 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000504154.6 |
| cDNA (HGVS): | c.2726G>A |
| Protein (HGVS): | p.Gly909Asp |
| VAF / Read Depth: | VAF 0.54 · DP – |
| Phased GT: | 0/1 (Unphased) |

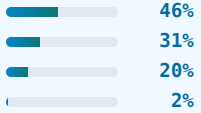
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0009% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.4 (Raw: 0.859 · Top 0.46%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAHD1 FAHD1 chr16:1,827,542 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.715 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables hydrolase activity, acting on carbon-carbon bonds, in ketonic substances; oxaloacetate decarboxylase activity; and oxaloacetate tautomerase activity. Involved in oxaloacetate metabolic process. Located in cytosol; mitochondrion; and nucleoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs200357596 p.Ala102Ser | MIS, p.Ala102Ser | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 22.1 (Cactus) | 0.71 / 0.24 | chr16:1827542 AF: 0.0037% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000427358.5 |
| cDNA (HGVS): | c.304G>T |
| Protein (HGVS): | p.Ala102Ser |
| VAF / Read Depth: | VAF 0.49 · DP – |
| Phased GT: | 0/1 (Unphased) |

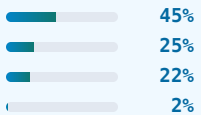
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0037% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.1 (Raw: 0.710 · Top 0.62%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SUMF1 SUMF1 chr3:4,362,213 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.641 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an enzyme that catalyzes the hydrolysis of sulfate esters by oxidizing a cysteine residue in the substrate sulfatase to an active site 3-oxoalanine residue, which is also known as C-alpha-formylglycine. Mutations in this gene cause multiple sulfatase deficiency, a lysosomal storage disorder. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2009]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs376126385 p.Asn352Lys | MIS, p.Asn352Lys | Heterozygous | Unphased | Uncertain significance | Tier1 | 23.9 | – | 25.9 (AlphaMissense) | 0.64 / 1.00 | chr3:4362213 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Multiple sulfatase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000272902.10 |
| cDNA (HGVS): | c.1056C>G |
| Protein (HGVS): | p.Asn352Lys |
| VAF / Read Depth: | VAF 0.66 · DP – |
| Phased GT: | 0/1 (Unphased) |

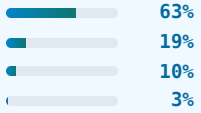
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#2161926 ↗ |
| gnomAD v4 AF: | 0.0009% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 25.9 (Raw: 1.256 · Top 0.26%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## MID1IP1 MID1IP1 chrX:38,805,491 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.630 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable enzyme activator activity. Predicted to be involved in several processes, including negative regulation of microtubule depolymerization; positive regulation of fatty acid biosynthetic process; and positive regulation of ligase activity. Predicted to be located in microtubule cytoskeleton. Predicted to be active in cytosol and nucleus. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs143008624 p.Gly182Asp | MIS, p.Gly182Asp | Heterozygous | Unphased | Benign | Tier1 | 26.1 | – | 27.1 (AlphaMissense) | 0.63 / 0.87 | chrX:38805491 AF: 0.3491% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000614558.3 |
| cDNA (HGVS): | c.545G>A |
| Protein (HGVS): | p.Gly182Asp |
| VAF / Read Depth: | VAF 0.48 · DP – |
| Phased GT: | 0/1 (Unphased) |

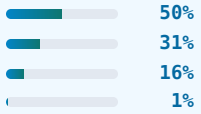
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#711083 ↗ |
| gnomAD v4 AF: | 0.3491% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.1 (Raw: 1.393 · Top 0.19%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## GUSB GUSB chr7:65,960,973 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.560 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a hydrolase that degrades glycosaminoglycans, including heparan sulfate, dermatan sulfate, and chondroitin-4,6-sulfate. The enzyme forms a homotetramer that is localized to the lysosome. Mutations in this gene result in mucopolysaccharidosis type VII. Alternative splicing results in multiple transcript variants. There are many pseudogenes of this locus in the human genome.[provided by RefSeq, May 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1236992554<br>p.Trp627Leu | MIS, p.Trp627Leu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.2 | — | 24.9 (Cactus) | 0.56 / 0.25 | chr7:65960973<br>AF: 0.0003% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000304895.9 |
| cDNA (HGVS): | c.1880G>T |
| Protein (HGVS): | p.Trp627Leu |
| VAF / Read Depth: | VAF 0.53 · DP — |
| Phased GT: | 0/1 (Unphased) |

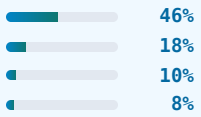
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0003% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.9 (Raw: 1.103 · Top 0.32%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PGK1 PGK1 chrX:78,114,140 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.543 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a glycolytic enzyme that catalyzes the conversion of 1,3-diphosphoglycerate to 3-phosphoglycerate. The encoded protein may also act as a cofactor for polymerase alpha. Additionally, this protein is secreted by tumor cells where it participates in angiogenesis by functioning to reduce disulfide bonds in the serine protease, plasmin, which consequently leads to the release of the tumor blood vessel inhibitor angiostatin. The encoded protein has been identified as a moonlighting protein based on its ability to perform mechanistically distinct functions. Deficiency of the enzyme is associated with a wide range of clinical phenotypes hemolytic anemia and neurological impairment. Pseudogenes of this gene have been defined on chromosomes 19, 21 and the X chromosome. [provided by RefSeq, Jan 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| chrX:78114140<br>p.Lys133Glu | MIS, p.Lys133Glu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.7 | — | 24.3 (AlphaMissense) | 0.54 / 0.57 | chrX:78114140<br>AF: 0.0008% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000373316.5 |
| cDNA (HGVS): | c.397A>G |
| Protein (HGVS): | p.Lys133Glu |
| VAF / Read Depth: | VAF 0.58 · DP — |
| Phased GT: | 0/1 (Unphased) |

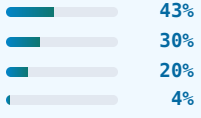
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0008% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.3 (Raw: 0.993 · Top 0.38%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CHST14 CHST14 chr15:40,472,259 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.540 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the HNK-1 family of sulfotransferases. The encoded protein transfers sulfate to the C-4 hydroxyl of N-acetylgalactosamine residues in dermatan sulfate. Mutations in this gene have been associated with adducted thumb-clubfoot syndrome.[provided by RefSeq, Mar 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1463514561 p.Gln349Pro | MIS, p.Gln349Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 24.3 | — | 22.9 (Cactus) | 0.54 / 0.20 | chr15:40472259 AF: 0.0004% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000306243.7 |
| cDNA (HGVS): | c.1046A>C |
| Protein (HGVS): | p.Gln349Pro |
| VAF / Read Depth: | VAF 0.52 · DP — |
| Phased GT: | 0/1 (Unphased) |

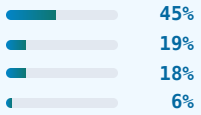
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#3229028 ↗ |
| gnomAD v4 AF: | 0.0004% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.9 (Raw: 0.798 · Top 0.51%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TMEM43 TMEM43 chr3:14,139,244 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.533 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to the TMEM43 family. Defects in this gene are the cause of familial arrhythmogenic right ventricular dysplasia type 5 (ARVD5), also known as arrhythmogenic right ventricular cardiomyopathy type 5 (ARVC5). Arrhythmogenic right ventricular dysplasia is an inherited disorder, often involving both ventricles, and is characterized by ventricular tachycardia, heart failure, sudden cardiac death, and fibrofatty replacement of cardiomyocytes. This gene contains a response element for PPAR gamma (an adipogenic transcription factor), which may explain the fibrofatty replacement of the myocardium, a characteristic pathological finding in ARVC. [provided by RefSeq, Oct 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs199526104 p.Trp316Ser | MIS, p.Trp316Ser | Heterozygous | Paternal | Conflicting classifications of pathogenicity | Tier1 | 32.0 | — | 26.5 (AlphaMissense) | 0.53 / 0.76 | chr3:14139244 AF: 0.0126% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**
Complex / Multifactorial OMIM #604400 ↗

**TMEM43-related disorder**
Complex / Multifactorial OMIM #604400 ↗

**Auditory neuropathy, autosomal dominant 3**
Autosomal Dominant OMIM #604400 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000306077.5 |
| cDNA (HGVS): | c.947G>C |
| Protein (HGVS): | p.Trp316Ser |
| VAF / Read Depth: | VAF 0.44 · DP — |
| Phased GT: | 1\|0 (Paternal) |

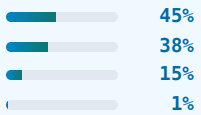
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#46155 ↗ |
| gnomAD v4 AF: | 0.0126% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.5 (Raw: 1.335 · Top 0.22%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLC27A3 SLC27A3 chr1:153,776,015 Multisystem 🔥 Tier-1 / Pathogenic Gene REVEL: 0.411 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to a family of integral membrane proteins and encodes a protein that is involved in lipid metabolism. The increased expression of this gene in human neural stem cells derived from induced pluripotent stem cells suggests that it plays an important role in early brain development. Naturally occurring mutations in this gene are associated with autism spectrum disorders. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Feb 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs756978807 p.Leu173Pro | MIS, p.Leu173Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 26.9 (AlphaMissense) | 0.41 / 0.94 | chr1:153776015 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000624995.4**
cDNA (HGVS): **c.518T>C**
Protein (HGVS): **p.Leu173Pro**
VAF / Read Depth: **VAF 0.23 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0001%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **26.9** (Raw: 1.371 · Top 0.21%)
Primary Driver: **AlphaMissense**

54%
24%
17%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ACSM2B ACSM2B chr16:20,540,665 Multisystem 🔥 Tier-1 / Pathogenic Gene REVEL: 0.401 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables benzoate-CoA ligase activity. Predicted to be involved in acyl-CoA metabolic process and fatty acid biosynthetic process. Located in mitochondrion. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs139253186 p.Tyr540Asn | MIS, p.Tyr540Asn | Heterozygous | Unphased | Not reviewed | Tier1 | 27.6 | – | 24.4 (AlphaMissense) | 0.40 / 0.58 | chr16:20540665 AF: 0.4705% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: **ENST00000329697.10**
cDNA (HGVS): **c.1618T>A**
Protein (HGVS): **p.Tyr540Asn**
VAF / Read Depth: **VAF 0.54 · DP –**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.4705%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.4** (Raw: 1.012 · Top 0.37%)
Primary Driver: **AlphaMissense**

44%
20%
20%
15%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## DMAP1 DMAP1 chr1:44,218,751 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.377 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a subunit of several, distinct complexes involved in the repression or activation of transcription. The encoded protein can independently repress transcription and is targeted to replication foci throughout S phase by interacting directly with the N-terminus of DNA methyltransferase 1. During late S phase, histone deacetylase 2 is added to this complex, providing a means to deacetylate histones in transcriptionally inactive heterochromatin following replication. The encoded protein is also a component of the nucleosome acetyltransferase of H4 complex and interacts with the transcriptional corepressor tumor susceptibility gene 101 and the pro-apoptotic death-associated protein 6, among others. Alternatively spliced transcript variants encoding the same protein have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| **chr1:44218751** p.Glu239Gly | MIS, p.Glu239Gly | Heterozygous | Unphased | Not reviewed | Tier1 | 29.7 | – | 23.4 (Cactus) | 0.38 / 0.38 | chr1:44218751 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000372289.7 |
| cDNA (HGVS): | c.716A>G |
| Protein (HGVS): | p.Glu239Gly |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0/1 (Unphased) |

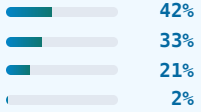
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0001% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.4 (Raw: 0.854 · Top 0.46%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PLK4 PLK4 chr4:127,881,151 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.356 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polo family of serine/threonine protein kinases. The protein localizes to centrioles, complex microtubule-based structures found in centrosomes, and regulates centriole duplication during the cell cycle. Three alternatively spliced transcript variants that encode different protein isoforms have been found for this gene. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs149003893 p.Gly6Glu | MIS, p.Gly6Glu | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 28.5 | – | 29.8 (AlphaMissense) | 0.36 / 0.99 | chr4:127881151 AF: 0.1341% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Inborn genetic diseases**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000270861.10 |
| cDNA (HGVS): | c.17G>A |
| Protein (HGVS): | p.Gly6Glu |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

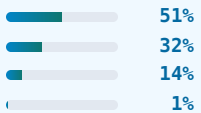
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#436338 ↗ |
| gnomAD v4 AF: | 0.1341% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.8 (Raw: 1.555 · Top 0.11%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB1 HLA-DRB1 chr6:32,582,135 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.311 Showing 3 of 5 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB1 belongs to the HLA class II beta chain paralogs. The class II molecule is a heterodimer consisting of an alpha (DRA) and a beta chain (DRB), both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa. It is encoded by 6 exons. Exon one encodes the leader peptide; exons 2 and 3 encode the two extracellular domains; exon 4 encodes the transmembrane domain; and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Hundreds of DRB1 alleles have been described and some alleles have increased frequencies associated with certain diseases or conditions. For example, DRB1*1302 has been related to acute and chronic hepatitis B virus persistence. There are multiple pseudogenes of this gene. [provided by RefSeq, Jul 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs16822805 p.Ala103Pro | MIS, p.Ala103Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 0.0 | – | 20.2 (AlphaMissense) | 0.21 / 0.78 | chr6:32584172 AF: 0.1762% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

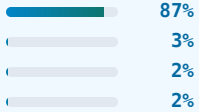
Transcript: ENST00000360004.6
cDNA (HGVS): c.307G>C
Protein (HGVS): p.Ala103Pro
VAF / Read Depth: VAF 0.53 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#691505 ↗
gnomAD v4 AF: 0.1762%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.2 (Raw: 0.573 · Top 0.95%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs17878857 p.Ala102Gly | MIS, p.Ala102Gly | Heterozygous | Unknown | Not reviewed | Tier2 | 12.2 | – | 11.4 (Cactus) | 0.31 / 0.11 | chr6:32584174 AF: 0.0002% |
| rs17881965 p.Gln99Glu | MIS, p.Gln99Glu | Heterozygous | Unphased | Not reviewed | Tier2 | 0.0 | – | 2.3 (Cactus) | 0.28 / 0.06 | chr6:32584184 AF: 0.0004% |

## CEP63 CEP63 chr3:134,537,268 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.276 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein with six coiled-coil domains. The protein is localized to the centrosome, a non-membraneous organelle that functions as the major microtubule-organizing center in animal cells. Several alternatively spliced transcript variants have been found, but their biological validity has not been determined. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs114108011 p.Gln185His | MIS, p.Gln185His | Heterozygous | Maternal | Benign | Tier1 | 34.0 | 0.93 | 38.6 (Splicing) | 0.28 / 0.54 | chr3:134537268 AF: 2.0823% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

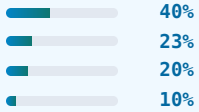
Transcript: ENST00000675561.1
cDNA (HGVS): c.555G>C
Protein (HGVS): p.Gln185His
VAF / Read Depth: VAF 0.74 · DP –
Phased GT: 0|1 (Maternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.93
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#128709 ↗
gnomAD v4 AF: 2.0823%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 38.6 (Raw: 2.279 · Top 0.01%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## WDHD1 WDHD1 chr14:55,002,157 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.271 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene contains multiple N-terminal WD40 domains and a C-terminal high mobility group (HMG) box. WD40 domains are found in a variety of eukaryotic proteins and may function as adaptor/regulatory modules in signal transduction, pre-mRNA processing and cytoskeleton assembly. HMG boxes are found in many eukaryotic proteins involved in chromatin assembly, transcription and replication. Alternative splicing results in two transcript variants encoding different isoforms. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs374497720 p.Val210Ala | MIS, p.Val210Ala | Heterozygous | Unphased | Not reviewed | Tier1 | 25.8 | – | 24.5 (AlphaMissense) | 0.27 / 0.58 | chr14:55002157 AF: 0.0190% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

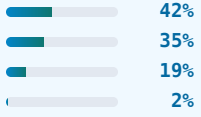
Transcript: ENST00000360586.8
cDNA (HGVS): c.629T>C
Protein (HGVS): p.Val210Ala
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0190%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 24.5 (Raw: 1.042 · Top 0.35%)
Primary Driver: AlphaMissense




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-B HLA-B chr6:31,356,717 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.251 Showing 3 of 4 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-B belongs to the HLA class I heavy chain paralogues. This class I molecule is a heterodimer consisting of a heavy chain and a light chain (beta-2 microglobulin). The heavy chain is anchored in the membrane. Class I molecules play a central role in the immune system by presenting peptides derived from the endoplasmic reticulum lumen. They are expressed in nearly all cells. The heavy chain is approximately 45 kDa and its gene contains 8 exons. Exon 1 encodes the leader peptide, exon 2 and 3 encode the alpha1 and alpha2 domains, which both bind the peptide, exon 4 encodes the alpha3 domain, exon 5 encodes the transmembrane region and exons 6 and 7 encode the cytoplasmic tail. Polymorphisms within exon 2 and exon 3 are responsible for the peptide binding specificity of each class one molecule. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. Hundreds of HLA-B alleles have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9266161 p.Leu105Pro | MIS, p.Leu105Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 22.5 | – | 20.0 (AlphaMissense) | 0.20 / 0.79 | chr6:31356717 AF: 0.0006% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

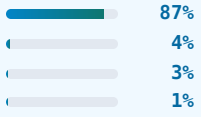
Transcript: ENST00000412585.7
cDNA (HGVS): c.314T>C
Protein (HGVS): p.Leu105Pro
VAF / Read Depth: VAF 0.55 · DP –
Phased GT: 0/1 (Unphased)

**3. SPLICEAI & CLINICAL REVIEW**

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0006%

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 20.0 (Raw: 0.559 · Top 1.00%)
Primary Driver: AlphaMissense




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9266144 p.Arg180Gly | MIS, p.Arg180Gly | Homozygous | N/A | Not reviewed | Tier2 | 0.0 | – | 10.6 (AlphaMissense) | 0.25 / 0.22 | chr6:31356248 AF: 0.0002% |
| rs1131212 p.Gln94His | MIS, p.Gln94His | Homozygous | N/A | Not reviewed | Tier2 | 7.1 | – | 11.3 (AlphaMissense) | 0.24 / 0.28 | chr6:31356749 AF: 0.2441% |

## TEK TEK chr9:27,185,615 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.208 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a receptor that belongs to the protein tyrosine kinase Tie2 family. The encoded protein possesses a unique extracellular region that contains two immunoglobulin-like domains, three epidermal growth factor (EGF)-like domains and three fibronectin type III repeats. The ligand angiopoietin-1 binds to this receptor and mediates a signaling pathway that functions in embryonic vascular development. Mutations in this gene are associated with inherited venous malformations of the skin and mucous membranes. Alternative splicing results in multiple transcript variants. Additional alternatively spliced transcript variants of this gene have been described, but their full-length nature is not known. [provided by RefSeq, Feb 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs189543659 p.Asn438Ile | MIS, p.Asn438Ile | Heterozygous | Unphased | Not reviewed | Tier1 | 24.3 | — | 21.9 (Cactus) | 0.21 / 0.21 | chr9:27185615 AF: 0.0011% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000380036.10 |
| cDNA (HGVS): | c.1313A>T |
| Protein (HGVS): | p.Asn438Ile |
| VAF / Read Depth: | VAF 0.50 · DP — |
| Phased GT: | 0/1 (Unphased) |

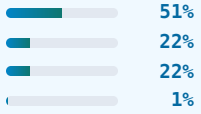
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0011% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.9 (Raw: 0.695 · Top 0.65%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NPC2 NPC2 chr14:74,480,701 Multisystem 🔥 Tier-1 / Pathogenic Gene

REVEL: 0.207 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein containing a lipid recognition domain. The encoded protein may function in regulating the transport of cholesterol through the late endosomal/lysosomal system. Mutations in this gene have been associated with Niemann-Pick disease, type C2 and frontal lobe atrophy. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs140130028 c.441+1G>A | SPL | Heterozygous | Maternal | Conflicting classifications of pathogenicity | Tier1 | 23.2 | 0.98 | 26.3 (Splicing) | 0.21 / — | chr14:74480701 AF: 0.6316% |

### 1. INHERITANCE & OMIM PATHOLOGY

**NPC2-related disorder**

Complex / Multifactorial OMIM #257220; 607625 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000555619.6 |
| cDNA (HGVS): | c.441+1G>A |
| Protein (HGVS): | — |
| VAF / Read Depth: | VAF 0.38 · DP — |
| Phased GT: | 0\|1 (Maternal) |

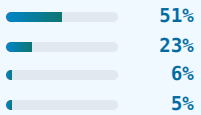
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.10 DL:0.98 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#100734 ↗ |
| gnomAD v4 AF: | 0.6316% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## 8. Rare & Private Variants (gnomAD AF ≤ 1%) — Tier-1 & Top High-Impact Genes

Rare coding and regulatory variants (gnomAD v4 allele frequency ≤ 1.0%), prioritizing Tier-1 / Pathogenic and high-CADD/AVI rare alleles.

**Selected Scope:** 32 Genes (32 Tier-1/Pathogenic)
Filtered from 743 matching genes in view

### TAT TAT chr16:71,576,247 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Min gnomAD AF: 0.0017% · CADD 36.0** Showing 2 of 2 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This nuclear gene encodes a mitochondrial protein tyrosine aminotransferase which is present in the liver and catalyzes the conversion of L-tyrosine into p-hydroxyphenylpyruvate. Mutations in this gene cause tyrosinemia (type II, Richner-Hanhart syndrome), a disorder accompanied by major skin and corneal lesions, with possible cognitive disability. A regulator gene for tyrosine aminotransferase is X-linked. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs118203914<br>p.Arg57Ter | STG, p.Arg57Ter | Heterozygous | Maternal | Pathogenic | Tier1 | 36.0 | – | 38.8 (Protein Termination) | – / – | chr16:71576247<br>AF: 0.0017% |

**1. INHERITANCE & OMIM PATHOLOGY**

**Tyrosinemia type II**
Complex / Multifactorial OMIM #276600 ↗

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000355962.5 |
| cDNA (HGVS): | c.169C>T |
| Protein (HGVS): | p.Arg57Ter |
| VAF / Read Depth: | VAF 0.46 · DP – |
| Phased GT: | 0\|1 (Maternal) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#402 ↗ |
| gnomAD v4 AF: | 0.0017% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 38.8 (Raw: 2.285 · Top 0.01%)
Primary Driver: Protein Termination

66%
24%
8%
1%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs572677025<br>c.*409G>A | UT3 | Heterozygous | Unphased | Uncertain significance | Tier2 | 0.8 | – | 5.8 (Splicing) | – / – | chr16:71567735<br>AF: 0.4009% |

### OGFOD3 OGFOD3 chr17:82,411,454 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Min gnomAD AF: 0.6286% · CADD 34.0** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable several functions, including L-ascorbic acid binding activity; dioxygenase activity; and iron ion binding activity. Located in membrane. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs150938774<br>c.380+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 34.0 | 0.96 | 33.8 (Splicing) | – / – | chr17:82411454<br>AF: 0.6286% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000313056.10 |
| cDNA (HGVS): | c.380+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.42 DL:0.96 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4429500 ↗ |
| gnomAD v4 AF: | 0.6286% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 33.8 (Raw: 1.882 · Top 0.04%)
Primary Driver: Splicing

54%
27%
9%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB5 HLA-DRB5 chr6:32,520,043 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0000% · CADD 33.0 Showing 3 of 25 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB5 belongs to the HLA class II beta chain paralogues. This class II molecule is a heterodimer consisting of an alpha (DRA) and a beta (DRB) chain, both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa and its gene contains 6 exons. Exon one encodes the leader peptide, exons 2 and 3 encode the two extracellular domains, exon 4 encodes the transmembrane domain and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. There are multiple pseudogenes of this gene. [provided by RefSeq, Feb 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs774541765<br>c.100+17G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.7 | 0.68 | 18.9 (Splicing) | – / – | chr6:32530108<br>AF: 0.0027% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

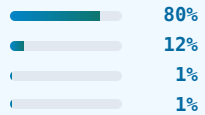
Transcript: ENST00000374975.4
cDNA (HGVS): c.100+17G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.46 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.68 DL:0.01
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0027%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 18.9 (Raw: 0.503 · Top 1.28%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs201576424<br>c.101-1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.93 | 28.0 (Splicing) | – / – | chr6:32522175<br>AF: 0.0063% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

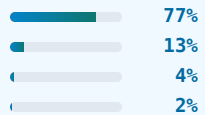
Transcript: ENST00000374975.4
cDNA (HGVS): c.101-1G>A
Protein (HGVS): –
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.65 AL:0.93 DG:0.00 DL:0.00
ClinVar Review: criteria provided
ClinVar VCV: VCV#4326162 ↗
gnomAD v4 AF: 0.0063%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.0 (Raw: 1.444 · Top 0.16%)
Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs796784269<br>c.371-393T>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 2.8 | 0.94 | 16.6 (Splicing) | – / – | chr6:32520044<br>AF: 0.1426% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Rheumatoid Arthritis Susceptibility**
Complex / Polygenic OMIM #604305 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

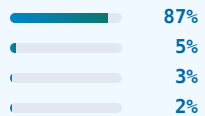
Transcript: ENST00000374975.4
cDNA (HGVS): c.371-393T>G
Protein (HGVS): –
VAF / Read Depth: VAF 0.38 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.94 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.1426%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 16.6 (Raw: 0.386 · Top 2.21%)
Primary Driver: Splicing



| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NIPSNAP2 NIPSNAP2 chr7:55,994,989 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0075% · CADD 33.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the NipSnap family of proteins that may be involved in vesicular transport. The encoded protein is localized to mitochondria and plays a role in oxidative phosphorylation. A pseudogene of this gene is located on the long arm of chromosome 2. Alternatively spliced transcript variants encoding multiple isoforms have been observed for this gene. [provided by RefSeq, Feb 2011]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs748801121 c.712+1G>T | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 33.0 | 0.97 | 33.8 (Splicing) | – / – | chr7:55994989 AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000322090.8 |
| cDNA (HGVS): | c.712+1G>T |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.50 · DP – |
| Phased GT: | 0/1 (Unphased) |

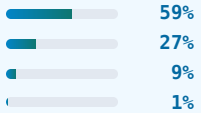
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.44 DL:0.97 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0075% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 33.8 (Raw: 1.881 · Top 0.04%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SLIT2 SLIT2 chr4:20,567,262 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0009% · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the slit family of secreted glycoproteins, which are ligands for the Robo family of immunoglobulin receptors. Slit proteins play highly conserved roles in axon guidance and neuronal migration and may also have functions during other cell migration processes including leukocyte migration. Members of the slit family are characterized by an N-terminal signal peptide, four leucine-rich repeats, nine epidermal growth factor repeats, and a C-terminal cysteine knot. Proteolytic processing of this protein gives rise to an N-terminal fragment that contains the four leucine-rich repeats and five epidermal growth factor repeats and a C-terminal fragment that contains four epidermal growth factor repeats and the cysteine knot. Both full length and cleaved proteins are secreted extracellularly and can function in axon repulsion as well as other specific processes. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2015]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1020555095 p.Gly909Asp | MIS, p.Gly909Asp | Heterozygous | Unphased | Not reviewed | Tier1 | 32.0 | – | 23.4 (Cactus) | 0.74 / 0.35 | chr4:20567262 AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000504154.6 |
| cDNA (HGVS): | c.2726G>A |
| Protein (HGVS): | p.Gly909Asp |
| VAF / Read Depth: | VAF 0.54 · DP – |
| Phased GT: | 0/1 (Unphased) |

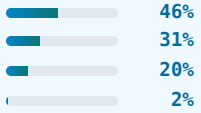
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0009% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.4 (Raw: 0.859 · Top 0.46%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## AMPD1 AMPD1 chr1:114,677,435 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0075% · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Adenosine monophosphate deaminase 1 catalyzes the deamination of AMP to IMP in skeletal muscle and plays an important role in the purine nucleotide cycle. Two other genes have been identified, AMPD2 and AMPD3, for the liver- and erythocyte-specific isoforms, respectively. Deficiency of the muscle-specific enzyme is apparently a common cause of exercise-induced myopathy and probably the most common cause of metabolic myopathy in the human. Alternatively spliced transcript variants encoding different isoforms have been identified in this gene.[provided by RefSeq, Feb 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs764950035 p.Glu435Gly | MIS, p.Glu435Gly | Heterozygous | Unphased | Uncertain significance | Tier1 | 32.0 | – | 28.2 (AlphaMissense) | 0.95 / 0.92 | chr1:114677435 AF: 0.0075% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Muscle AMP deaminase deficiency**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

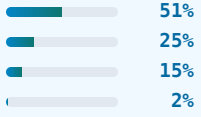
Transcript: ENST00000520113.7
cDNA (HGVS): c.1304A>G
Protein (HGVS): p.Glu435Gly
VAF / Read Depth: VAF 0.49 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#2173226 ↗
gnomAD v4 AF: 0.0075%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 28.2 (Raw: 1.454 · Top 0.15%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TMEM43 TMEM43 chr3:14,139,244 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0126% · CADD 32.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to the TMEM43 family. Defects in this gene are the cause of familial arrhythmogenic right ventricular dysplasia type 5 (ARVD5), also known as arrhythmogenic right ventricular cardiomyopathy type 5 (ARVC5). Arrhythmogenic right ventricular dysplasia is an inherited disorder, often involving both ventricles, and is characterized by ventricular tachycardia, heart failure, sudden cardiac death, and fibrofatty replacement of cardiomyocytes. This gene contains a response element for PPAR gamma (an adipogenic transcription factor), which may explain the fibrofatty replacement of the myocardium, a characteristic pathological finding in ARVC. [provided by RefSeq, Oct 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs199526104 p.Trp316Ser | MIS, p.Trp316Ser | Heterozygous | Paternal | Conflicting classifications of pathogenicity | Tier1 | 32.0 | – | 26.5 (AlphaMissense) | 0.53 / 0.76 | chr3:14139244 AF: 0.0126% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**
Complex / Multifactorial OMIM #604400 ↗

**TMEM43-related disorder**
Complex / Multifactorial OMIM #604400 ↗

**Auditory neuropathy, autosomal dominant 3**
Autosomal Dominant OMIM #604400 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

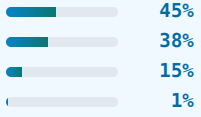
Transcript: ENST00000306077.5
cDNA (HGVS): c.947G>C
Protein (HGVS): p.Trp316Ser
VAF / Read Depth: VAF 0.44 · DP –
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#46155 ↗
gnomAD v4 AF: 0.0126%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.5 (Raw: 1.335 · Top 0.22%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## LDHAL6A LDHAL6A chr11:18,475,640 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Min gnomAD AF: 0.1631% · CADD 31.0** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables L-lactate dehydrogenase (NAD+) activity. Predicted to be involved in lactate metabolic process and pyruvate catabolic process. Located in cytoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs145833201<br>c.592+1G>A | SPL | Heterozygous | Unphased | Not reviewed | Tier1 | 31.0 | 0.69 | 23.3 (Splicing) | – / – | chr11:18475640<br>AF: 0.1631% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000280706.3 |
| cDNA (HGVS): | c.592+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.36 · DP – |
| Phased GT: | 0/1 (Unphased) |

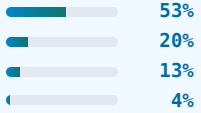
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.69 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4370113 ↗ |
| gnomAD v4 AF: | 0.1631% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.3 (Raw: 0.845 · Top 0.47%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## DMAP1 DMAP1 chr1:44,218,751 Multisystem 🔥 Tier-1 / Pathogenic Gene

**Min gnomAD AF: 0.0001% · CADD 29.7** Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a subunit of several, distinct complexes involved in the repression or activation of transcription. The encoded protein can independently repress transcription and is targeted to replication foci throughout S phase by interacting directly with the N-terminus of DNA methyltransferase 1. During late S phase, histone deacetylase 2 is added to this complex, providing a means to deacetylate histones in transcriptionally inactive heterochromatin following replication. The encoded protein is also a component of the nucleosome acetyltransferase of H4 complex and interacts with the transcriptional corepressor tumor susceptibility gene 101 and the pro-apoptotic death-associated protein 6, among others. Alternatively spliced transcript variants encoding the same protein have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| chr1:44218751<br>p.Glu239Gly | MIS, p.Glu239Gly | Heterozygous | Unphased | Not reviewed | Tier1 | 29.7 | – | 23.4 (Cactus) | 0.38 / 0.38 | chr1:44218751<br>AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000372289.7 |
| cDNA (HGVS): | c.716A>G |
| Protein (HGVS): | p.Glu239Gly |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0/1 (Unphased) |

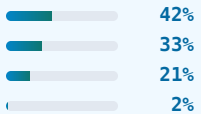
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0001% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 23.4 (Raw: 0.854 · Top 0.46%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

**PLK4** PLK4 chr4:127,881,151 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.1341% · CADD 28.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the polo family of serine/threonine protein kinases. The protein localizes to centrioles, complex microtubule-based structures found in centrosomes, and regulates centriole duplication during the cell cycle. Three alternatively spliced transcript variants that encode different protein isoforms have been found for this gene. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs149003893 p.Gly6Glu | MIS, p.Gly6Glu | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 28.5 | – | 29.8 (AlphaMissense) | 0.36 / 0.99 | chr4:127881151 AF: 0.1341% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Inborn genetic diseases**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000270861.10 |
| cDNA (HGVS): | c.17G>A |
| Protein (HGVS): | p.Gly6Glu |
| VAF / Read Depth: | VAF 0.45 · DP – |
| Phased GT: | 0/1 (Unphased) |

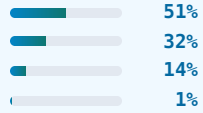
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#436338 ↗ |
| gnomAD v4 AF: | 0.1341% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 29.8 (Raw: 1.555 · Top 0.11%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-DRB1 HLA-DRB1 chr6:32,582,135 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene Min gnomAD AF: 0.0000% · CADD 28.0 Showing 3 of 30 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-DRB1 belongs to the HLA class II beta chain paralogs. The class II molecule is a heterodimer consisting of an alpha (DRA) and a beta chain (DRB), both anchored in the membrane. It plays a central role in the immune system by presenting peptides derived from extracellular proteins. Class II molecules are expressed in antigen presenting cells. The beta chain is approximately 26-28 kDa. It is encoded by 6 exons. Exon one encodes the leader peptide; exons 2 and 3 encode the two extracellular domains; exon 4 encodes the transmembrane domain; and exon 5 encodes the cytoplasmic tail. Within the DR molecule the beta chain contains all the polymorphisms specifying the peptide binding specificities. Hundreds of DRB1 alleles have been described and some alleles have increased frequencies associated with certain diseases or conditions. For example, DRB1*1302 has been related to acute and chronic hepatitis B virus persistence. There are multiple pseudogenes of this gene. [provided by RefSeq, Jul 2020]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs16822805<br>p.Ala103Pro | MIS, p.Ala103Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 0.0 | – | 20.2 (AlphaMissense) | 0.21 / 0.78 | chr6:32584172<br>AF: 0.1762% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.307G>C |
| Protein (HGVS): | p.Ala103Pro |
| VAF / Read Depth: | VAF 0.53 · DP – |
| Phased GT: | 0/1 (Unphased) |

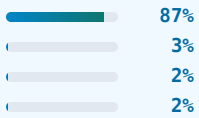
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#691505 ↗ |
| gnomAD v4 AF: | 0.1762% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 20.2 (Raw: 0.573 · Top 0.95%)

Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28723976<br>c.371-297A>G | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 7.4 | 0.58 | 12.9 (Splicing) | – / – | chr6:32582135<br>AF: 0.7402% |

**1. INHERITANCE & OMIM PATHOLOGY**

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

**2. TRANSCRIPT, PHASING & DEPTH**

| | |
|---|---|
| Transcript: | ENST00000360004.6 |
| cDNA (HGVS): | c.371-297A>G |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.41 · DP – |
| Phased GT: | 0/1 (Unphased) |

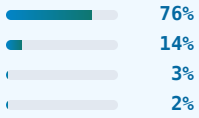
**3. SPLICEAI & CLINICAL REVIEW**

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.58 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.7402% |

**4. ALPHAGENOME 18-MODALITY ATLAS**

AlphaGenome AVI (Phred): 12.9 (Raw: 0.248 · Top 5.08%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9269730<br>c.788-526C>T | INT | Homozygous | N/A | Not reviewed | Tier3 | 2.6 | – | 3.3 (CAGE) | – / – | chr6:32579630<br>AF: 0.0000% |

## SLC27A3 SLC27A3 chr1:153,776,015 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0001% · CADD 27.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene belongs to a family of integral membrane proteins and encodes a protein that is involved in lipid metabolism. The increased expression of this gene in human neural stem cells derived from induced pluripotent stem cells suggests that it plays an important role in early brain development. Naturally occurring mutations in this gene are associated with autism spectrum disorders. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Feb 2017]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs756978807 p.Leu173Pro | MIS, p.Leu173Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 26.9 (AlphaMissense) | 0.41 / 0.94 | chr1:153776015 AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000624995.4
cDNA (HGVS): c.518T>C
Protein (HGVS): p.Leu173Pro
VAF / Read Depth: VAF 0.23 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0001%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.9 (Raw: 1.371 · Top 0.21%)
Primary Driver: AlphaMissense

54%
24%
17%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAHD1 FAHD1 chr16:1,827,542 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0037% · CADD 27.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables hydrolase activity, acting on carbon-carbon bonds, in ketonic substances; oxaloacetate decarboxylase activity; and oxaloacetate tautomerase activity. Involved in oxaloacetate metabolic process. Located in cytosol; mitochondrion; and nucleoplasm. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs200357596 p.Ala102Ser | MIS, p.Ala102Ser | Heterozygous | Unphased | Not reviewed | Tier1 | 27.2 | – | 22.1 (Cactus) | 0.71 / 0.24 | chr16:1827542 AF: 0.0037% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

Transcript: ENST00000427358.5
cDNA (HGVS): c.304G>T
Protein (HGVS): p.Ala102Ser
VAF / Read Depth: VAF 0.49 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0037%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.1 (Raw: 0.710 · Top 0.62%)
Primary Driver: Cactus

45%
25%
22%
2%

🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ACSM2B ACSM2B chr16:20,540,665 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.4705% · CADD 27.6 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Enables benzoate-CoA ligase activity. Predicted to be involved in acyl-CoA metabolic process and fatty acid biosynthetic process. Located in mitochondrion. [provided by Alliance of Genome Resources, Jul 2025]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs139253186 p.Tyr540Asn | MIS, p.Tyr540Asn | Heterozygous | Unphased | Not reviewed | Tier1 | 27.6 | — | 24.4 (AlphaMissense) | 0.40 / 0.58 | chr16:20540665 AF: 0.4705% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

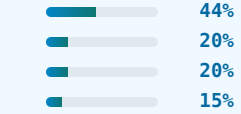
Transcript: ENST00000329697.10
cDNA (HGVS): c.1618T>A
Protein (HGVS): p.Tyr540Asn
VAF / Read Depth: VAF 0.54 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.4705%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.4 (Raw: 1.012 · Top 0.37%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## POLG POLG chr15:89,323,460 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.1619% · CADD 26.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Mitochondrial DNA polymerase is heterotrimeric, consisting of a homodimer of accessory subunits plus a catalytic subunit. The protein encoded by this gene is the catalytic subunit of mitochondrial DNA polymerase. The encoded protein contains a polyglutamine tract near its N-terminus that may be polymorphic. Defects in this gene are a cause of progressive external ophthalmoplegia with mitochondrial DNA deletions 1 (PEOA1), sensory ataxic neuropathy dysarthria and ophthalmoparesis (SANDO), Alpers-Huttenlocher syndrome (AHS), and mitochondrial neurogastrointestinal encephalopathy syndrome (MNGIE). Two transcript variants encoding the same protein have been found for this gene. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs121918054 p.Gly737Arg | MIS, p.Gly737Arg | Heterozygous | Paternal | Pathogenic/Likely pathogenic | Tier1 | 26.3 | — | 30.3 (AlphaMissense) | 0.94 / 0.94 | chr15:89323460 AF: 0.1619% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Spinocerebellar ataxia with epilepsy**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Sensory ataxic neuropathy, dysarthria, and ophthalmoparesis**
Complex / Multifactorial OMIM #203700; 607459; 613662; 157640; 258450 ↗

**Progressive external ophthalmoplegia with mitochondrial DNA deletions, autosomal dominant 1**
Autosomal Dominant OMIM #203700; 607459; 613662; 157640; 258450 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

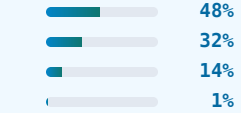
Transcript: ENST00000268124.11
cDNA (HGVS): c.2209G>C
Protein (HGVS): p.Gly737Arg
VAF / Read Depth: VAF 0.61 · DP —
Phased GT: 1|0 (Paternal)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#13513 ↗
gnomAD v4 AF: 0.1619%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 30.3 (Raw: 1.582 · Top 0.09%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## WDHD1 WDHD1 chr14:55,002,157 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0190% · CADD 25.8 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene contains multiple N-terminal WD40 domains and a C-terminal high mobility group (HMG) box. WD40 domains are found in a variety of eukaryotic proteins and may function as adaptor/regulatory modules in signal transduction, pre-mRNA processing and cytoskeleton assembly. HMG boxes are found in many eukaryotic proteins involved in chromatin assembly, transcription and replication. Alternative splicing results in two transcript variants encoding different isoforms. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs374497720 p.Val210Ala | MIS, p.Val210Ala | Heterozygous | Unphased | Not reviewed | Tier1 | 25.8 | – | 24.5 (AlphaMissense) | 0.27 / 0.58 | chr14:55002157 AF: 0.0190% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

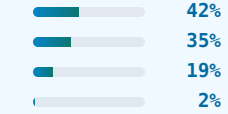
Transcript: ENST00000360586.8
cDNA (HGVS): c.629T>C
Protein (HGVS): p.Val210Ala
VAF / Read Depth: VAF 0.50 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0190%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.5 (Raw: 1.042 · Top 0.35%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## MID1IP1 MID1IP1 chrX:38,805,491 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.3491% · CADD 26.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Predicted to enable enzyme activator activity. Predicted to be involved in several processes, including negative regulation of microtubule depolymerization; positive regulation of fatty acid biosynthetic process; and positive regulation of ligase activity. Predicted to be located in microtubule cytoskeleton. Predicted to be active in cytosol and nucleus. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs143008624 p.Gly182Asp | MIS, p.Gly182Asp | Heterozygous | Unphased | Benign | Tier1 | 26.1 | – | 27.1 (AlphaMissense) | 0.63 / 0.87 | chrX:38805491 AF: 0.3491% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

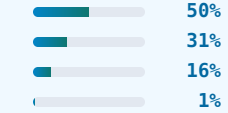
Transcript: ENST00000614558.3
cDNA (HGVS): c.545G>A
Protein (HGVS): p.Gly182Asp
VAF / Read Depth: VAF 0.48 · DP –
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, single submitter
ClinVar VCV: VCV#711083 ↗
gnomAD v4 AF: 0.3491%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.1 (Raw: 1.393 · Top 0.19%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## FAAP24 FAAP24 chr19:32,974,193 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.5937% · CADD 25.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 FAAP24 is a component of the Fanconi anemia (FA) core complex (see MIM 227650), which plays a crucial role in DNA damage response (Ciccia et al., 2007 [PubMed 17289582]).[supplied by OMIM, Mar 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs36017455 p.Ser126Phe | MIS, p.Ser126Phe | Heterozygous | Unphased | Benign | Tier1 | 25.7 | — | 27.0 (AlphaMissense) | — / 0.69 | chr19:32974193 AF: 0.5937% |

### 1. INHERITANCE & OMIM PATHOLOGY

**FAAP24-related disorder**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

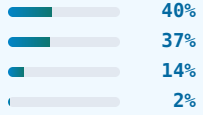
Transcript: ENST00000588258.6
cDNA (HGVS): c.377C>T
Protein (HGVS): p.Ser126Phe
VAF / Read Depth: VAF 0.44 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#784831 ↗
gnomAD v4 AF: 0.5937%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.0 (Raw: 1.385 · Top 0.20%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CTH CTH chr1:70,415,987 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.8026% · CADD 25.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a cytoplasmic enzyme in the trans-sulfuration pathway that converts cystathione derived from methionine into cysteine. Glutathione synthesis in the liver is dependent upon the availability of cysteine. Mutations in this gene cause cystathioninuria. Alternative splicing of this gene results in three transcript variants encoding different isoforms. [provided by RefSeq, Jun 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs28941785 p.Thr67Ile | MIS, p.Thr67Ile | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 25.5 | — | 27.8 (AlphaMissense) | 0.93 / 0.81 | chr1:70415987 AF: 0.8026% |

### 1. INHERITANCE & OMIM PATHOLOGY

**CTH-related disorder**
Complex / Multifactorial OMIM #219500 ↗

**Cystathioninuria**
Complex / Multifactorial OMIM #219500 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

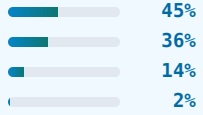
Transcript: ENST00000370938.8
cDNA (HGVS): c.200C>T
Protein (HGVS): p.Thr67Ile
VAF / Read Depth: VAF 0.46 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided, conflicting classifications
ClinVar VCV: VCV#2939 ↗
gnomAD v4 AF: 0.8026%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 27.8 (Raw: 1.434 · Top 0.17%)
Primary Driver: AlphaMissense




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## CHST14 CHST14 chr15:40,472,259 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0004% · CADD 24.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a member of the HNK-1 family of sulfotransferases. The encoded protein transfers sulfate to the C-4 hydroxyl of N-acetylgalactosamine residues in dermatan sulfate. Mutations in this gene have been associated with adducted thumb-clubfoot syndrome.[provided by RefSeq, Mar 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1463514561 p.Gln349Pro | MIS, p.Gln349Pro | Heterozygous | Unphased | Uncertain significance | Tier1 | 24.3 | – | 22.9 (Cactus) | 0.54 / 0.20 | chr15:40472259 AF: 0.0004% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Cardiovascular phenotype**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000306243.7 |
| cDNA (HGVS): | c.1046A>C |
| Protein (HGVS): | p.Gln349Pro |
| VAF / Read Depth: | VAF 0.52 · DP – |
| Phased GT: | 0/1 (Unphased) |

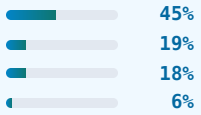
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, single submitter |
| ClinVar VCV: | VCV#3229028 ↗ |
| gnomAD v4 AF: | 0.0004% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 22.9 (Raw: 0.798 · Top 0.51%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## TEK TEK chr9:27,185,615 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0011% · CADD 24.3 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a receptor that belongs to the protein tyrosine kinase Tie2 family. The encoded protein possesses a unique extracellular region that contains two immunoglobulin-like domains, three epidermal growth factor (EGF)-like domains and three fibronectin type III repeats. The ligand angiopoietin-1 binds to this receptor and mediates a signaling pathway that functions in embryonic vascular development. Mutations in this gene are associated with inherited venous malformations of the skin and mucous membranes. Alternative splicing results in multiple transcript variants. Additional alternatively spliced transcript variants of this gene have been described, but their full-length nature is not known. [provided by RefSeq, Feb 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs189543659 p.Asn438Ile | MIS, p.Asn438Ile | Heterozygous | Unphased | Not reviewed | Tier1 | 24.3 | – | 21.9 (Cactus) | 0.21 / 0.21 | chr9:27185615 AF: 0.0011% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000380036.10 |
| cDNA (HGVS): | c.1313A>T |
| Protein (HGVS): | p.Asn438Ile |
| VAF / Read Depth: | VAF 0.50 · DP – |
| Phased GT: | 0/1 (Unphased) |

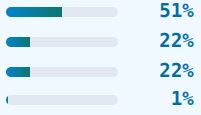
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0011% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.9 (Raw: 0.695 · Top 0.65%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## SUMF1 SUMF1 chr3:4,362,213 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0009% · CADD 23.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes an enzyme that catalyzes the hydrolysis of sulfate esters by oxidizing a cysteine residue in the substrate sulfatase to an active site 3-oxoalanine residue, which is also known as C-alpha-formylglycine. Mutations in this gene cause multiple sulfatase deficiency, a lysosomal storage disorder. Alternative splicing results in multiple transcript variants. [provided by RefSeq, Sep 2009]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs376126385<br>p.Asn352Lys | MIS, p.Asn352Lys | Heterozygous | Unphased | Uncertain significance | Tier1 | 23.9 | — | 25.9 (AlphaMissense) | 0.64 / 1.00 | chr3:4362213<br>AF: 0.0009% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Multiple sulfatase deficiency**

Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

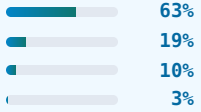
Transcript: **ENST00000272902.10**
cDNA (HGVS): **c.1056C>G**
Protein (HGVS): **p.Asn352Lys**
VAF / Read Depth: **VAF 0.66 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided, single submitter**
ClinVar VCV: VCV#2161926 ↗
gnomAD v4 AF: **0.0009%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **25.9** (Raw: 1.256 · Top 0.26%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PGK1 PGK1 chrX:78,114,140 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0008% · CADD 22.7 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a glycolytic enzyme that catalyzes the conversion of 1,3-diphosphoglycerate to 3-phosphoglycerate. The encoded protein may also act as a cofactor for polymerase alpha. Additionally, this protein is secreted by tumor cells where it participates in angiogenesis by functioning to reduce disulfide bonds in the serine protease, plasmin, which consequently leads to the release of the tumor blood vessel inhibitor angiostatin. The encoded protein has been identified as a moonlighting protein based on its ability to perform mechanistically distinct functions. Deficiency of the enzyme is associated with a wide range of clinical phenotypes hemolytic anemia and neurological impairment. Pseudogenes of this gene have been defined on chromosomes 19, 21 and the X chromosome. [provided by RefSeq, Jan 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| chrX:78114140<br>p.Lys133Glu | MIS, p.Lys133Glu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.7 | — | 24.3 (AlphaMissense) | 0.54 / 0.57 | chrX:78114140<br>AF: 0.0008% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

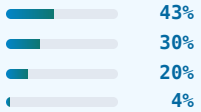
Transcript: **ENST00000373316.5**
cDNA (HGVS): **c.397A>G**
Protein (HGVS): **p.Lys133Glu**
VAF / Read Depth: **VAF 0.58 · DP —**
Phased GT: **0/1 (Unphased)**

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: **AG:0.00 AL:0.00 DG:0.00 DL:0.00**
ClinVar Review: **criteria provided**
gnomAD v4 AF: **0.0008%**

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): **24.3** (Raw: 0.993 · Top 0.38%)
Primary Driver: **AlphaMissense**




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## NPC2 NPC2 chr14:74,480,701 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.6316% · CADD 23.2 | Showing 1 of 1 variants | OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a protein containing a lipid recognition domain. The encoded protein may function in regulating the transport of cholesterol through the late endosomal/lysosomal system. Mutations in this gene have been associated with Niemann-Pick disease, type C2 and frontal lobe atrophy. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs140130028<br>c.441+1G>A | SPL | Heterozygous | Maternal | Conflicting classifications of pathogenicity | Tier1 | 23.2 | 0.98 | 26.3 (Splicing) | 0.21 / – | chr14:74480701<br>AF: 0.6316% |

### 1. INHERITANCE & OMIM PATHOLOGY

**NPC2-related disorder**

Complex / Multifactorial OMIM #257220; 607625 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000555619.6 |
| cDNA (HGVS): | c.441+1G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.38 · DP – |
| Phased GT: | 0\|1 (Maternal) |

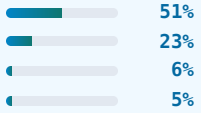
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.10 DL:0.98 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#100734 ↗ |
| gnomAD v4 AF: | 0.6316% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 26.3 (Raw: 1.312 · Top 0.23%)

Primary Driver: Splicing




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## HLA-B HLA-B chr6:31,356,717 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0000% · CADD 22.5 | Showing 3 of 5 variants | OMIM ↗ NCBI ↗ GeneCards ↗

📖 HLA-B belongs to the HLA class I heavy chain paralogues. This class I molecule is a heterodimer consisting of a heavy chain and a light chain (beta-2 microglobulin). The heavy chain is anchored in the membrane. Class I molecules play a central role in the immune system by presenting peptides derived from the endoplasmic reticulum lumen. They are expressed in nearly all cells. The heavy chain is approximately 45 kDa and its gene contains 8 exons. Exon 1 encodes the leader peptide, exon 2 and 3 encode the alpha1 and alpha2 domains, which both bind the peptide, exon 4 encodes the alpha3 domain, exon 5 encodes the transmembrane region and exons 6 and 7 encode the cytoplasmic tail. Polymorphisms within exon 2 and exon 3 are responsible for the peptide binding specificity of each class one molecule. Typing for these polymorphisms is routinely done for bone marrow and kidney transplantation. Hundreds of HLA-B alleles have been described. [provided by RefSeq, Jul 2008]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9266161<br>p.Leu105Pro | MIS, p.Leu105Pro | Heterozygous | Unphased | Not reviewed | Tier1 | 22.5 | – | 20.0 (AlphaMissense) | 0.20 / 0.79 | chr6:31356717<br>AF: 0.0006% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000412585.7 |
| cDNA (HGVS): | c.314T>C |
| Protein (HGVS): | p.Leu105Pro |
| VAF / Read Depth: | VAF 0.55 · DP – |
| Phased GT: | 0/1 (Unphased) |

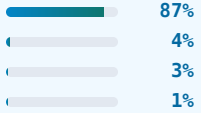
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0006% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 20.0 (Raw: 0.559 · Top 1.00%)

Primary Driver: AlphaMissense




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| rs9266162<br>p.Leu105Val | MIS, p.Leu105Val | Heterozygous | Unphased | Not reviewed | Tier2 | 8.9 | – | 9.0 (AlphaMissense) | 0.19 / 0.17 | chr6:31356718<br>AF: 0.0000% |
| rs9266144<br>p.Arg180Gly | MIS, p.Arg180Gly | Homozygous | N/A | Not reviewed | Tier2 | 0.0 | – | 10.6 (AlphaMissense) | 0.25 / 0.22 | chr6:31356248<br>AF: 0.0002% |

## GUSB GUSB chr7:65,960,973 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0003% · CADD 22.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a hydrolase that degrades glycosaminoglycans, including heparan sulfate, dermatan sulfate, and chondroitin-4,6-sulfate. The enzyme forms a homotetramer that is localized to the lysosome. Mutations in this gene result in mucopolysaccharidosis type VII. Alternative splicing results in multiple transcript variants. There are many pseudogenes of this locus in the human genome.[provided by RefSeq, May 2014]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1236992554 p.Trp627Leu | MIS, p.Trp627Leu | Heterozygous | Unphased | Not reviewed | Tier1 | 22.2 | — | 24.9 (Cactus) | 0.56 / 0.25 | chr7:65960973 AF: 0.0003% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

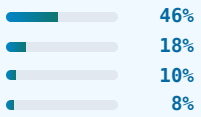
Transcript: ENST00000304895.9
cDNA (HGVS): c.1880G>T
Protein (HGVS): p.Trp627Leu
VAF / Read Depth: VAF 0.53 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.00 DL:0.00
ClinVar Review: criteria provided
gnomAD v4 AF: 0.0003%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 24.9 (Raw: 1.103 · Top 0.32%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## VPS35L VPS35L chr16:19,629,820 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.3238% · CADD 22.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 Involved in Golgi to plasma membrane transport and endocytic recycling. Located in endosome. Part of protein-containing complex. Implicated in Ritscher-Schinzel syndrome. [provided by Alliance of Genome Resources, Jun 2026]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs144908771 p.Thr518= | SYN, p.Thr518= | Heterozygous | Unphased | Likely benign | Tier1 | 22.5 | 0.64 | 21.4 (Cactus) | — / — | chr16:19629820 AF: 0.3238% |

### 1. INHERITANCE & OMIM PATHOLOGY

**VPS35L-related disorder**
Complex / Multifactorial

**Ritscher-Schinzel syndrome**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

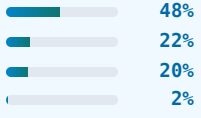
Transcript: ENST00000417362.7
cDNA (HGVS): c.1554G>A
Protein (HGVS): p.Thr518=
VAF / Read Depth: VAF 0.45 · DP —
Phased GT: 0/1 (Unphased)

### 3. SPLICEAI & CLINICAL REVIEW

SpliceAI Δ: AG:0.00 AL:0.00 DG:0.64 DL:0.01
ClinVar Review: criteria provided, multiple submitters, no conflicts
ClinVar VCV: VCV#2672622 ↗
gnomAD v4 AF: 0.3238%

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 21.4 (Raw: 0.657 · Top 0.73%)
Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## APOB APOB chr2:21,016,593 Heart & Cardiovascular 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.1326% · CADD 9.1 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene product is the main apolipoprotein of chylomicrons and low density lipoproteins (LDL), and is the ligand for the LDL receptor. It occurs in plasma as two main isoforms, apoB-48 and apoB-100: the former is synthesized exclusively in the gut and the latter in the liver. The intestinal and the hepatic forms of apoB are encoded by a single gene from a single, very long mRNA. The two isoforms share a common N-terminal sequence. The shorter apoB-48 protein is produced after RNA editing of the apoB-100 transcript at residue 2180 (CAA->UAA), resulting in the creation of a stop codon, and early translation termination. Mutations in this gene or its regulatory region cause hypobetalipoproteinemia, normotriglyceridemic hypobetalipoproteinemia, and hypercholesterolemia due to ligand-defective apoB, diseases affecting plasma cholesterol and apoB levels. [provided by RefSeq, Dec 2019]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs72653073<br>p.Leu1060= | SYN, p.Leu1060= | Heterozygous | Unphased | Conflicting classifications of pathogenicity | Tier1 | 9.1 | – | 11.5 (PhastCons 470) | – / – | chr2:21016593<br>AF: 0.1326% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Familial Hypercholesterolemia 2**
Autosomal Dominant OMIM #144010 ↗

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

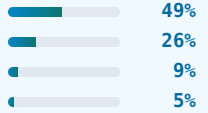
| | |
|---|---|
| Transcript: | ENST00000233242.5 |
| cDNA (HGVS): | c.3178T>C |
| Protein (HGVS): | p.Leu1060= |
| VAF / Read Depth: | VAF 0.61 · DP – |
| Phased GT: | 0/1 (Unphased) |

### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, conflicting classifications |
| ClinVar VCV: | VCV#334158 ↗ |
| gnomAD v4 AF: | 0.1326% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 11.5 (Raw: 0.199 · Top 7.16%)
Primary Driver: PhastCons 470




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## UCK2 UCK2 chr1:165,828,740 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.3704% · CADD 6.9 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a pyrimidine ribonucleoside kinase. The encoded protein (EC 2.7.1.48) catalyzes phosphorylation of uridine and cytidine to uridine monophosphate (UMP) and cytidine monophosphate (CMP), respectively.[provided by RefSeq, Oct 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs562989479<br>c.99+808G>T | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 6.9 | 0.57 | 19.2 (Splicing) | – / – | chr1:165828740<br>AF: 0.3704% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

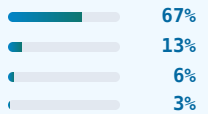
| | |
|---|---|
| Transcript: | ENST00000367879.9 |
| cDNA (HGVS): | c.99+808G>T |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.32 · DP – |
| Phased GT: | 0/1 (Unphased) |

### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.57 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.3704% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 19.2 (Raw: 0.517 · Top 1.20%)
Primary Driver: Splicing




Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## PLB1 PLB1 chr2:28,461,598 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0012% · CADD 3.5 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 This gene encodes a membrane-associated phospholipase that displays lysophospholipase and phospholipase A2 activities through removal of sn-1 and sn-2 fatty acids of glycerophospholipids. In addition, it displays lipase and retinyl ester hydrolase activities. The encoded protein is highly conserved and is composed of a large, glycosylated extracellular domain composed of four tandem homologous domains, followed by a hydrophobic segment that anchors the enzyme to the membrane and a short C-terminal cytoplasmic tail. This gene has been identified as a candidate rheumatoid arthritis risk gene. [provided by RefSeq, Jul 2016]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs56116810<br>c.-114+4401G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 3.5 | – | 3.6 (Cactus) | – / – | chr2:28461598<br>AF: 0.0012% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000416713.5 |
| cDNA (HGVS): | c.-114+4401G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.50 · DP – |
| Phased GT: | 0/1 (Unphased) |

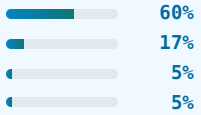
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided |
| gnomAD v4 AF: | 0.0012% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 3.6 (Raw: 0.009 · Top 43.90%)

Primary Driver: Cactus




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

## ADAMTS7 ADAMTS7 chr15:78,790,059 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.1293% · CADD 0.2 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene is a member of the ADAMTS (a disintegrin and metalloproteinase with thrombospondin motifs) family. Members of this family share several distinct protein modules, including a propeptide region, a metalloproteinase domain, a disintegrin-like domain, and a thrombospondin type 1 (TS) motif. Individual members of this family differ in the number of C-terminal TS motifs, and some have unique C-terminal domains. The encoded preproprotein is proteolytically processed to generate the mature enzyme. This enzyme contains two C-terminal TS motifs and may regulate vascular smooth muscle cell (VSMC) migration. Mutations in this gene may be associated with susceptibility to coronary artery disease. [provided by RefSeq, Feb 2016]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs142754252<br>c.1029-221G>A | INT | Heterozygous | Unphased | Not reviewed | Tier1 | 0.2 | 0.65 | 13.8 (Splicing) | – / – | chr15:78790059<br>AF: 0.1293% |

### 1. INHERITANCE & OMIM PATHOLOGY

Complex / multifactorial quantitative trait or regulatory modifier.

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000388820.5 |
| cDNA (HGVS): | c.1029-221G>A |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.42 · DP – |
| Phased GT: | 0/1 (Unphased) |

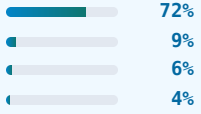
### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.65 AL:0.00 DG:0.02 DL:0.00 |
| ClinVar Review: | criteria provided |
| ClinVar VCV: | VCV#4406248 ↗ |
| gnomAD v4 AF: | 0.1293% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

AlphaGenome AVI (Phred): 13.8 (Raw: 0.277 · Top 4.16%)

Primary Driver: Splicing




🧬 Ref vs. Alt Track Predictions ↗ LitVar2 ↗

**ATM** ATM chr11:108,250,699 Multisystem 🔥 Tier-1 / Pathogenic Gene

Min gnomAD AF: 0.0001% · CADD 0.0 Showing 1 of 1 variants OMIM ↗ NCBI ↗ GeneCards ↗

📖 The protein encoded by this gene belongs to the PI3/PI4-kinase family. This protein is an important cell cycle checkpoint kinase that phosphorylates; thus, it functions as a regulator of a wide variety of downstream proteins, including tumor suppressor proteins p53 and BRCA1, checkpoint kinase CHK2, checkpoint proteins RAD17 and RAD9, and DNA repair protein NBS1. This protein and the closely related kinase ATR are thought to be master controllers of cell cycle checkpoint signaling pathways that are required for cell response to DNA damage and for genome stability. Mutations in this gene are associated with ataxia telangiectasia, an autosomal recessive disorder. [provided by RefSeq, Aug 2010]

| VARIANT / HGVS | CONSEQUENCE | ZYGOSITY / DP | PHASE | CLINVAR | TIER | CADD | SPLICEAI | ALPHAGENOME AVI | REVEL / AM | COORDINATE & AF |
|---|---|---|---|---|---|---|---|---|---|---|
| rs1565381646<br>c.1236-2del | SPL | Heterozygous | Unknown | Pathogenic/Likely pathogenic | Tier1 | – | – | – | – / – | chr11:108250699<br>AF: 0.0001% |

### 1. INHERITANCE & OMIM PATHOLOGY

**Hereditary cancer-predisposing syndrome**
Complex / Multifactorial

**Familial cancer of breast**
Complex / Multifactorial

**Ataxia-telangiectasia syndrome**
Complex / Multifactorial

Quadrant: **Curated**

### 2. TRANSCRIPT, PHASING & DEPTH

| | |
|---|---|
| Transcript: | ENST00000675843.1 |
| cDNA (HGVS): | c.1236-2del |
| Protein (HGVS): | – |
| VAF / Read Depth: | VAF 0.17 · DP – |
| Phased GT: | ./. (Unknown) |

### 3. SPLICEAI & CLINICAL REVIEW

| | |
|---|---|
| SpliceAI Δ: | AG:0.00 AL:0.00 DG:0.00 DL:0.00 |
| ClinVar Review: | criteria provided, multiple submitters, no conflicts |
| ClinVar VCV: | VCV#574589 ↗ |
| gnomAD v4 AF: | 0.0001% |

### 4. ALPHAGENOME 18-MODALITY ATLAS

Evaluated via CADD / SpliceAI / REVEL ensemble.

LitVar2 Literature ↗